

Academic Regulations & Detailed Syllabus

CHOICE BASED CREDIT SYSTEM

R22

ELECTRICAL AND ELECTRONICS ENGINEERING

Bachelor of Technology (B.Tech)

B. Tech. - Regular Four Year Degree Programme
(For batches admitted from the academic year 2022 - 2023)
&

B. Tech. - Lateral Entry Scheme
(For batches admitted from the academic year 2023 - 2024)



**MARRI
LAXMAN
REDDY**

GROUP OF INSTITUTIONS

MLRInstitute of Technology

(Autonomous)

Laxman Reddy Avenue, Dundigal

Hyderabad – 500043, Telangana State

www.mlrinstitutions.ac.in, Email: director@mlrinstitutions.ac.in

ACADEMIC REGULATIONS (R22)

B. Tech. - Regular Four Year Degree Programme (For batches admitted from the academic year 2022-23)

1.0 For pursuing four year Under Graduate Degree Programme of study in Engineering & Technology (UGP in E&T) offered by MLR Institute of Technology under Autonomous status is herein referred to as MLRIT (Autonomous):

All the rules specified here in approved by the Academic Council will be in force and applicable to students admitted from the Academic Year 2022-23 onwards. Any reference to “Institute” or “College” in these rules and regulations shall stand for M L R Institute of Technology (Autonomous).

All the rules and regulations, specified hereafter shall be read as a whole for the purpose of interpretation as and when a doubt arises, the interpretation of the Chairman, Academic Council is final. As per the requirements of statutory bodies, the Principal, M L R Institute of Technology shall be the chairman Academic Council.

2.0 Eligibility for Admission

- 2.1 Admission to the undergraduate(UG) programme shall be made either on the basis of the merit rank obtained by the qualified student in entrance test conducted by the Telangana State Government(EAMCET)or the College or on the basis of any other order of merit approved by the University/TSHE, subject to reservations as prescribed by the government from time to time.
- 2.2 The medium of instructions for the entire undergraduate programme in Engineering & Technology will be **English** only.

3.0 B.Tech. Programme Structure

- 3.1 A student after securing admission shall complete the B.Tech. programme in a minimum period of **four** academic years (8 semesters), and a maximum period of **eight** academic years (16 semesters) starting from the date of commencement of first year first semester, failing which student shall forfeit seat in B.Tech course. Each student shall secure 160 credits (with CGPA ≥ 5) required for the completion of the undergraduate programme and award of the B.Tech.degree.
- 3.2 **UGC/AICTE** specified definitions/descriptions are adopted appropriately for various terms and abbreviations used in these academic regulations/ norms, which are listed below.

3.2.1 Semester Scheme

Each undergraduate programme is of 4 academic years (8 semesters) with the academic year divided into two semesters of 22 weeks (≥ 90 instructional days) each and in each semester - ‘Continuous Internal Evaluation (CIE)’ and ‘Semester End Examination (SEE)’ under Choice Based Credit System (CBCS) and Credit Based Semester System (CBSS) indicated by UGC, and curriculum/course structure suggested by AICTE are followed.

3.2.2 Credit Courses

All subjects/courses are to be registered by the student in a semester to earn credits which shall be assigned to each subject/course in an L:T:P:C(lecture periods: tutorial periods: practical periods: credits) structure based on the following general pattern.

- One credit for one hour/ week/ semester for Theory/ Lecture (L) courses or Tutorials.
- One credit for two hours/ week/ semester for Laboratory/ Practical (P)courses.

Courses like Environmental Science, Constitution of India, Intellectual Property Rights, and Gender Sensitization Lab are mandatory courses. These courses will not carry any credits.

3.2.3 Subject Course Classification

All subjects/ courses offered for the undergraduate programme in E&T (B.Tech. degree programmes) are broadly classified as follows. The College has followed almost all the guidelines issued by AICTE/UGC.

S. No.	Broad Course Classification	CourseGroup/ Category	Course Description
1	Foundation Courses (FnC)	BS–Basic Sciences	Includes Mathematics, Physics and Chemistry Subjects.
2		ES–Engineering Sciences	Includes Fundamental Engineering Subjects.
3		HS–Humanities and Social Sciences	Includes subjects related to Humanities, Social Sciences and Management.
4	Core Courses (CoC)	PC–Professional Core.	Includes core subjects related to the parent discipline/ department/ branch of Engineering.
5	Elective Courses (E&C)	PE–Professional Electives	Includes elective subjects related to the parent discipline/ department/ branch of Engineering.
6		OE–Open Electives	Elective subjects which include inter-disciplinary subjects or subjects in an area outside the parent discipline/ department/ branch of Engineering.
7	Core Courses	Project Work	B.Tech. Project or UG Project or UG Major Project or Project Stage I & II.
8		Industry Training/Internship/ Industry Oriented Mini-project/ Mini- Project/ Skill Development Courses.	Industry Training/ Internship/ Industry Oriented Mini-Project/ Mini-Project/ Skill Development Courses.

9		Seminar	Seminar/ Colloquium based on core contents related to parent discipline/ department/ branch of Engineering.
10	Minor Courses	-	1 or 2 Credit Courses (subset of HS)
11	Mandatory Courses (MC)	-	Mandatory Courses (non-credit)

4.0 **Course Registration**

- 4.1 A ‘faculty advisor or counselor’ shall be assigned to a group of 20 students, who will advise the students about the undergraduate programme, its course structure and curriculum, choice/option for subjects/courses, based on their competence, progress, pre-requisites and interest.
- 4.2 The academic section of the college invites ‘registration forms’ from students before the beginning of the semester through ‘on-line registration’, ensuring ‘date and time stamping’. The on-line registration requests for any ‘current semester’ shall be **completed before the commencement of SEEs (Semester End Examinations) of the ‘preceding semester’**.
- 4.3 A student can apply for **on-line** registration, **only after** obtaining the ‘**written approval**’ from faculty advisor/counselor, which should be submitted to the college academic section through the Head of the Department. A copy of it shall be retained with the Head of the Department, Faculty Advisor/ Counselor and the student.
- 4.4 A student may be permitted to register for all the subjects/ courses in a semester as specified in the course structure with maximum additional subject(s)/course(s) limited to 6 Credits (any 2 elective subjects), based on **progress** and SGPA/CGPA, and completion of the ‘**pre-requisites**’ as indicated for various subjects/courses, in the department course structure and syllabus contents.
- 4.5 Choice for ‘**additional subjects/ courses**’, not more than any 2 elective subjects in any Semester, must be clearly indicated, which needs the specific approval and signature of the Faculty Advisor/Mentor/HOD.
- 4.6 If the student submits ambiguous choices or multiple options or erroneous entries during **on-line** registration for the subject(s) / course(s) under a given/ specified course group/ category as listed in the course structure, only the first mentioned subject/ course in that category will be taken into consideration.
- 4.7 Subject/ course options exercised through **on-line** registration are final and **cannot** be changed or inter-changed; further, alternate choices also will not be considered. However, if the subject/ course that has already been listed for registration by the Head of the Department in a semester could not be offered due to any inevitable or unexpected reasons, then the students shall be allowed to have alternate choice either for a new subject (subject to offering of such a subject), or for another existing subject (subject to availability of seats). Such alternate arrangements will be made by the Head of the Department, with due notification and time-framed schedule, within **a week** after the commencement of class-work for that semester.
- 4.8 Dropping of subjects/courses may be permitted, only after obtaining prior approval from

the faculty advisor/ counselor ‘within a period of 15 days’ from the beginning of the current semester.

4.9 Open Electives: The students have to choose three Open Electives(OE-I,II&III) from the list of Open Electives given by other departments. However, the student can opt for an Open Elective subject offered by his own (parent) department, if the student has not registered and not studied that subject under any category (Professional Core, Professional Electives, Mandatory Courses etc.) offered by parent department in any semester. Open Elective subjects already studied should not repeat/should not match with any category (Professional Core, Professional Electives, Mandatory Courses etc.) of subjects even in the forthcoming semesters.

4.10 Professional Electives: The students have to choose six Professional Electives (PE-I to VI) from the list of professional electives given.

5.0 Subjects/ courses to be offered

5.1 A subject/ course may be offered to the students, **only if** a minimum of 15 students opt for it.

5.2 More than **one faculty member** may offer the **same subject** (lab/ practical may be included with the corresponding theory subject in the same semester) in any semester. However, selection of choice for students will be based on-‘**first come first serve** basis and CGPA criterion’ (i.e. the first focus shall be on early **on-line entry** from the student for registration in that semester, and the second focus, if needed, will be on CGPA of the student).

5.3 If more entries for registration of a subject come into picture, then the Head of the Department concerned shall decide, whether or not to offer such a subject/course for **two (or multiple)sections**.

5.4 In case of options coming from students of other departments/ branches/ disciplines (not considering **open electives**), first **priority** shall be given to the student of the ‘**parent department**’.

6.0 Attendance requirements:

6.1 A student shall be eligible to appear for the semester end examinations, if the student acquires a minimum of 75% of attendance in aggregate of all the subjects/ courses (including attendance in mandatory courses like Environmental Science, Constitution of India, Intellectual Property Rights, and Gender Sensitization Lab) for that semester. **Two periods** of attendance for each theory subject shall be considered, if the student appears for the mid-term examination of that subject.

6.2 Shortage of attendance in aggregate upto 10% (65% and above, and below 75%) in each semester may be condoned by the college academic committee on genuine and valid grounds, based on the student’s representation with supporting evidence.

6.3 A stipulated fee shall be payable for condoning of shortage of attendance.

6.4 Shortage of attendance below 65% in aggregate shall in **NO** case be condoned.

6.5 **Students whose shortage of attendance is not condoned in any semester are not eligible to take their end examination of that semester. They get detained and their**

registration for that semester shall stand cancelled, including all academic credentials (internal marks etc.) of that semester. **They will not be promoted to the next semester.** They may seek re-registration for all those subjects registered in that semester in which the student is detained, by seeking re-admission into that semester as and when offered; if there are any professional electives and/ or open electives, the same may also be re-registered if offered. However, if those electives are not offered in later semesters, then alternate electives may be chosen from the **same** set of elective subjects offered under that category.

- 6.6** A student fulfilling the attendance requirement in the present semester shall not be eligible for readmission into the same class.

7.0 Academic Requirements

The following academic requirements have to be satisfied, in addition to the attendance requirements mentioned in Item No.6.

- 7.1** A student shall be deemed to have satisfied the academic requirements and earned the credits allotted to each subject/ course, if student secures not less than 35% (21 marks out of 60 marks) in the semester end examination, and a minimum of 40%(40 marks out of 100 marks) in the sum total of the CIE(Continuous Internal Evaluation) and SEE(Semester End Examination) taken together; in terms of letter grades, this implies securing ‘C’ grade or above in that subject/course.
- 7.2** A student shall be deemed to have satisfied the academic requirements and earned the credits allotted to Real-time Research Project (or) Field Based Research Project (or) Industry Oriented Mini Project (or) Internship (or) Seminar, if the student secures not less than 40% marks (i.e. 40 out of 100 allotted marks) in each of them. The student is deemed to have failed, if he (i) does not submit a report on Industry Oriented Mini Project/Internship, or (ii) not make a presentation of the same before the evaluation committee as per schedule, or (iii) secures less than 40% marks in Real-time Research Project (or) Field Based Research Project (or) Industry Oriented Mini Project (or) Internship evaluations.

A student may reappear once for each of the above evaluations, when they are scheduled again; if the student fails in such ‘one reappearance’ evaluation also, the student has to reappear for the same in the next subsequent semester, as and when it is scheduled.

7.3 Promotion Rules

S. No.	Promotion	Conditions to be fulfilled
1	First year first semester to first year second semester	Regular course of study of first year first semester.
2	First year second semester to Second year first semester	(i) Regular course of study of first year second semester.

		(ii) Must have secured at least 20 credits out of 40 credits i.e., 50% credits up to first year second semester from all the relevant regular and supplementary examinations, whether the student takes those examinations or not.
3.	Second year first semester to Second year second semester	Regular course of study of second year first semester.
4	Second year second semester to Third year first semester	(i) Regular course of study of second year second semester. (ii) Must have secured at least 48 credits out of 80 credits i.e., 60% credits up to second year second semester from all the relevant regular and supplementary examinations, whether the student takes those examinations or not.
5	Third year first semester to Third year second semester	Regular course of study of third year first semester.
6	Third year second semester to Fourth year first semester	(i) Regular course of study of third year second semester. (ii) Must have secured at least 72 credits out of 120 credits i.e., 60% credits up to third year second semester from all the relevant regular and supplementary examinations, whether the student takes those examinations or not.
7	Fourth year first semester to Fourth year second semester	Regular course of study of fourth year first semester.

7.4 A student(i) shall register for all courses/subjects covering 160 credits as specified and listed in the course structure, (ii) fulfills all the attendance and academic requirements for 160 credits, (iii) earn all 160 credits by securing $SGPA \geq 5.0$ (in each semester), and $dCGPA \geq 5$ (at the end of 8 semesters), (iv) **passes all the mandatory courses**, to successfully complete the undergraduate programme. The performance of the student in these 160 credits shall be considered for the calculation of the final CGPA (**at the end of undergraduate programme**), and shall be indicated in the grade card / marks memo of IV-year II semester.

7.5 If a student registers for 'extra subjects' (in the parent department or other departments/branches of Engg.) other than those listed subjects totaling to 160 credits as specified in the course structure of this department, the performances in those 'extra subjects' (although evaluated and graded using the same procedure as that of the required 160 credits) will not be considered while calculating the SGPA and CGPA. For such 'extra subjects' registered, percentage of marks and letter grade alone will be indicated in the grade card / marks memo as a performance measure, subject to completion of the attendance and academic requirements as stated in regulations Items 6 and 7.1–7.4 above.

7.6 A student eligible to appear in the semester end examination for any subject/

course, but absent from it or failed (thereby failing to secure 'C' grade or above) may reappear for that subject/ course in the supplementary examination as and when conducted. In such cases, internal marks (CIE) assessed earlier for that subject/course will be carried over, and added to the marks to be obtained in the SEE supplementary examination for evaluating performance in that subject.

- 7.7 A student **detained in a semester due to shortage of attendance may be re-admitted in the same semester in the next academic year for fulfilment of academic requirements.** The academic regulations under which a student has been re-admitted shall be applicable. Further, no grade allotments or SGPA/ CGPA calculations will be done for the entire semester in which the student has been detained.
- 7.8 A student detained **due to lack of credits, shall be promoted to the next academic year only after acquiring the required number of academic credits.** The academic regulations under which the student has been readmitted shall be applicable to him.

8.0 ***Evaluation - Distribution and Weightage of Marks***

- 8.1 The performance of a student in every subject/course (including practicals and Project Stage – I & II) will be evaluated for 100 marks each, with 40 marks allotted for CIE (Continuous Internal Evaluation) and 60 marks for SEE (Semester End-Examination).
- 8.2 In CIE, for theory subjects, during a semester, there shall be two mid-term examinations. Each Mid-Term examination consists of two parts i) **Part – A** for 10 marks, ii) **Part – B** for 20 marks with a total duration of 2 hours as follows:

1. Mid Term Examination for 30 marks:
 - a. Part-A: Objective/quiz paper for 10 marks.
 - b. Part-B: Descriptive paper for 20 marks.

The objective/quiz paper is set with multiple choice, fill-in the blanks and match the following type of questions for a total of 10 marks. The descriptive paper shall contain 6 full questions out of which, the student has to answer 4 questions, each carrying 5 marks. The **average of the two Mid Term Examinations** shall be taken as the final marks for Mid Term Examination (for 30 marks).

The remaining 10 marks of Continuous Internal Evaluation are distributed as:

2. Assignment for 5 marks. (**Average of 2 Assignments** each for 5 marks)
3. Subject Viva-Voce/PPT/Poster Presentation/Case Study on a topic in the concerned subject for 5 marks

While the first mid-term examination shall be conducted on 50% of the syllabus, the second mid-term examination shall be conducted on the remaining 50% of the syllabus.

Five (5) marks are allocated for assignments (as specified by the subject teacher concerned). The first assignment should be submitted before the conduct of the first mid-term examination, and the second

assignment should be submitted before the conduct of these second mid-term examination. The average of the two assignments shall be taken as the final marks for assignment (for 5 marks).

Subject Viva-Voce/PPT/Poster Presentation/ Case Study on a topic in the subject concerned for 5 marks before II Mid-Term Examination.

- The Student, in each subject, shall have to earn 35% of marks (i.e. 14 marks out of 40 marks) in CIE, 35% of marks (i.e. 21 marks out of 60) in SEE and Over all 40% of marks (i.e. 40 marks out of 100 marks) both CIE and SEE marks put together.

The student is eligible to write Semester End Examination of the concerned subject, if the student scores $\geq 35\%$ (14 marks) of 40 Continuous Internal Examination (CIE) marks.

In case, the student appears for Semester End Examination (SEE) of the concerned subject but not scored minimum 35% of CIE marks (14 marks out of 40 internal marks), his performance in that subject in SEE shall stand cancelled in spite of appearing the SEE.

There is NO Computer Based Test (CBT) for R22 regulations.

The details of the end semester question paper pattern are as follows:

8.2.1 The semester end examinations (SEE), for theory subjects, will be conducted for 60 marks consisting of two parts viz. i) **Part- A** for 10 marks, ii) **Part - B** for 50 marks.

- Part-A is a compulsory question which consists of ten sub-questions from all units carrying equal marks.
- Part-B consists of five questions (numbered from 2 to 6) carrying 10 marks each. Each of these questions is from each unit and may contain sub-questions. For each question there will be an “either” “or” choice, which means that there will be two questions from each unit and the students should answer either of the two questions.
- The duration of Semester End Examination is 3 hours.

8.3 For practical subjects there shall be a Continuous Internal Evaluation (CIE) during the semester for 40 marks and 60 marks for semester end examination. Out of the 40 marks for internal evaluation :

1. A write-up on day-to-day experiment in the laboratory (in terms of aim, components/procedure, expected outcome) which shall be evaluated for 10 marks
2. **10 marks for viva-voce** (or) tutorial (or) case study (or) application (or) poster presentation of the course concerned.
3. Internal practical examination conducted by the laboratory teacher concerned shall be evaluated for 10 marks.

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4. The remaining 10 marks are for Laboratory Project, which consists of the Design(or)Software/HardwareModelPresentation(or)AppDevelopment(or)PrototypePresentation submission which shall be evaluated after completion of laboratory course and before semester end practical examination.

The Semester End Examination shall be conducted with an external examiner and the laboratory teacher. The external examiner shall be appointed from the other colleges which will be decided by the principal of the College.

In the Semester End Examination held for 3 hours, total 60 marks are divided and allocated as shown below:

1. 10 marks for write-up
 2. 15 for experiment/program
 3. 15 for evaluation of results
 4. 10 marks for presentation on another experiment/program in the same laboratory course and
 5. 10 marks for viva-voce on concerned laboratory course.
- The Student, in each subject, shall have to earn 35% of marks (i.e. 14 marks out of 40 marks) in CIE, 35% of marks (i.e. 21 marks out of 60) in SEE and Overall 40% of marks (i.e. 40 marks out of 100 marks) both CIE and SEE marks put together.

The student is eligible to write Semester End Examination of the concerned subject, if the student scores $\geq 35\%$ (14 marks) of 40 Continuous Internal Examination (CIE) marks.

In case, the student appears for Semester End Examination (SEE) of the concerned subject but not scored minimum 35% of CIE marks (14 marks out of 40 internal marks), his performance in that subject in SEE shall stand cancelled in spite of appearing the SEE.

8.4 The evaluation of courses having ONLY internal marks in I Year I Semester and II Year II Semester is as follows:

1. I Year I, II Semester course (ex., Elements of CE/ME/EEE/ECE/CSE/Seminar & etc): The internal evaluation is for 50 marks and it shall take place during I Mid-Term examination and II Mid-Term examination. The average marks of two MID-Term examinations is the final for 50 marks. Students shall have to earn 40%, i.e. 20 marks out of 50 marks from average of the two examinations. There shall be NO external evaluation. The student is deemed to have failed, if he (i) is absent as per schedule, or (ii) secures less than 40% marks in this course. Seminar evaluation conducted through day to day evaluation of the students in presentation/abstract.
2. II Year II Semester Real-Time (or) Field-based Research Project course: The internal evaluation is for 50 marks and it shall take place during I Mid-Term examination and II Mid-Term examination. The average marks

of two Mid-Term examinations is the final for 50 marks. Students shall have to earn 40%, i.e. 20 marks out of 50 marks from an average of the two examinations. There shall be NO external evaluation. The student is deemed to have failed, if he (i) does not submit a report on the Project, or (ii) does not make a presentation of the same before the internal committee as per schedule, or (ii) secures less than 40% marks in this course.

- 8.5** There shall be an Industry training (or) Internship (or) Industry oriented Mini-project (or) Skill Development Courses (or) Paper presentation in reputed journal (or) Industry Oriented Mini Project in collaboration with an industry of their specialization. Students shall register for this immediately after II-Year II Semester Examinations and pursue it during summer vacation/semester break & during III Year without effecting regular course work. Internship at reputed organization (or) Skill development courses (or) Paper presentation in reputed journal (or) Industry Oriented Mini Project shall be submitted in a report form and presented before the committee in III-year II semester before end semester examination. It shall be evaluated for 100 external marks. The committee consists of an External Examiner, Head of the Department, Supervisor of the Industry Oriented Mini Project (or) Internship etc, Internal Supervisor and a Senior Faculty Member of the Department. There shall be **NO internal marks** for Industry Training (or) Internship (or) Mini-Project (or) Skill Development Courses (or) Paper Presentation in reputed journal (or) Industry Oriented Mini Project.
- 8.6** The UG project shall be initiated at the end of the IV Year I Semester and the duration of the project work is one semester. The student must present Project Stage – I during IV Year I Semester before II Mid examinations, in consultation with his Supervisor, the title, objective and plan of action of his Project work to the departmental committee for approval before commencement of IV Year II Semester. Only after obtaining the approval of the departmental committee, the student can start this project work.
- 8.7** UG project work shall be carried out in two stages: Project Stage – I for approval of project before Mid-II examinations in IV Year I Semester and Project Stage – II during IV Year II Semester. Student has to submit project work report at the end of IV Year II Semester. The project shall be evaluated for 100 marks before commencement of SEE Theory examinations.
- 8.8** For Project Stage – I, the departmental committee consisting of Head of the Department, project supervisor and a senior faculty member shall approve the project work to begin before II Mid-Term examination of IV Year I Semester. The student is deemed to be not eligible to register for the Project work, if he does not submit a report on Project Stage - I or does not make a presentation of the same before the evaluation committee as per schedule.

A student who has failed may reappear once for the above evaluation, when it is scheduled again; if he fails in such 'one reappearance' evaluation also, he has to reappear for the same in the next subsequent semester, as and when it is scheduled.

- 8.9** For Project Stage – II, the external examiner shall evaluate the project work for 60 marks and the internal project committee shall evaluate it for 40 marks. Out of 40 internal marks, the departmental committee consisting of Head of the Department, Project Supervisor and a Senior Faculty Member shall evaluate the project work for 20 marks and Project Supervisor shall evaluate for 20 marks. The topics for Industry Oriented Mini Project/Internship/SDC etc. and the main Project shall be different from the topic already taken. The student is deemed to have failed, if he (i) does not submit a report on the Project, or (ii) does not make a presentation of the same before the External Examiner as per schedule, or (iii) secures less than 40% marks in the sum total of the CIE and SEE taken together.

For conducting viva-voce of project, Principal/COE selects an external examiner from the list of experts in the relevant branch submitted by the Principal of the College.

A student who has failed, may reappear once for the above evaluation, when it is scheduled again; if student fails in such 'one reappearance' evaluation also, he has to reappear for the same in the next subsequent semester, as and when it is scheduled.

- 8.10** A student shall be given only one time chance to re-register for a maximum of two subjects in a semester:
- If the internal marks secured by a student in the Continuous Internal Evaluation marks for 40 (Sum of average of two mid-term examinations consisting of Objective & descriptive parts, Average of two Assignments & Subject Viva-voce/PPT/Poster presentation/Case Study on a topic in the concerned subject) are less than 35% and failed in those subjects.

A student must re-register for the failed subject(s) for 40 marks within four weeks of commencement of the class work in next academic year.

In the event of the student taking this chance, his Continuous Internal Evaluation marks for 40 and Semester End Examination marks for 60 obtained in the previous attempt stand cancelled.

- 8.11 For mandatory non-credit Audit courses,** The internal evaluation is for 50 marks and it shall take place during I Mid-Term examination and II Mid-Term examination. The average marks of two MID-Term examinations is the final for 50 marks. Students shall have to earn 40%, i.e. 20 marks out of 50 marks from an average of the two examinations. There shall be NO external evaluation. The student is deemed to have failed, if he (i) is absent as per schedule, or (ii) secures less than 40% marks in this course.

- 8.12** For Mandatory Non-Credit Courses offered in a Semester, after securing $\geq 65\%$ attendance and has secured not less than 40% of marks in the sum Total. Then the student is **PASS** and will be qualified for the award of the degree. No marks or Letter Grade shall be

allotted for these courses/activities. However, for non credit courses ‘**Satisfactory**’ or ‘**Unsatisfactory**’ shall be indicated instead of the letter grade and this will not be counted for the computation of SGPA/CGPA.

9.0 **Grading Procedure**

- 9.1 Grades will be awarded to indicate the performance of students in each Theory Subject, Laboratory/Practicals/ Industry-Oriented Mini Project/Internship/SDC and Project Stage. Based on the percentage of marks obtained (Continuous Internal Evaluation plus Semester End Examination, both taken together) as specified in item 8 above, a corresponding letter grade shall be given.
- 9.2 As a measure of the performance of a student, a 10-point absolute grading system using the following letter grades (as per UGC/AICTE guidelines) and corresponding percentage of marks shall be followed:

% of Marks Secured in a Subject/Course (Class Intervals)	Letter Grade (UGC Guidelines)	Grade Points
Greater than or equal to 90%	O (Outstanding)	10
80 and less than 90%	A ⁺ (Excellent)	9
70 and less than 80%	A (Very Good)	8
60 and less than 70%	B ⁺ (Good)	7
50 and less than 60%	B (Average)	6
40 and less than 50%	C (Pass)	5
Below 40%	F (FAIL)	0
Absent	Ab	0

- 9.3 A student who has obtained an ‘**F**’ grade in any subject shall be deemed to have ‘**failed**’ and is required to reappear as a ‘supplementary student’ in the semester end examination, as and when offered. In such cases, internal marks in those subjects will remain the same as those obtained earlier.
- 9.4 To a student who has not appeared for an examination in any subject, ‘**Ab**’ grade will be allocated in that subject, and he is deemed to have ‘**Failed**’. A student will be required to reappear as a ‘supplementary student’ in the semester end examination, as and when offered next. In this case also, the internal marks in those subjects will remain the same as those obtained earlier.
- 9.5 A letter grade does not indicate any specific percentage of marks secured by the student, but it indicates only the range of percentage of marks.
- 9.6 A student earns Grade Point (GP) in each subject/course, on the basis of the letter grade secured in that subject/ course. The corresponding ‘Credit Points’ (CP) are computed by multiplying the grade point with credits for that particular subject/course.

Credit Points (CP) = Grade Point (GP) x Credits For a course

- 9.7 A student passes the subject/ course only when GP ≥ 5 (‘C’ grade or above)

- 9.8** The Semester Grade Point Average (SGPA) is calculated by dividing the sum of credit points (ΣCP) secured from all subjects/ courses registered in a semester, by the total number of credits registered during that semester. SGPA is rounded off to two decimal places. SGPA is thus computed as

$$SGPA = \{ \sum_{i=1}^N C_i G_i \} / \{ \sum_{i=1}^N C_i \} \dots \text{For each Semester,}$$

where 'i' is the subject indicator index (considering all subjects in a semester), 'N' is the no. of subjects '**registered**' for the semester (as specifically required and listed under the course structure of the parent department), C_i is the no. of credits allotted to the i^{th} subject, and G_i represents the grade points (GP) corresponding to the letter grade awarded for that i^{th} subject.

- 9.9** The Cumulative Grade Point Average (CGPA) is a measure of the overall cumulative performance of a student in all semesters considered for registration. The CGPA is the ratio of the total credit points secured by a student in **all** registered courses (of 160) in **all** semesters, and the total number of credits registered in **all** the semesters. CGPA is rounded off to two decimal places. CGPA is thus computed from the I year I semester onwards at the end of each semester as per the formula

$$CGPA = \{ \sum_{j=1}^M C_j G_j \} / \{ \sum_{j=1}^M C_j \} \dots \text{for all S Semesters registered}$$

(i.e., up to and inclusive of S Semesters, $S \geq 2$),

where 'M' is the TOTAL no. of Subjects (as specifically required and listed under the Course Structure of the parent Department) the Student has 'REGISTERED' from the 1st Semester onwards up to and inclusive of the Semester S (obviously $M > N$), 'j' is the Subject indicator index (takes into account all Subjects from 1 to S Semesters), C_j is the no. of Credits allotted to the jth Subject, and G_j represents the Grade Points (GP) corresponding to the Letter Grade awarded for that jth Subject. After registration and completion of I Year I Semester however, the SGPA of that Semester itself may be taken as the CGPA, as there are no cumulative effects.

Illustration of calculation of SGPA:

Course/Subject	Credits	Letter Grade	Grade Points	Credit Points
Course 1	4	A	8	$4 \times 8 = 32$
Course 2	4	O	10	$4 \times 10 = 40$
Course 3	4	C	5	$4 \times 5 = 20$
Course 4	3	B	6	$3 \times 6 = 18$
Course 5	3	A+	9	$3 \times 9 = 27$
Course 6	3	C	5	$3 \times 5 = 15$
	21			152

$$SGPA = 152/21 = 7.24$$

Illustration of Calculation of CGPA up to 3rd Semester:

Semester	Course/ Subject Title	Credits Allotted	Letter Grade Secured	Corresponding Grade Point (GP)	Credit Points (CP)
I	Course 1	3	A	8	24
I	Course 2	3	O	10	30
I	Course 3	3	B	6	18
I	Course 4	4	A	8	32
I	Course 5	3	A+	9	27
I	Course 6	4	C	5	20
II	Course 7	4	B	6	24
II	Course 8	4	A	8	32
II	Course 9	3	C	5	15
II	Course 10	3	O	10	30
II	Course 11	3	B+	7	21
II	Course 12	4	B	6	24
II	Course 13	4	A	8	32
II	Course 14	3	O	10	30
III	Course 15	2	A	8	16
III	Course 16	1	C	5	5
III	Course 17	4	O	10	40
III	Course 18	3	B+	7	21
III	Course 19	4	B	6	24
III	Course 20	4	A	8	32
III	Course 21	3	B+	7	21
	Total Credits	69		Total Credit Points	518

CGPA = 518/69 = 7.51

The calculation process of CGPA illustrated above will be followed for each subsequent semester until 8th semester. The CGPA obtained at the end of 8th semester will become the final CGPA secured for entire B.Tech. programme.

- 9.10** For merit ranking or comparison purposes or any other listing, **only** the ‘**rounded off**’ values of the CGPAs will be used.
- 9.11** SGPA and CGPA of a semester will be mentioned in the semester Memorandum of Grades if all subjects of that semester are passed in first attempt. Otherwise the SGPA and CGPA shall be mentioned only on the Memorandum of Grades in which sitting he

passed his last exam in that semester. However, mandatory courses will not be taken into consideration.

10.0 Passing Standards

- 10.1** A student shall be declared successful or 'passed' in a semester, if he secures a $GP \geq 5$ ('C' grade or above) in every subject/course in that semester (i.e. when the student gets an SGPA ≥ 5.0 at the end of that particular semester); and he shall be declared successful or 'passed' in the entire undergraduate programme, only when he gets a CGPA ≥ 5.00 ('C' grade or above) for the award of the degree as required.
- 10.2** After the completion of each semester, a grade card or gradesheet shall be issued to all the registered students of that semester, indicating the letter grades and credits earned. It will show the details of the courses registered (course code, title, no. of credits, grade earned, etc.) and credits earned. **There is NO exemption of credits in any case.**

11.0 Declaration of results

- 11.1** Computation of SGPA and CGPA are done using the procedure listed in 9.6 to 9.9.
- 11.2** For final percentage of marks equivalent to the computed final CGPA, the following formula may be used.
- $$\% \text{ of Marks} = (\text{final CGPA} - 0.5) \times 10$$

12.0 Award of Degree

- 12.1** A student who registers for all the specified subjects/ courses as listed in the course structure and secures the required number of 160 credits (with CGPA ≥ 5.0), within 8 academic years from the date of commencement of the first academic year, shall be declared to have '**qualified**' for the award of B.Tech. degree in the branch of Engineering selected at the time of admission.
- 12.2** A student who qualifies for the award of the degree as listed in item 12.1 shall be placed in the following classes.
- 12.3** A student with final CGPA (at the end of the undergraduate programme) > 8.00 , and fulfilling the following conditions - shall be placed in '**First Class with Distinction**'.
- However, he
- (i) Should have passed all the subjects/courses in '**First Appearance**' within the first 4 academic years (or 8 sequential semesters) from the date of commencement of first year first semester.
 - (ii) Should not have been detained or prevented from writing the semester end examinations in any semester due to shortage of attendance or any other reason.
- A student not fulfilling any of the above conditions with final CGPA > 8 shall be placed in '**First Class**'.
- 12.4** Students with final CGPA (at the end of the undergraduate programme) ≥ 7.0 but $<$

8.00 shall be placed in '**First Class**'.

12.5 Students with final CGPA (at the end of the undergraduate programme) ≥ 6.00 but < 7.00 , shall be placed in '**Second Class**'.

12.6 All other students who qualify for the award of the degree (as per item 12.1), with final CGPA (at the end of the undergraduate programme) ≥ 5.00 but < 6 , shall be placed in '**pass class**'.

12.7 A student with final CGPA (at the end of the undergraduate programme) < 5.00 will not be eligible for the award of the degree.

12.8 Students fulfilling the conditions listed under item 12.3 alone will be eligible for award of '**Gold Medal**'.

12.9 Award of 2-Year B.Tech. Diploma Certificate

1. A student is awarded 2-Year UG Diploma Certificate in the concerned engineering branch on completion of all the academic requirements and earned all the 80 credits (with in 4 years from the date of admission) upto B. Tech. – II Year – II Semester, if the student wants to exit the 4-Year B.Tech. program. **The student once opted and awarded for 2-Year UG Diploma Certificate, the student will not be permitted to join in B. Tech. III Year – I Semester and continue for completion of remaining years of study for 4-Year B. Tech. Degree.**
2. A student may be permitted to take one year break after completion of II Year – II Semester or B. Tech. – III Year – II Semester (with Principal permission through the HOD of the department well in advance) and can re-enter the course in **next Academic Year in the same college** and complete the course on fulfilling all the academic credentials within a stipulated duration i.e. double the duration of the course (Ex. within 8 Years for 4-Year program).

13.0 Withholding of results

13.1 If the student has not paid the fees to the College at any stage, or has dues pending due to any reason whatsoever, or if any case of indiscipline is pending, the result of the student may be withheld, and the student will not be allowed to go into the next higher semester. The award or issue of the degree may also be withheld in such cases.

14.0 Transitory Regulations

A. For students detained due to shortage of attendance:

1. A Student who has been detained in I year of R18 Regulations due to lack of attendance, shall be permitted to join I year I Semester of R22 Regulations and he is required to complete the study of B.Tech./B. Pharmacy programme within the stipulated period of eight academic years from the date of first admission in I Year.
2. A student who has been detained in any semester of II, III and IV years of R18 regulations for want of attendance, shall be permitted to join the corresponding semester of R22 Regulations and is required to complete the study of B.Tech./B. Pharmacy within

the stipulated period of eight academic years from the date of first admission in I Year. The R22 Academic Regulations under which a student has been readmitted shall be applicable to that student from that semester. See rule (C) for further Transitory Regulations.

3. For students detained due to shortage of credits: A student of MLR18 Regulations who has been detained due to lack of credits, shall be promoted to the next semester of R22 Regulations only after acquiring the required number of credits as per the corresponding regulations of his/her first admission. The total credits required are 160 including both MLR18 & R22 regulations. The student is required to complete the study of B.Tech. within the stipulated period of eight academic years from the year of first admission. The R22 Academic Regulations are applicable to a student from the year of readmission. See rule (C) for further Transitory Regulations.

B. For readmitted students in R22 Regulations:

4. A student who has failed in any subject under any regulation has to pass those subjects in the same regulations.
5. The maximum credits that a student acquires for the award of degree, shall be the sum of the total number of credits secured in all the regulations of his/her study including R22 Regulations. **There is NO exemption of credits in any case.**
6. If a student is readmitted to R22 Regulations and has any subject with 80% of syllabus common with his/her previous regulations, that particular subject in R22 Regulations will be substituted by another subject to be suggested by the Concerned BOS.

15.0 Student Transfers

- 15.1 There shall be no branch transfers after the completion of admission process.
- 15.2 There shall be no transfers from one college/stream to another within the constituent colleges and units of Jawaharlal Nehru Technological University Hyderabad.
- 15.3 The students seeking transfer to colleges affiliated to JNTUH from various other Universities/institutions have to pass the failed subjects which are equivalent to the subjects of JNTUH, and also pass the subjects of JNTUH which the students have not studied at the earlier institution. Further, though the students have passed some of the subjects at the earlier institutions, if the same subjects are prescribed in different semesters of JNTUH, the students have to study those subjects in JNTUH in spite of the fact that those subjects are repeated.
- 15.4 The transferred students from other Universities/Institutions to College who are on rolls are to be provided one chance to write the CBT (for internal marks) in the **equivalent subject(s)** as per the clearance letter issued by the University.
- 15.5 The college has to provide one chance to write the internal examinations in the **equivalent subject(s)** to the students transferred from other universities/institutions to JNTUH autonomous affiliated colleges who are on rolls, as per the clearance (equivalence) letter issued by the University.

16.0 Malpractice Prevention Committee

A malpractice prevention committee shall be constituted to examine and punish the students who involve in malpractice / indiscipline in examinations. The committee shall consist of:

- a) Controller of examinations - Chairman
- b) Addl. Controller of examinations.- Member Convener
- c) Subject expert - member
- d) Head of the department of which the student belongs to. - Member
- e) The invigilator concerned - member

The committee shall conduct the meeting after taking explanation of the student and punishment will be awarded by following the malpractice rules meticulously.

Any action on the part of candidate at the examination like trying to get undue advantage in the performance at examinations or trying to help another, or derive the same through unfair means is punishable according to the provisions contained hereunder. The involvement of the Staff who are in charge of conducting examinations, valuing examination papers and preparing / keeping records of documents relating to the examinations, in such acts (inclusive of providing incorrect or misleading information) that infringe upon the course of natural justice to one and all concerned at the examination shall be viewed seriously and will be recommended for appropriate punishment after thorough enquiry.

17.0 Scope

- 17.1 The academic regulations should be read as a whole, for the purpose of any interpretation.
- 17.2 In case of any doubt or ambiguity in the interpretation of the above rules, the decision of the Chairman of Academic Council is final.
- 17.3 The College may change or amend the academic regulations, course structure or syllabi at any time, and the changes or amendments made shall be applicable to all students with effect from the dates notified by the college authorities.
- 17.4 Where the words “he”, “him”, “his”, occur in the regulations, they include “she”, “her”, “hers”.

**ACADEMIC REGULATIONS FOR B.TECH (LATERAL ENTRY
SCHEME)
FROM THE AY 2023-24**

1. **Eligibility for the award of B.Tech Degree (LES)**

The LES students after securing admission shall pursue a course of study for not less than three academic years and not more than six academic years.

2. The students shall register for 120 credits and secure 120 credits with CGPA ≥ 5 from II year to IV-year B.Tech. programme (LES) for the award of B.Tech. degree.

3. The students, who fail to fulfil the requirement for the award of the degree in six academic years from the year of admission, shall forfeit their seat in B.Tech.

4. The attendance requirements of B. Tech. (Regular) shall be applicable to B.Tech. (LES).

5. **Promotion rule**

S. No	Promotion	Conditions to be fulfilled
1	Second year first semester to second year second semester	Regular course of study of second year first semester.
2	Second year second semester to third year first semester	(i) Regular course of study of second year second semester. (ii) Must have secured at least 24 credits out of 40 credits i.e., 60% credits up to second year second semester from all the relevant regular and supplementary examinations, whether the student takes those examinations or not.
3	Third year first semester to third year second semester	Regular course of study of third year first semester.
4	Third year second semester to fourth year first semester	i) Regular course of study of third year second semester. ii) Must have secured at least 48 credits out of 80 credits i.e., 60% credits up to third year second semester from all the relevant regular and supplementary examinations, whether the student takes those examinations or not.
5	Fourth year first semester to fourth year second semester	Regular course of study of fourth year first semester.

6. All the other regulations as applicable to B. Tech. 4-year degree course(Regular) will hold good for B. Tech. (Lateral EntryScheme).

7. LES students are not eligible for 2-Year B. Tech. Diploma Certificate.

Malpractices Rules

Disciplinary Action For / Improper Conduct in Examinations

	Nature of Malpractices/ Improper conduct	Punishment
	If the student:	
1. (a)	Possesses or keeps accessible in examination hall, any paper, note book, programmable calculators, cell phones, pager, palm computers or any other form of material concerned with or related to the subject of the examination (theory or practical) in which student is appearing but has not made use of (material shall include any marks on the body of the student which can be used as an aid in the subject of the examination).	Expulsion from the examination hall and cancellation of the performance in that subject only.
(b)	Gives assistance or guidance or receives it from any other student orally or by any other body language methods or communicates through cell phones with any student or persons in or outside the exam hall in respect of any matter.	Expulsion from the examination hall and cancellation of the performance in that subject only of all the students involved. In case of an outsider, he will be handed over to the police and a case is registered against him.
2.	Has copied in the examination hall from any paper, book, programmable calculators, palm computers or any other form of material relevant to the subject of the examination (theory or practical) in which the student is appearing.	Expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the student has already appeared including practical examinations and project work and shall not be permitted to appear for the remaining examinations of the subjects of that semester/year. The hall ticket of the student is to be cancelled
3.	Impersonates any other student in connection with the examination.	The student who has impersonated shall be expelled from examination hall. The student is also debarred and forfeits the seat. The performance of the original student who has been impersonated, shall be cancelled in all the subjects of the examination (including practicals and project work) already appeared and shall not be allowed to appear for examinations of the remaining subjects of that semester/year. The student is also debarred for two consecutive semesters from class work and all college examinations. The continuation of the course by the student is subject to the academic regulations in connection with forfeiture of seat. If the imposter is an outsider, he will be handed over to the police and a case is registered against him.

4.	Smuggles in the answer book or additional sheet or takes out or arranges to send out the question paper during the examination or answer book or additional sheet, during or after the examination.	Expulsion from the examination hall and cancellation of performance in that subject and all the other subjects the student has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that semester/year. The student is also debarred for two consecutive semesters from class work and all college examinations. The continuation of the course by the student is subject to the academic regulations in connection with forfeiture of seat.
5.	Uses objectionable, abusive or offensive language in the answer paper or in letters to the examiners or writes to the examiner requesting him to award pass marks.	Cancellation of the performance in that subject.
6.	Refuses to obey the orders of the chief superintendent/COE/ACoE/any officer on duty or misbehaves or creates disturbance of any kind in and around the examination hall or organizes a walk out or instigates others to walk out, or threatens the officer or any person on duty in or outside the examination hall of any injury to his person or to any of his relations whether by words, either spoken or written or by signs or by visible representation, assaults the officer-in-charge, or any person on duty in or outside the examination hall or any of his relations, or indulges in any other act of misconduct or mischief which results in damage to or destruction of property in the examination hall or any part of the college campus or engages in any other act which in the opinion of the officer on duty amounts to use of unfair means or misconduct or has the tendency to disrupt the orderly conduct of the examination.	In case of students of the college, they shall be expelled from examination halls and cancellation of their performance in that subject and all other subjects the student(s) has (have) already appeared and shall not be permitted to appear for the remaining examinations of the subjects of that semester/year. The students also are debarred and forfeit their seats. In case of outsiders, they will be handed over to the police and a police case is registered against them.

7.	Leaves the exam hall taking away answer script or intentionally tears off the script or any part thereof inside or outside the examination hall.	Expulsion from the examination hall and cancellation of performance in that subject and all the other subjects the student has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that semester/year. The student is also debarred for two consecutive semesters from class work and all college examinations. The continuation of the course by the student is subject to the academic regulations in connection with forfeiture of seat.
8.	Possesses any lethal weapon or firearm in the examination hall.	Expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the student has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that semester/year. The student is also debarred and forfeits the seat.
9.	If student of the college, who is not a student for the particular examination or any person not connected with the college indulges in any malpractice or improper conduct mentioned in clause 6 to 8.	Expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the student has already appeared including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that semester/year. The student is also debarred and forfeits the seat. Person(s) who do not belong to the college will be handed over to the police and, a police case will be registered against them.
10.	Comes in a drunken condition to the examination hall.	Expulsion from the examination hall and cancellation of the performance in that subject and all other subjects the student has already appeared for including practical examinations and project work and shall not be permitted for the remaining examinations of the subjects of that semester/year.
11.	Copying detected on the basis of internal evidence, such as, during valuation or during special scrutiny.	Cancellation of the performance in that subject and all other subjects the student has appeared for including practical examinations and project work of that semester/year examinations.
12.	If any malpractice is detected which is not covered in the above clauses 1 to 11 shall be reported to the Principal for further action to award a suitable punishment.	

Malpractices identified by squad or special invigilators

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1. Punishments to the students as per the above guidelines.
 2. Punishment for staff: (if the squad reports that the staff is also involved in encouraging malpractices)
 - a. A show-cause notice shall be issued to the staff.
 - b. Impose a suitable fine on the staff.

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COURSE STRUCTURE
B. TECH – Electrical and Electronics Engineering

REGULATIONS: AR22

I YEAR I SEMESTER									
Code	Course	Category	Periods per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		CIE	SEE	Total
A6BS01	Linear Algebra and Calculus	BSC	3	1	0	4	40	60	100
A6BS07	Applied Physics	BSC	3	1	0	4	40	60	100
A6BS09	Engineering Chemistry	BSC	3	1	0	4	40	60	100
A6ME01	Engineering Graphics	ESC	1	0	4	3	40	60	100
A6BS08	Applied Physics Lab	BSC	0	0	3	1.5	40	60	100
A6BS10	Engineering Chemistry Lab	ESC	0	0	3	1.5	40	60	100
A6ME03	Engineering workshop Practices	ESC	0	1	2	2	40	60	100
A6BS11	Environmental Science	MC	3	0	0	0	50		50
TOTAL			13	4	12	20	330	420	750
I YEAR II SEMESTER									
Code	Course	Category	Periods per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		CIE	SEE	Total
A6BS02	Numerical Methods and Integral Transforms	BSC	3	1	0	4	40	60	100
A6HS01	English for Skill Enhancement	HSMC	3	0	0	3	40	60	100
A6EE01	Electrical Circuits-1	ESC	3	1	0	4	40	60	100
A6CS02	Programming for Problem Solving	ESC	3	0	0	3	40	60	100
A6HS02	English Language and Communication Skills Lab	HSMC	0	0	3	1.5	40	60	100
A6CS03	Programming for Problem Solving Lab	ESC	0	0	3	1.5	40	60	100
A6EE02	Electrical Circuits Lab	ESC	0	0	2	1	40	60	100
A6EE03	Applied Python Programming Laboratory	ESC	0	0	2	1	40	60	100
A6HS04	Seminar	HSMC	0	0	2	1	50		50
TOTAL			12	2	12	20	370	480	850

II B.Tech I Semester									
Course Code	Course Title	Course Area	Hours per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		Internal (CIE)	External (SEE)	Total
A6EE04	Electrical Circuits – II	ESC	3	-	-	3	40	60	100
A6EC01	Electronic Devices and circuits	ESC	3	-	-	3	40	60	100
A6EE05	Electrical Machines – I	PCC	3	1	-	4	40	60	100
A6EE06	Electromagnetic Fields	PCC	3	-	-	3	40	60	100
A6BS06	Vector calculus and Complex Analysis	BSC	3	1	-	4	40	60	100
A6EC02	Electronic Devices and circuits Lab	ESC	-	-	2	1	40	60	100
A6EE07	Electrical Machines –I Lab	PCC	-	-	2	1	40	60	100
A6EE08	Electrical Simulation tools Laboratory	PCC	-	-	2	1	40	60	100
Total			15	2	6	20	320	480	800
Mandatory Course (Non-Credit)									
A6HS06	Constitution of India	HSMC	2	-	-	-	50	-	50

II B.Tech IISemester									
Course Code	Course Title	Course Area	Hours per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		Internal (CIE)	External (SEE)	Total
A6EC12	Digital System Design	PCC	3	-	-	3	40	60	100
A6EE09	Electrical Machines – II	PCC	3	-	-	3	40	60	100
A6EE10	Power Systems – I	PCC	3	-	-	3	40	60	100
A6EC06	Signals and systems	PCC	3	-	-	3	40	60	100
A6EE11	Electrical Measurements and Instrumentation	PCC	3	-	-	3	40	60	100
A6EC16	Digital System Design Lab	PCC	-	-	2	1	40	60	100
A6EE12	Electrical Machines - II Lab	PCC	-	-	2	1	40	60	100
A6EE13	Electrical Measurements and Instrumentation Lab	PCC	-	-	2	1	40	60	100
A6EE14	Real time research project	PWC			4	2	50		50
Total			15	-	10	20	370	480	850
Mandatory Course (Non-Credit)									
A6HS05	Gender sensitization	HSMC	-	-	2	-	50	-	50

III B.Tech I Semester									
Course Code	Course Title	Course Area	Hours per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		Internal (CIE)	External (SEE)	Total
A6EE15	Power Electronics	PCC	3	1	-	4	40	60	100
A6EE16	Power Systems – II	PCC	3	-	-	3	40	60	100
A6EE17	Linear Control Systems	PCC	3	-	-	3	40	60	100
	Professional Elective –1	PEC	3	-	-	3	40	60	100
	Business Economics and Financial Analysis	HSMC	3	-	-	3	40	60	100
A6EE18	Control Systems Lab	PCC	-	-	2	1	40	60	100
A6EE19	Power Electronics Lab	PCC	-	-	2	1	40	60	100
A6HS03	Advanced English Communication skills Lab	HSMC	-	-	2	1	40	60	100
A6EE20	Independent study	PWC	-	-	-	1		100	100
Total			15	1	6	20	320	580	900
Mandatory Course (Non-Credit)									
A6HS10	Human values and Professional Ethics		2	-	-	-	50	-	50
III B.Tech II Semester									
Course Code	Course Title	Course Area	Hours per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		Internal (CIE)	External (SEE)	Total
A6EE21	Switch gear and protection	PCC	3	-	-	3	40	60	100
A6EC21	Microprocessors and Microcontrollers	PCC	3	-	-	3	40	60	100
A6EE22	Electric Drives	PCC	3	-	-	3	40	60	100
	Professional Elective –2	PEC	3	-	-	3	40	60	100
	Open Elective – 1	OEC	3	-	-	3	40	60	100
A6EE23	Electric Drives Lab	PCC	-	-	2	1	40	60	100
A6EC22	Microprocessors and Microcontrollers Lab	PCC	-	-	2	1	40	60	100
A6EE24	Power Systems and Protection Lab	PCC	-		2	1	40	60	100
A6EE25	Mini project/Internship	PWC	-	-	4	2		100	100
Total			15	-	10	20	320	580	900
Mandatory Course (Non-Credit)									
A6BS11	Environmental Science	HSMC	2	-	-	-	50	-	50

IV B.Tech I Semester									
Course Code	Course Title	Course Area	Hours per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		Internal (CIE)	External (SEE)	Total
A6EE26	Power System Operation and control	PCC	3	1	-	4	40	60	100
A6EE27	Utilization of Electrical Energy	PCC	3	-	-	3	40	60	100
	Professional Elective-3	PEC	3	-	-	3	40	60	100
	Professional Elective-4	PEC	3	-	-	3	40	60	100
	Open Elective – 2	OEC	3	-	-	3	40	60	100
A6EE28	Power Systems Simulation Lab	PCC	-	-	2	1	40	60	100
A6EE29	Research Project Phase-I	PWC	-	-	6	3	100		100
Total			15	1	8	20	340	360	700

IV B.Tech II Semester									
Course Code	Course Title	Course Area	Hours per Week			Credits	Scheme of Examination Maximum Marks		
			L	T	P		Internal (CIE)	External (SEE)	Total
	Professional Elective -5	PEC	3	-	-	3	40	60	100
	Professional Elective -6	PEC	3	-	-	3	40	60	100
	Open Elective-3	OEC	3	-	-	3	40	60	100
A6EE30	Research Project Phase-II	PWC	-	-	22	9+2	40	60	100
Total			9	-	22	20	160	240	400

PROFESSIONAL ELECTIVES			
PROFESSIONAL ELECTIVE-I		PROFESSIONAL ELECTIVE-II	
A6EE31	Wind and Solar Energy Systems	A6EE35	Linear and Digital IC Applications
A6EE32	AI Techniques in Electrical Engineering	A6EE36	Industrial Automation
A6EE33	Special Electrical Machines	A6EE37	Solar Power Batteries
A6EE34	Digital Signal Processing	A6EE38	Electrical Energy Conservation and Auditing
PROFESSIONAL ELECTIVE-III		PROFESSIONAL ELECTIVE-IV	
A6EE39	Modelling and Analysis of Electrical Machines	A6EE43	HVDC Transmission Systems
A6EE40	Machine Learning Applications to Electrical Engineering	A6EE44	Power System Dynamics and Control
A6EE41	Electrical Distribution Systems	A6EE45	Soft Computing Techniques
A6EE42	Flexible Alternating Current Transmission Systems	A6EE46	High Voltage Engineering
PROFESSIONAL ELECTIVE-V		PROFESSIONAL ELECTIVE-VI	
A6EE47	Power System Reliability	A6EE51	Reactive Power Control & Management
A6EE48	Smart Grid Technologies	A6EE52	Advanced Control Systems
A6EE49	Advanced Electric Drives	A6EE53	Power Quality
A6EE50	Electric Hybrid Vehicles	A6EE54	Power Electronic Applications to Renewable Energy Systems

OPEN ELECTIVE COURSE-I			
S. No.	Course Code	Course Name	Offering Department
1	A6EE55	Electrical Wiring and Safety Measures	Electrical &Electronics Engineering
2	A6EE56	Electrical Materials	
OPEN ELECTIVE COURSE-II			
S. No.	Course Code	Course Name	Offering Department
1	A6EE57	Neural Networks and Fuzzy Logic	Electrical &Electronics Engineering
2	A6EE58	Energy Storage Systems	
OPEN ELECTIVE COURSE-III			
S. No.	Course Code	Course Name	Offering Department
1	A6EE59	Solar Energy and Applications	Electrical &Electronics Engineering
2	A6EE65	Instrumentation and Control	

I B.TECH I SEMESTER SYLLABUS

I Semester: Common for All (ECE/EEE/MECH/AERO/ CSE/IT/AI&ML/CS/DS)								
Course Code:	Category	Hours / Week			Credits	Maximum Marks		
A6BS01	BSC	L	T	P	C	CIA	SEE	Total
		3	1	-	4	40	60	100
Contact Classes: 44	Tutorial Classes: 08	Practical Classes:--			Total Classes: 52			
Course Objectives								
To learn								
1. Concept of Rank of a matrix, Consistency and solving system of linear equations.								
2. Concept of eigen values, eigen vectors and diagonalization of the matrix.								
3. The concept of differential equations and solvethem using appropriate methods.								
4. Evaluate multiple integrals and improper integrals								
5. The partial derivatives of several variable functions.								
UNIT-I	MATRICES AND THEIR APPLICATIONS						Classes: 08	
Real matrices: Symmetric-skew-symmetric and orthogonal matrices –Complex matrices: Hermitian, Skew – Hermitian and Unitary matrices –Elementary row and column transformations –Elementary matrix-Finding rank of a matrix by reducing to Echelon form and Normal form-Finding the inverse of a matrix using elementary row/column transformations (Gauss-Jordan method)-Consistency of system of linear equations (homogeneous and non-homogeneous) using the rank of a matrix –Solving m n and n n linear system of equations by Gauss Elimination, Gauss siedel Method								
UNIT-II	EIGEN VALUES, EIGEN VECTORS						Classes: 08	
Eigen values and Eigen vectors and its properties(without proof),Cayley-Hamilton theorem (Statement and verification)-Finding inverse and powers of a matrix by Cayley-Hamilton theorem, Diagonalisation of matrices. The n-dimensional vector space(only definition and examples)-linear dependence of vectors-Basis and dimension								
UNIT-III	ORDINARY DIFFERENTIAL EQUATIONS AND THEIR APPLICATIONS						Classes: 10	
Introduction- Exact and reducible to Exact differential equations-Newton’s Law of cooling-Law of Growth and Decay. Linear differential equations of second and higher order with constant coefficients - Non-Homogeneous term of the type $Q(x) = e^{ax}$, $\sin ax$, $\text{Cos}ax$, $e^{ax}v(x)$, $x^n v(x)$ - Method of variation of parameters L-C-R Circuits.								
UNIT-IV	MULTIPLE INTEGRALS, BETA AND GAMMA FUNCTIONS						Classes: 10	
Double and triple integrals (Cartesian and polar), Change of order of integration in double integrals, Change of variables (Cartesian to polar) in double integrals. Finding the area and volume of a region using double and triple integral. Beta- Gamma Functions and their Properties-Relation between them- Evaluation of improper integrals using Gamma and Beta functions.								
UNIT-V	CALCULUS OF SEVERAL VARIABLES						Classes: 08	
Limit, Continuity - Partial derivative- Partial derivatives of higher order -Total derivative - Chain rule,Jacobians-functional dependence & independence. Applications: Maxima and Minima of functions of two variables without constraints and Lagrange’s method (with constraints).								

Text Books:

1. Ervin Kreyszig, Advanced Engineering Mathematics, 9th Edition, John Wiley & Sons, 2006.
2. B.S.Grewal, Higher Engineering Mathematics, Khanna publishers, 36th Edition, 2010.

Reference Books:

1. G.B.Thomas, calculus and analytical geometry,9th Edition, Pearson Reprint 2006.
2. N.P Bali and Manish Goyal ,A Text of Engineering Mathematics,Laxmi publications,2008.
3. E.L.Ince, Ordinary differential Equations,Dover publications,1958.

Web references:

1. https://www.efunda.com/math/math_home/math.cfm
2. <https://www.ocw.mit.edu/resources/#Mathematics>
3. <https://www.sosmath.com/>
4. <https://www.mathworld.wolfram.com/>

E -Text Books:

1. <https://www.e-booksdirectory.com/details.php?ebook=10166>

MOOCS Course:

1. <https://swayam.gov.in/>
2. <https://onlinecourses.nptel.ac.in/>

Course Outcomes

at the end of the course, student will be able to:

1. Solve the system of linear equations using rank of the matrices.
2. Find the Eigen values and Eigen vectors of a matrix
3. Identify the different types of differential equations and solve them using appropriate methods.
4. Evaluate the improper integrals using beta and gamma functions.
5. Find the Maxima and Minima of several variable functions.

APPLIED PHYSICS**R22**

I-Year B.Tech : CSE, IT, CSE-Data Science, CSE-Cyber Security, CSE-AIML, CSIT,EEE								
Course Code:	Category	Hours / Week			Credits	Maximum Marks		
A6BS07	BSC	L	T	P	C	CIA	SEE	Total
		3	1	0	4	40	60	100
Course Objectives: The course should enable the students to: 1. Learn the basic principles of quantum physics and its applications 2. Understand the formation of energy bands and atomic structure in solids for material classification 3. Understand the underlying mechanism involved in construction and working properties of different types of semiconductor devices 4. Learn the basic principles of laser and optical fiber in information technology 5. Identify the importance of nanoscale and various fabrication and characterization techniques and quantum computations in engineering applications Course Outcomes: At the end of the course students will be able to: 1. Analyze the microscopic properties of materials using principles of quantum physics for engineering applications 2. Explain the behavior of different electronic materials based on the concepts of band theory 3. Apply the knowledge of Solar PV cells for choice of materials in efficient alternate energy generation 4. Gain the knowledge of production of laser and usage of fibers in fiber optic communication technology 5. Comprehend the knowledge of quantum physics in quantum computation for secure information Technology								
UNIT-I	Quantum Physics					Classes: 10		
Black body radiation, Stefan-Boltzmann’s law, Planck’s radiation law (Qualitative treatment), Photoelectric effect, Waves and Particles, de Broglie Hypothesis, Matter Waves, Davisson and Germer’s experiment, Heisenberg’s Uncertainty Principle, Schrodinger’s Time Independent Wave Equation, Physical Significance of the wave Function, Particle in One Dimensional Potential Box.								
UNIT-II	Electronic properties of Materials & Band theory of solids					Classes: 10		
Electronic properties of Materials: Classical free electron theory and Quantum free electron theories of metals, success and drawbacks, Bloch theorem, Kronig-Penny model (Qualitative treatment), E-k diagram, effective mass of electron. Band theory of solids: Origin of energy band formation in solids, Fermi energy level, Fermi-Dirac distribution law, Classification of materials as conductors, insulators and semiconductors.								

UNIT-III	Semi-conductors & Semiconductor Devices	Classes: 08
Semiconductors: Intrinsic and Extrinsic Semiconductors, formation of PN junction diode and its V-I characteristics, Direct and Indirect band gap semiconductors, Hall effect and its applications.		
Semiconductor Devices: Construction, working and V-I characteristics of PIN Diode, LED, Solar cell and their applications.		
UNIT-IV	Laser & Fiber Optics	Classes: 12
Laser: Characteristics of Laser, Absorption, Spontaneous and Stimulated emission of radiations. Lasing actions-Pumping mechanism, Meta stable state and Population inversion, Nd-YAG laser, CO2 laser, Applications of lasers in different fields.		
Fiber Optics: Structure of fibers, Total Internal Reflection, Acceptance angle – Numerical Aperture, Types of fibers- SI and GI fibers, Single and Multimode fibers - SMSI, MMSI, MMGI, Fiber Optic Communication system, Signal Degradation - Attenuation mechanism, Dispersion, Applications of fibers in different fields.		
UNIT-V	Physics of Quantum computing & Quantum gates	Classes: 10
Physics of Quantum computing: Idea of classical bits and qubits, advantages with qubits over classical bits, Bloch vector representation of state of qubits.		
Quantum gates- Single qubit logic gates- Pauli X, Y, Z and Hadmard gate in matrix form, Two level gates-CNOT and SWAP gates and representation in matrix form, Comments on No cloning theorem, Entanglement, Quantum Teleportation – Basic Idea, Quantum Key distribution protocol - BB84 protocol.		
Text Books:		
1. P.K Palanisamy, Engineering Physics, Sitech Publications, 2013, IVth Ed. 2. Nielsen M. A., I. L Chung, Quantum Computation & Quantum Information, Cambridge Univ. Press. 3. M. N. Avadhanulu, P.G. Kshirsagar & TVS Arun Murthy” A Text book of Engineering Physics”- S. Chand Publications, 11th Edition 2019.		
Reference Books:		
1. Quantum Physics, H.C. Verma, TBS Publication, 2 nd Edition 2012 2. Fundamenttals of Physics – Halliday, Resnick and Walker, John Wiley & Sons, 11 th Edition, 2018. 3. Engineering Physics by Shatendra Sharma and Jyotsna Sharma, Pearson Publication,2019 4. B.K Pandey and Chaturvedi, Engineering Physics, Cengage Learning, 2 nd Edition, 2022		

ENGINEERING CHEMISTRY R22

IB.TECH-I/IISEMESTER:AERO, MECH, EEE								
Course Code:	Category	Hours / Week			Credits	Maximum Marks		
A6BS09	BSC	L	T	P	C	CIE	SEE	Total
		3	1	0	4	40	60	100
Contact Classes:50	Tutorial Classes: 0	Practical Classes: 0			Total Classes: 50			
Course Outcomes: At the end of the course students will be able to: 1. Illustrate the types of hard and soft water, treatment of drinking and industrial water. 2. Demonstrate the working principles of batteries and fuel cells. 3. Apply the methods of metal finishing in solving corrosion related problems. 4. Explain the chemistry of various fuels and their combustion. 5. Identify the properties, application of lubricants and mechanism of lubrication.								
UNIT-I	WATER AND ITS TREATMENT						Classes: 10	
Introduction to hardness of water – Estimation of hardness of water by complexometric method and related numerical problems. Potable water and its specifications - Steps involved in the treatment of potable water - Disinfection of potable water by chlorination and break - point chlorination. Defluoridation - Determination of Fion by ion- selective electrode method. Boiler troubles: Sludges, Scales and Caustic embrittlement. Internal treatment of Boiler feed water - Calgon conditioning - Phosphate conditioning - Colloidal conditioning, External treatment methods - Softening of water by ion- exchange processes. Desalination of water – Reverse osmosis.								
UNIT-II	BATTERY CHEMISTRY AND FUEL CELLS						Classes:10	
Introduction - Classification of batteries- primary, secondary and reserve batteries with examples. Basic requirements for commercial batteries. Construction, working and applications of: Zn-air, Lead acid and Lithium ion battery, Applications of Li-ion battery to electrical vehicles. Fuel Cells- Differences between battery and a fuel cell, Construction and applications of Methanol Oxygen fuel cell, Molten carbonate fuel cells and Solid oxide fuel cell.								
UNIT-III	CORROSION AND ITS CONTROL						Classes: 10	
Introduction-causes and effects -Mechanism of chemical and electrochemical Corrosion-Types of corrosion - Galvanic, water-line and pitting corrosion.Factors affecting rate of corrosion-Corrosion control methods- Cathodic protection (Sacrificial anodic, Impressed current cathodic) Protective coatings-Metallic coatings-Hot dipping methods-Galvanization-Tinning-Cementation(sherardizing)- Electroplating of Cu-Electrolessplating of Ni-Organic coatings-Paints (constituents and functions).								
UNIT-IV	ENERGY SOURCES						Classes: 10	

Introduction - Fuels- Characteristics of good fuels - Classification of fuels with examples – coal – analysis of coal – proximate and ultimate analysis and their significance. Composition, Characteristics and uses of gasoline, diesel, kerosene oils - Knocking- octane and cetane rating – Aviation fuel: Composition and Characteristics- LPG and CNG: Composition, Characteristics and Uses. Biodiesel – Transesterification, advantages. Combustion - Calorific value – Gross Calorific Value and Net Calorific Value. Dulong's formula- Calculation of air quantities required for combustion of a fuel – Numerical problems.

UNIT-V

LUBRICANTS

Classes: 10

Introduction - Lubricant: Definition – Classification with examples: Solid, gas and semisolid lubricants – Mechanism of Lubrication –Thin film, Thick film and Extreme pressure-Characteristics of good lubricant – Important properties of lubricant: viscosity, cloud- pour point, flash Point, fire point, demulsibility, volatility - Lubricant Additives – Advantages - Grading of lubricants.

Text Books:

1. Text Book of Engineering Chemistry by Dr. Bharathi Kumari.Y, Dr. Jyothsna, V.G.S Book Links.
2. Text Book of Engineering Chemistry by C. Parameshwaramurthy, C.V. Agarwal, B.S. Publications.
3. A text book of Engineering Chemistry by M. Thirumala Chary, E. Laxminarayana and K. Shashikala, Pearson Publications, 2021.
4. Textbook of Engineering Chemistry by Jaya Shree Anireddy, Wiley Publications.

Reference Books:

1. Sashichawla, A Textbook of Engineering Chemistry, Dhanapath Rai and sons, 2003.
2. P.C. Jain and M. Jain, Engineering Chemistry, 15/e, Dhanapat Rai & Sons, Delhi, 2014.

ENGINEERING GRAPHICS

FOR AERO, ECE, EEE & MECH								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6ME01	ESC	L	T	P	C	CIA	SEE	Total
		1	-	4	3	40	60	100
COURSE OUTCOMES: At the end of the course the student should be able to: 1 Explain various commands and create drawing in AutoCAD. 2 Sketch the various curves used in engineering applications 3 Prepare orthographic projections of lines, planes by visualizing them in different positions. 4 Solve the problem of projections of solids and development of surfaces for industrial needs. 5 Construct the isometric view into orthographic views and viceversa.								
UNIT-I	INTRODUCTION							
Introduction to Engineering Graphics: Principles and their significance, INTRODUCTION TO COMPUTER AIDED DRAFTING: Initial Setup Commands, Draw Commands, Modify Commands, 2D Drawings-Simple Exercises.								
UNIT-II	ENGINEERING CURVES							
Engineering Curves: Ellipse, Parabola, and Hyperbola (General Method only). Special curves: Cycloids, Involute.								
UNIT-III	ORTHOGRAPHIC PROJECTION							
Principles of Orthographic Projections – conventions – first and third angle projections. Projections of points- Projection of lines inclined to both the planes. (First angle projection only) PROJECTIONS OF PLANES: Projections of regular planes inclined to both planes.								
UNIT-IV	PROJECTION OF SOLIDS AND DEVELOPMENT OF SURFACES							
PROJECTION OF SOLIDS: Regular Solids inclined to both planes DEVELOPMENT OF SURFACE/SOLIDS: Theory of development, development of lateral surface with base.								
UNIT-V	ISOMETRIC DRAWINGS							
Divisions of pictorial projection, theory of Isometric Drawing- Isometric view and Isometric projections. Drawing Isometric circles, Dimensioning Isometric objects. Conversion of Isometric view to Orthographic views and Orthographic views to isometric view.								
Text Books: 1 Bhatt N.D., Panchal V.M. & Ingle P.R., (2014), Engineering Drawing, Charotar PublishingHouse. 2 Agrawal B. & Agrawal C. M. (2012), Engineering Graphics, TMHPublication.								

Reference Books:

1. D.M. Kulkarni, A.P.Rastogi, A.K. Sarka “Engineering Graphics with AutoCAD” PHI publications,2013.
2. Narayana, K.L. & P Kannaiah (2008), Text book on Engineering Drawing, Scitech Publishers.
3. Shah, M.B. &Rana B.C. (2008), Engineering Drawing and Computer Graphics, PearsonEducation.

APPLIED PHYSICS LAB**R22****I-Year B.Tech :** CSE, IT, CSE-Data Science, CSE-Cyber Security, CSE-AIML, CSIT,EEE

Course Code:	Category	Hours / Week			Credits	Maximum Marks		
A6BS08	BSC	L	T	P	C	CIA	SEE	Total
		0	0	3	1.5	40	60	100

Course Objectives:**The course should enable the students to:**

1. Understand the temperature and other dependent properties of semiconductor devices and their usages in different applications
2. To know the Laser and Fibre optic technologies and its applications in real time scenario
3. Understand the electromagnetic properties using experimental knowledge
4. Understand the method of least squares fitting

Course Outcomes:**By the end of the course students will be able to:**

1. **Analyze** the electric properties of semiconductor materials by determining energy gap of semiconductors, charge carrier concentration in Semiconductors using Hall effect and threshold voltage of LEDs, photo current in Photo diodes, solar cell, and temperature effect on resistance using thermistor.
2. **Identify** the optical properties of light such as diffraction phenomenon using grating material for calculation of the wavelength of Laser and acceptance angle, NA of optical fiber using OFC and determine the value of Plank's constant using a light source and interference by using Newton's rings
3. **Analyze** the electromagnetic properties of a current carrying coil by using Stewart Gee's experiment
4. **Analyze** the least squares fitting method for data analysis using experimental data of Tensional Pendulum

APPLIED PHYSICS LABORATORY**LIST OF EXPERIMENTS**

Experiment-1	ENERGY GAP OF P-N JUNCTION DIODE: To determine the energy gap of a given semiconductor diode
Experiment-2	SOLAR CELL: To study the V-I and V-P characteristics and determine the fill factor of solar cell
Experiment-3	LIGHT EMITTING DIODE: To study the characteristics of LED by plotting V-I graph and determine the threshold value of given LEDs
Experiment-4	HALL EFFECT: To determine Hall co-efficient and charge carrier concentration of a given semiconductor
Experiment-5	PIN PHOTO DIODE: To study the V-I Characteristics of Photo Diode with respect to intensity of light
Experiment-6	OPTICAL FIBRE: To determine the numerical aperture and acceptance angle of an optical fiber

Experiment-7	LASER: To determine the wavelength of a given laser source by using diffraction grating method
Experiment-8	NEWTON'S RINGS: To determine the radius of curvature of a given Plano convex lens by forming Newton's rings
Experiment-9	THERMISTOR: To study the variation of resistance with respect to temperature using thermistor
Experiment-10	Understanding the method of least squares - Torsional pendulum as an example
Experiment-11	PLANCK'S CONSTANT: To determine value of planck's constant by measuring radiation in fixed spectral range
Experiment-12	STEWART GEE'S EXPERIMENT: To study the variation of magnetic field along the axis of a circular coil and calculation of magnetic flux

Note: Students have to perform any 8 experiments

Reference Books:	
1.	"Applied Physics Lab Manual"- Dr. Radhika Devi, Mr. A V Laxman Rao, N. Noel
2.	S. Balasubramanian, M.N Srinivasan "A Text book of practical Physics" – S Chand Publishers, 2017.

ENGINEERING CHEMISTRY LAB
R22

I B. TECH I/II SEM: AERO, MECH, EEE

Course Code:	Category	Hours / Week			Credits	Maximum Marks		
A6BS10	BSC	L	T	P	C	CIE	SEE	Total
		0	0	3	1.5	40	60	100
Contact Classes: 0	Tutorial Classes: 0	Practical Classes: 39			Total Classes: 39			

Course Outcomes:

At the end of the course students will be able to:

1. **Estimate** the hardness, alkalinity and chloride content in water to check its suitability for drinking.
2. **Measure** the physical properties of lubricants.
3. **Demonstrate** the digital and instrumental methods of analysis
4. **Synthesize** the various polymeric compounds.

LIST OF EXPERIMENTS

Experiment-1	Determination of total hardness of water by complexometric method using EDTA
Experiment-2	Determination of Alkalinity of given water sample
Experiment-3	Estimation of Chloride content of water by Argentometry.
Experiment-4	Estimation of the concentration of an acid by Conductometry
Experiment-5	Estimation of amount of ferrous ion by potentiometry using potassium dichromate
Experiment-6	Estimation of HCl by potentiometry
Experiment-7	Preparation of Bakelite
Experiment-8	Preparation of Thiokol Rubber
Experiment-9	Estimation of acid value of given lubricant oil
Experiment-10	Estimation of Viscosity of lubricant oil using Ostwald's Viscometer.
Experiment-11	Determination of rate of corrosion of mild steel in the presence and absence of inhibitor
Experiment-12	Estimation of Surface tension of a given liquids using Stalagmometer

VIRTUAL LABS

Experiment-13	Electroplating of copper on an article of iron
Experiment-14	Batteries for electrical vehicles
Experiment-15	Learn the specific charge/discharge characteristics of a Lead Acid battery through
Experiment-16	Determine the moisture, volatile, fixed carbon content and content in a given coal sample by proximate analysis.

Reference Books:

- 1) Senior practical physical chemistry, B. D. Khosla, A. Gulati and V. Garg (R. Chand and amp; Co., Delhi)
- 2) An introduction to practical chemistry, K. K. Sharma and D. S. Sharma (Vikas publishing, N. Delhi)
- 3) Vogel's textbook of practical organic chemistry 5th edition.
- 4) Text book on Experiments and calculations in Engineering chemistry- S. S. Dara.

ENGINEERING WORKSHOP PRACTICES

I Semester / II Semester(AERO, MECH,EEE)								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6ME03	ESC	L	T	P	C	CIE	SEE	Total
		0	1	2	2	40	60	100
COURSE OBJECTIVES: 1. To Study of different hand operated tools, uses and their demonstration. 2. To gain a good basic working knowledge required for the production of various engineering products. 3. To provide hands on experience about use of different engineering materials, tools, equipment's and processes those are common in the engineering field. 4. It explains the construction, function, use and application of different working tools, equipment and machines. 5. To study commonly used carpentry joints 6. Identify and use marking out tools, hand tools, measuring equipment and to work to prescribed tolerances. 7. To develop a right attitude, team working, precision and safety at work place								
COURSE OUTCOMES: Student will be able to: 1. Assemble different components 2. Identify and apply suitable tools for different trades of Engineering processes. 3. Practice on manufacturing of components using workshop trades including Soldering, Carpentry, Fitting and Tin smithy & Fabricate Components with their own hands. 4. Apply basic electrical engineering knowledge for house wiring practice. 5. Learn the safety precautions for various operations in basic trade operations.								

WEEKS	TRADES FOR EXERCISES
	Fitting
Week 1	Filing Four Sides of MS Work piece
Week 2	L Fit
	Carpentry
Week 3	Half Lap Joint
Week 4	Dove Tail Joint
	Tin Smithy
Week 5	Prepare a Rectangular Tray
Week 6	Prepare A Square Tin
	Electrical
Week 7	Series and Parallel Connection One Way Switch

Week 8	Two Way Switch Connection Stair Case Wiring
	Electronics
Week 9	Soldering - Series Connection & Parallel Connection
Week 10	Desoldering & Construction of Wheat stone bridge
Week 11	Blacksmithy : Round to S-Hook
Week 12	Blacksmithy : Round to Chisel
WEEK	TRADES FOR DEMONSTRATION & EXPOSURE
Week 13	Introduction to Plumbing
TEXT BOOKS	
1. Work shop Manual - P. Kannaiah/ K.L. Narayana/ Scitech. 2. Rao P.N., "Manufacturing Technology", Tata McGraw Hill House, Vol. I and Vol. II. 3. Workshop Manual / K. Venugopal / Anuradha.	
REFERENCE BOOKS:	
1. Hajra Choudhury S.K., Hajra Choudhury A.K. and Nirjhar Roy S.K., "Elements of Workshop Technology", Media promoters and publishers private limited, Mumbai, Vol. I 2008 and Vol. II 2010. 2. Kalpakjian S. and Steven S. Schmid, "Manufacturing Engineering and Technology", Pearson Education India Edition, 4 th edition, 2002. 3. Workshop Manual / Venkat Reddy/ BSP	

ENVIRONMENTAL SCIENCE						R22		
IB.TECH-I/I SEMESTER: COMMON TO ALL BRANCHES								
CourseCode:	Category	Hours/Week			Credits	MaximumMarks		
A6BS11	BSC	L	T	P	C	CIE	SEE	Total
		3	0	0	0	50	-	50
ContactClasses: 30	Tutorial Classes:0	PracticalClasses:--0			TotalClasses:30			
COURSE OUTCOMES:								
On Successful completion of this course, Students will be able to								
1. Identify the consequences of human actions on the web of life, global economy and quality of human life. 2. Evaluate the strategies for scientific, social, economic and legal environmental protection. 3. Study the impact of conservation of biodiversity. 4. Analyze the reasons for environmental pollution. 5. Assess the environmental impact of air, water, biological and socio-economical aspects and risk assessment towards sustainable future.								
UNIT-I	ECOSYSTEMS						CLASSES:7	
Ecosystems: Definition, Scope and Importance of ecosystem. Classification, structure and function of an ecosystem, Food chains, food web and ecological pyramids.								
Flow of energy, Biogeochemical cycles, Bioaccumulation, Biomagnification, ecosystem value, services and carrying capacity.								
UNIT-II	NATURAL RESOURCES						CLASSES: 8	
Natural Resources: Classification of Resources: Living and Non-Living resources, water resources: use and over utilization of surface and ground water, floods and droughts, Dams: benefits and problems-case studies. Mineral resources: use and exploitation, environmental effects of extracting and using mineral resources. Land resources: Forest resources.								
Energy resources: Growing energy needs, renewable and non-renewable energy sources, use of alternate energy source, case studies.								
UNIT-III	BIODIVERSITY AND BIOTIC RESOURCES						CLASSES: 5	
Biodiversity and Biotic Resources: Introduction, Definition, genetic, species and ecosystem diversity. Value of biodiversity; consumptive use, productive use, social, ethical, aesthetic and optional values.								
India as a mega diversity nation: Hot spots of biodiversity. Threats to biodiversity: habitat loss, poaching of wildlife, man-wildlife conflicts; conservation of biodiversity: In-Situ and Ex-situ conservation. National Biodiversity Act-Case studies.								
UNIT-IV	ENVIRONMENTAL POLLUTION AND CONTROL TECHNOLOGIES						CLASSES:5	
Environmental Pollution and Control Technologies: Environmental Pollution: Classification of pollution, Air Pollution, Water Pollution and drinking water quality standards. Soil Pollution Impact of modern agriculture, Noise Pollution Health hazards, standards. Concepts of bioremediation.								
Global Environmental Problems and Global Efforts: Ozone depletion and Ozone depleting								

substances (ODS) Concepts of Bioremediation International conventions /Protocols: Earth summit, Kyoto protocol and Montréal Protocol; NAPCC-GoI Initiatives, COP 24, COP25.

UNIT-V	ENVIRONMENTAL POLICY, LEGISLATION & EIA	CLASSES:5
Environmental Policy, Legislation & Environmental Impact Assessment (EIA): Environmental Protection act, Legal aspects Air Act 1981, Water Act, Forest Act, Wildlife Act, Municipal solid waste management, biomedical waste management hazardous waste management and handling rules. Environmental Impact Assessment: EIA structure, methods of baseline data acquisition. Strategies for risk assessment, Towards Sustainable Future: Concept of Sustainable Development, Urban Sprawl, Concept of Green Building.		
TEXT BOOKS: 1. Textbook of Environmental Studies for Undergraduate Courses by Erach Bharucha for University Grants Commission. 2. Environmental Studies by R. Rajagopalan, Oxford University Press.		

I B.TECH II SEMESTER SYLLABUS

NUMERICAL METHODS AND INTEGRAL TRANSFORMS

R22

II Semester: Common for All (ECE/EEE/MECH/AERO/ CSE/IT/AI&ML/CS/DS)								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6BS02	BSC	L	T	P	C	CIE	SEE	Total
		3	1	-	4	40	60	100
Contact Classes: 44	Tutorial Classes: 08	Practical Classes: Nil			Total Classes: 52			
Course Objectives								
To learn								
1. Curve fitting and Interpolation techniques.								
2. Numerical techniques.								
3. Fourier series for periodic function								
4. Laplace transforms								
5. Concept and application of Fourier Transforms and Vector differentiation.								
UNIT-I	INTERPOLATION AND CURVE FITTING						Classes: 08	
INTERPOLATION: Finite differences: Forward, Backward and Central differences - Other difference operators and relations between them - Difference of a polynomial – Missing terms - Newton’s forward interpolation, Newton’s backward interpolation, Gauss’s forward and backward interpolation formulae. Interpolation with unequal intervals – Lagrange’s interpolation.								
CURVE FITTING: Method of least squares - Fitting a straight line, second degree parabola and non-linear curves of the form $y = a e^{b^x}$, $y = a x^b$, $y = a b^x$ by the method of least squares.								
UNIT-II	NUMERICAL TECHNIQUES						Classes: 08	
ROOT FINDING TECHNIQUES:								
Bisection method Regula-falsi method, Iteration method and Newton Raphson method.								
NUMERICAL INTEGRATION :								
Trapezoidal rule - Simpson’s one-third rule - Simpson’s three-eighth rule.								
NUMERICAL SOLUTION OF ORDINARY DIFFERENTIAL EQUATIONS:								
Taylor’s series method – Euler’s - modified Euler’s Method – Runge-Kutta method.								
UNIT-III	FOURIER SERIES						Classes:10	
Periodic function-Determination of Fourier Coefficients-Fourier Series-Even and Odd functions-Fourier series in arbitrary interval-Half range Fourier sine and cosine expansions.								
UNIT-IV	LAPLACE TRANSFORMS						Classes: 10	
Laplace transforms of elementary functions- First shifting theorem - Change of scale property – Multiplication by t^n - Division by t – Laplace transforms of derivatives and integrals – Unit step function – Second shifting theorem – Periodic function – Evaluation of integrals by Laplace transforms – Inverse Laplace transforms- Method of partial fractions – Other methods of finding inverse transforms – Convolution theorem –								

Applications of Laplace transforms to ordinary differential equations.		
UNIT-V	FOURIER TRANSFORMS AND VECTOR DIFFERENTIATION	Classes: 10
Fourier Integral theorem (Statement only)-Fourier Sine and Cosine Integrals, Fourier Transforms, Cosine and Sine transforms, properties, Inverse transforms. Vector functions, vector differentiation, gradient, directional derivative, divergence, curl and scalar potential.		
Text Books:		
1. Ervin Kreyszig, Advanced Engineering Mathematics, 9th Edition, John Wiley & Sons, 2006. 2. B.S.Grewal, Higher Engineering Mathematics, Khanna publishers, 36th Edition, 2010.		
Reference Books:		
1. G.B.Thomas, calculus and analytical geometry,9th Edition, Pearson Reprint 2006. 2. N.P Bali and Manish Goyal ,A Text of Engineering Mathematics,Laxmi publications,2008. 3. E.L.Ince, Ordinary differential Equations,Dover publications,1958.		
Web references:		
1. https://www.efunda.com/math/math_home/math.cfm 2. https://www.ocw.mit.edu/resources/#Mathematics 3. https://www.sosmath.com/ 4. https://www.mathworld.wolfram.com/		
E -Text Books:		
1.https://www.e-booksdirectory.com/details.php?ebook=10166		
MOOCS Course:		
1. https://swayam.gov.in/ 2. https://onlinecourses.nptel.ac.in/		

Course Outcomes:

At the end of the course, student will be able to:

1. Apply Curve fitting and Interpolation techniques.
2. Apply various numerical techniques
3. Find the Fourier series of the periodic functions.
4. Obtain the Laplace transforms of functions
5. Find Fourier transforms and apply vector differentiation techniques.

ENGLISH FOR SKILL ENHANCEMENT

I B.TECH : Common to All Branches

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6HS01	HSMC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100
Contact Classes: 64	Tutorial Classes: 0	Practical Classes: 00			Total Classes: 64			

OBJECTIVES:

Student will be able to:

1. Develop language proficiency with emphasis on Vocabulary, Grammar, Reading and Writing skills.
2. Apply the theoretical and practical components of English syllabus to study academic subjects more effectively and critically.
3. Analyze a variety of texts and interpret them to demonstrate in writing or speech.
4. Write/ compose clearly and creatively, and adjust writing style appropriately to the content, the context, and nature of the subject.
5. Develop language components to communicate effectively in formal and informal situations.

UNIT-I	'Toasted English' by R. K. Narayan				Classes: 06+06
Vocabulary	: The Concept of Word Formation -The Use of Prefixes and Suffixes - Acquaintance with Prefixes and Suffixes from Foreign Languages to form Derivatives – Synonyms and Antonyms				
Grammar	: Identifying Common Errors in Writing with Reference to Articles and Prepositions				
Reading:	: Reading and Its Importance- Techniques for Effective Reading				
Writing	: Sentence Structures – Use of Phrases and Clauses in Sentences – Importance of Proper Punctuation – Techniques for Writing precisely – Paragraph Writing – Types, Structures and Features of a Paragraph – Creating Coherence – Organizing Principles of Paragraphs in Documents.				
UNIT-II	'ApproJRD' by Sudha Murthy				Classes: 07+07
Vocabulary	: Words Often Misspelt – Homophones, Homonyms and Homographs				
Grammar	: Identifying Common Errors in Writing with Reference to Noun-pronoun Agreement and Subject-verb Agreement				
Reading	: Sub-Skills of Reading – Skimming and Scanning – Exercises for Practice				
Writing	: Nature and Style of Writing-Defining/Describing People, Objects, Places and Events – Classifying-Providing Examples or Evidence.				
UNIT-III	'Lessons from Online Learning' by F. Haider Alvi, Deborah Hurst et al				Classes: 07+07
Vocabulary	: Words Often Confused – Words from Foreign Languages and their Use in English.				

Grammar	:IdentifyingCommonErrorsinWritingwith ReferencetoMisplacedModifiersand Tenses	
Reading	:Sub-SkillsofReading–IntensiveReadingandExtensiveReading– ExercisesforPractice	
Writing	:FormatofaFormalLetter-WritingFormal LettersE.g.,LetterofComplaint,LetterofRequisition,EmailEtiquette,JobApplic ation withCV/Resume	
UNIT-IV	ArtandLiterature	Classes: 06+06
Vocabulary	: StandardAbbreviationsinEnglish	
Grammar	: RedundanciesandClichésinOralandWrittenCommunication	
Reading	:Survey,Question,Read,ReciteandReview(SQ3RMethod) – ExercisesforPractice	
Writing	:WritingPractices- EssayWriting-WritingIntroductionandConclusion- PrécisWriting	
UNIT-V	Go,KisstheWorld’bySubrotoBagchi	Classes: 06+06
Vocabulary	: TechnicalVocabularyandtheirUsage	
Grammar	: Common Errors in English (<i>Covering all the other aspects of grammar which were notcoveredin the previousunits</i>)	
Reading	: ReadingComprehension-Exercisesfor Practice	
Writing	: Technical Reports- Introduction – Characteristics of a Report – Categories of ReportsFormats- Structure of Reports (Manuscript Format) -Types of Reports - Writing aReport	
Text Books:		
1. “English: Language, Context and Culture” by Orient BlackSwan Pvt. Ltd, Hyderabad.2022.Print.		
Reference Books:		
1. EffectiveAcademicWriting byLissand Davis(OUP) 2. Richards,JackC.(2022)InterchangeSeries.Introduction,1,2,3.CambridgeUniversityPress 3. Wood,F.T.(2007). RemedialEnglishGrammar.Macmillan. 4. Chaudhuri,SantanuSinha.(2018).LearnEnglish:AFunBookofFunctionalLanguage,Gra mmarandVocabulary.(2 nd ed.,).SagePublicationsIndiaPvt.Ltd. 5. (2019).TechnicalCommunication.WileyIndiaPvt.Ltd. 6. Vishwamohan,Aysha.(2013).EnglishforTechnicalCommunicationforEngineeringStud ents.McGraw-Hill Education India Pvt.Ltd. 7. Swan,Michael.(2016).PracticalEnglishUsage.OxfordUniversityPress.FourthEdition		

Web References:
1. http://www.bbc.co.uk/learningenglish 2. http://learnenglish.britishcouncil.org 3. https://www.cambridgeenglish.org/learning-english/ 4. https://study.com/academy/subj/english.html 5.
E-Text Books:
1. https://www.pdfdrive.com/advanced-english-books.html
MOOC Courses:
1. http://nptel.ac.in/courses/109/106/109106067 2. https://www.britishcouncil.org.tr/en/english/mooc
Course Outcomes: By the end of this course, students will be able to: 1. Understand the importance of vocabulary and sentence structures. 2. Choose appropriate vocabulary and sentence structures for their oral and written communication. 3. Demonstrate their understanding of the rules of functional grammar. 4. Develop comprehension skills from the known and unknown passages. 5. Take an active part in drafting paragraphs, letters, essays, abstracts, précis and reports in various contexts. 6. Acquire basic proficiency in reading and writing modules of English.

ELECTRICAL CIRCUITS - I**I Year II Semester**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6EE01	PCC	L	T	P	C	CIA	SEE	Total
		3	1	-	4	40	60	100

Course Objectives:

The course should enable the students to:

- I. To understand and analyze basic electric parameters
- II. To introduce the electrical elements and analyze them
- III. To analyze two port networks.

Course Outcomes:

Upon successful completion of this course, student will be able to :

1. Evaluate voltage and current values in resistive circuits with independent sources.
2. Analyze the single phase ac circuits, the representation of alternative quantities
3. Evaluate the voltage and currents in electrical circuits using network theorems
4. Obtain equivalent network topologies and locus diagrams for electrical networks
5. Develop various two port network parameters and their inter relations.

SYLLABUS

UNIT-I	DC CIRCUITS	Classes: 10
Electrical circuit elements (R, L and C), ohms law, Types of sources, Source transformation - Voltage and current division rule - star to delta and delta to star transformation - Kirchhoff current and voltage laws - Mesh and Super mesh analysis, Node and super node analysis		
UNIT-II	AC CIRCUITS- RESONANCE	Classes: 10
Representation of sinusoidal waveform, peak ,average and r.m.s values, form factor, peak factor, phasor representation, real power, reactive power, apparent power, power factor. Analysis of single-phase ac circuits consisting of R, L, C series and parallel combinations . Series and parallel resonance circuits		
UNIT-III	NETWORK THEOREMS	Classes: 10
Superposition Theorem, Thevenin's and Norton Theorems, Reciprocity theorem, Maximum power transfer theorem, , Millimans theorem, Tellegans theorem and Compensation Theorem.		
UNIT-IV	NETWORK TOPOLOGY & LOCUS DIAGRAMS	Classes: 10

Definitions, incidence matrix, basic tie set and basic cut set matrices for planar networks. Elementary treatment of locus diagrams of RL, RC and RLC circuits (series and parallel combinations)		
UNIT-V	TWO PORT NETWORK PARAMETERS	Classes: 10
Impedance Parameters, Admittance Parameters, Transmission (ABCD) Parameters, Hybrid Parameters, Conversion of one Parameter to another, Conditions for Reciprocity, and Symmetry, Interconnection of Two Port networks in Series, Parallel and Cascaded configurations, illustrative problems		
Text Books:		
1. M. E. Van Valkenburg, "Network Analysis", Prentice Hall, 2006. 2. Engineering Circuit Analysis - William Hayt, Jack E. Kemmerly, S M Durbin, McGraw Hill Companies. 3. Network Analysis – A. Sudhakar, Shyam Mohan S Palli, 3rd Edition, Tata McGraw Hill Publications.		
Reference Books:		
1. D. Roy Choudhury, "Networks and Systems", New Age International Publications, 1998. 2. Electric Circuits - A. Chakrabarty, Dhanipat Rai & Sons. 3. Network Analysis - N.C Jagan and C. Lakhminarayana, BS publications.		

PROGRAMMING FOR PROBLEM SOLVING

I/I Semester: Common for all Branches								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6CS02	ESC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	40	60	100
Contact Classes: 64	Tutorial Classes: Nil	Practical Classes: Nil			Total Classes: 64			
Course Objectives : 1) To familiarize with the syntax and semantics of C programming language. 2) To learn the usage of structured programming approach in solving problems. 3) To use arrays, pointers, strings and structures in solving problems. 4) To understand how to solve problems related to matrices, Searching and sorting. 5) To understand how to use files to perform read and write operations.								
Course Outcomes : 1) Apply algorithmic thinking to understand, define and solve problems 2) Develop computer programs using programming constructs and control structures and to use arrays to develop C programs 3) Decompose a problem into functions to develop modular reusable code and to use pointers to solve complex problems. 4) Use Strings and structures to formulate algorithms and programs. 5) Use FILE to perform read and write operations.								
UNIT-I	INTRODUCTION - PROBLEM SOLVING AND ALGORITHMIC THINKING & INTRODUCTION TO C LANGUAGE						Classes: 12	
Algorithm -Definition, Characteristics of Algorithm.Constituents of algorithms: - Sequence, Selection and Repetition. Algorithm with Example: Roots Of a Quadratic Equations, Minimum And Maximum Numbers of a Given Set, Given number is prime number or not, given integer is palindrome or not, etc. Flowchart/Pseudo Codewith examples. Introduction To CLanguage: Structureof C Program, Data Types, data input and output statements, Operators, Pprecedenceand Associativity of operators , Evaluation of Expressions, Type Conversions In Expressions.								
UNIT-II	CONTROL STRUCTURES AND ARRAYS						Classes: 15	
Control structures: Decision statements; if and switch statement; Loop control statements: while, for and do while loops, Jump statements: break, continue, goto statements. Arrays: Concepts, One dimensional array, declaration and initialization of one dimensional arrays, two dimensional arrays, initialization and accessing, multi dimensional arrays								
UNIT-III	FUNCTIONS AND POINTERS						Classes: 17	
Functions: Function definition, Types of Functions: User defined and built-in Functions, Advantages of User Defined Functions. Parameter passing in functions: Call by value, Call by reference, Passing arrays to functions,								

Recursion as a different way of solving problems. Example programs, such as Finding Factorial, Fibonacci series, Towers of Hanoi etc.. Storage classes. Pointers: Pointer basics, pointer arithmetic, pointers to pointers, generic pointers, array of pointers, Functions returning pointers, Dynamic memory allocation.		
UNIT-IV	STRINGS AND USER DEFINED DATA TYPES	Classes: 10
Strings: Arrays of characters, variable length character strings, inputting character strings, character library functions, String Handling Functions, Arrays Of Strings Structures and Unions: Structure definition, initialization, accessing structures, nested structures, arrays of structures, Structures and functions, Self-referential structures, unions, typedef, enumerations.		
UNIT-V	FILE HANDLING , SEARCHING AND SORTING	Classes: 10
File Handling: Command Line Arguments, File Modes, Basic File Operations Read, Write And Append, Example Programs. Random Access Using fseek, ftell and rewind Functions. Basic Searching And Sorting Algorithms: Linear and Binary Search, Bubble Sort, Insertion Sort, Quick Sort.		
Text Books:		
1) B.A. Forouzan and R.F. Gilberg C Programming and Data Structures, Cengage Learning, (3rd Edition) 2) Byron Gottfried, "Programming with C", Schaum's Outlines Series, McGraw Hill Education, 3rd edition, 2017. 3) Programming in C. Balagurusamy Edition 3 Publisher Tata McGraw-Hill Publishing, 1990		
Reference Books:		
1) W. Kernighan Brian, Dennis M. Ritchie, "The C Programming Language", PHI Learning, 2nd Edition, 1988. 2) Yashavant Kanetkar, "Exploring C", BPB Publishers, 2nd Edition, 2003. 3) Schildt Herbert, "C: The Complete Reference", Tata McGraw Hill Education, 4th Edition, 2014.4. R. S. Bichkar, "Programming with C", Universities Press, 2nd Edition, 2012. 4) Dey Pradeep, Manas Ghosh, "Computer Fundamentals and Programming in C", Oxford University Press, 2nd Edition, 2006. 5) Stephen G. Kochan, "Programming in C", Addison-Wesley Professional, 4th Edition, 2014.		
Web References:		
1. https://en.wikipedia.org/wiki/Computational_thinking 2. https://nptel.ac.in/courses/106/104/106104128/ 3. https://en.cppreference.com/w/c/language 4. https://www.learn-c.org/		
E-Text Books:		

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1. https://slidelegend.com/queue/computational-thinking-for-the-modern-problem-solver_59d6f01e1723ddb0c7a0df47.html
 2. http://flowgorithm.altervista.org/#elf_11_Lw
 3. <http://www.freebookcentre.net/Language/Free-C-Programming-Books-Download.htm>

MOOC Course

- 1) <https://www.coursera.org/learn/computational-thinking-problem-solving>
- 2) https://onlinecourses.nptel.ac.in/noc18_cs33/preview
- 3) <https://www.alison.com/courses/Introduction-to-Programming-in-c>
- 4) <http://www.ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-s096-effective-programming-in-c-and-c-january-iap-2014/index.html>

ENGLISH LANGUAGE AND COMMUNICATION SKILLS LABORATORY

I B.TECH : Common to All Branches								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6HS02	HSMC	L	T	P	C	CIE	SEE	Total
		0	0	3	1.5	40	60	100
Contact Classes: 00	Tutorial Classes: 0	Practical Classes: 36				Total Classes: 36		
OBJECTIVES: Student will be able to: 1. To facilitate computer-assisted multi-media instruction enabling individualized and independent language learning 2. To sensitize the students to the nuances of English speech sounds, word accent, intonation and rhythm 3. To bring about a consistent accent and intelligibility in students’ pronunciation of English by providing an opportunity for practice in speaking 4. To improve the fluency of students in spoken English and neutralize the impact of dialects. 5. To train students to use language appropriately for public speaking, group discussions and interviews								
EXERCISE-I						Classes: 06+06		
CALL Lab: <i>Understand:</i> Listening Skill- Its importance – Purpose- Process- Types- Barriers- Effective Listening. <i>Practice:</i> Introduction to Phonetics – Speech Sounds – Vowels and Consonants – Minimal Pairs- Consonant Clusters- Past Tense Marker and Plural Marker- <i>Testing Exercises</i> ICS Lab: <i>Understand:</i> Spoken vs. Written language- Formal and Informal English. <i>Practice:</i> Ice-Breaking Activity and JAM Session- Situational Dialogues – Greetings – Taking Leave – Introducing Oneself and Others								
EXERCISE-II		‘ApproJRD’bySudhaMurthy				Classes: 07+07		
Vocabulary		: WordsOftenMisspelt–Homophones, HomonymsandHomographs						
Grammar		: IdentifyingCommonErrorsinWritingwithReferencetoNoun-pronoun AgreementandSubject-verbAgreement						
Reading		: Sub-SkillsofReading – Skimming andScanning– ExercisesforPractice						
Writing		: NatureandStyleofWriting-Defining/DescribingPeople,Objects,Placesand Events– Classifying-Providing ExamplesorEvidence.						

EXERCISE-III	‘Lessons from Online Learning’ by F. Haider Alvi, Deborah Hurst et al	Classes: 07+07
Vocabulary	: Words Often Confused – Words from Foreign Languages and their Use in English.	
Grammar	: Identifying Common Errors in Writing with Reference to Misplaced Modifiers and Tenses	
Reading	: Sub-Skills of Reading – Intensive Reading and Extensive Reading – Exercises for Practice	
Writing	: Format of a Formal Letter – Writing Formal Letters E.g., Letter of Complaint, Letter of Requisition, Email Etiquette, Job Application with CV/Resume	
EXERCISE-IV	Art and Literature	Classes: 06+06
Vocabulary	: Standard Abbreviations in English	
Grammar	: Redundancies and Clichés in Oral and Written Communication	
Reading	: Survey, Question, Read, Recite and Review (SQ3R Method) – Exercises for Practice	
Writing	: Writing Practices- Essay Writing- Writing Introduction and Conclusion- Précis Writing	
EXERCISE-V	Go, Kiss the World’ by Subroto Bagchi	Classes: 06+06
Vocabulary	: Technical Vocabulary and their Usage	
Grammar	: Common Errors in English (<i>Covering all the other aspects of grammar which were not covered in the previous units</i>)	
Reading	: Reading Comprehension- Exercises for Practice	
Writing	: Technical Reports- Introduction – Characteristics of a Report – Categories of Reports Formats- Structure of Reports (Manuscript Format) -Types of Reports - Writing a Report	
Text Books:		
2. “English: Language, Context and Culture” by Orient Black Swan Pvt. Ltd, Hyderabad. 2022. Print.		
Reference Books:		

8. Effective Academic Writing by Lissand Davis (OUP)
9. Richards, Jack C. (2022) Interchange Series. Introduction, 1, 2, 3. Cambridge University Press
10. Wood, F. T. (2007). Remedial English Grammar. Macmillan.
11. Chaudhuri, Santanu Sinha. (2018). Learn English: A Fun Book of Functional Language, Grammar and Vocabulary. (2nd ed.,). Sage Publications India Pvt. Ltd.
12. (2019). Technical Communication. Wiley India Pvt. Ltd.
13. Vishwamohan, Aysha. (2013). English for Technical Communication for Engineering Students. McGraw-Hill Education India Pvt. Ltd.
14. Swan, Michael. (2016). Practical English Usage. Oxford University Press. Fourth Edition

Web References:

6. <http://www.bbc.co.uk/learningenglish>
7. <http://learnenglish.britishcouncil.org>
8. <https://www.cambridgeenglish.org/learning-english/>
9. <https://study.com/academy/subj/english.html>
- 10.

E-Text Books:

2. <https://www.pdfdrive.com/advanced-english-books.html>

MOOC Courses:

3. <http://nptel.ac.in/courses/109/106/109106067>
4. <https://www.britishcouncil.org/tr/en/english/mooc>

Course Outcomes:

By the end of this course, students will be able to:

7. Understand the importance of vocabulary and sentence structures.
8. Choose appropriate vocabulary and sentence structures for their oral and written communication.
9. Demonstrate their understanding of the rules of functional grammar.
10. Develop comprehension skills from the known and unknown passages.
11. Take an active part in drafting paragraphs, letters, essays, abstracts, précis and reports in various contexts.
12. Acquire basic proficiency in reading and writing modules of English.

PROGRAMMING FOR PROBLEM SOLVING LAB

I - II Semester: Common to all branches								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6CS03	ESC	L	T	P	C	CIA	SEE	Total
		-	-	3	1.5	40	60	100
Contact Classes: Nil	Tutorial Classes: Nil	Practical Classes: 36			Total Classes:36			

COURSE OBJECTIVES:

- 1) To be familiarize with flowgorithm to solve simple problems
- 2) To develop programs to solve basic problems by understanding basic concepts in C like operators, control statements etc.
- 3) To develop modular, reusable and readable C Programs using the concepts like functions, arrays, strings, pointers and structures.

COURSE OUTCOMES:

At the end of the course, student will be able to

- 1) Solve simple mathematical problems using Flowgorithm.
- 2) Correct syntax errors as reported by the compilers and logical errors encountered at run time Develop programs by using decision making and looping constructs.
- 3) Implement real time applications using the concept of array, pointers, functions and structures.

Solve real world problems using matrices, searching and sorting.

LIST OF EXPERIMENTS	
Week-1	INTRODUCTION TO FLOGORITHM
1. Installation and working of Flowgorithm Software. 2. Write and implement basic arithmetic operations using Flowgorithm – sum, average, product, difference, quotient and remainder of given numbers etc. 3. Draw a flowchart to calculate area of Shapes (Square, Rectangle, Circle and Triangle). 4. Draw a flowchart to find the sum of individual digits of a 3 digit number 5. Draw a flowchart to read input name, marks of 5 subjects of a student and display the name of the student, the total marks scored, percentage scored.	

6. Draw a flowchart to find roots of a quadratic equation. 7. Draw a flowchart to find the largest and smallest among three entered numbers and also display whether the identified largest/smallest number is even or odd 8. Draw a flowchart to check whether the triangle is equilateral, isosceles or scalene triangle 9. Draw a flowchart to check whether a given number is palindrome or not.	
Week-2	BASIC DATA TYPES
1) Write a C program to find division of 1 integer 1 float numbers. 2) Write a C program to find division of 2 integer numbers. 3) Write a C program to find average of (1int,1float) numbers. 4) Write a C Program to Swap Numbers without Using Temporary Variable 5) Write a C Program to Swap two Numbers Using Temporary Variable 6) Write a C Write a program to read the values of x, y and z and print the results of the following expressions in one line. a) $(x+y+z) / (x-y-z)$ b) $(x+y+z) / 3$ c) $(x+y) * (x-y) * (y-z)$	
Week-3	OPERATORS
1) Write a C program to convert temperature from Fahrenheit to Celsius and vice versa($c=(f-32)/1.8$) 2) Write a C program to find area and perimeter of a circle.($area=\pi r^2$ perimeter= $2\pi r$) 3) Write a C program to calculate area and perimeter of a right angled triangle. a. ($Area=1/2*b*h$ perimeter= $w+h+sqrt(w*w+h*h)$) 4) Find the sum of natural numbers 1 to n.(read n as input)(Use formula $sum=n(n+1)/2$) 5) Write a C program to calculate Simple interest ($SI=PTR/100$) 6) Write a C program to calculate area and perimeter of a rectangle. $Area=l*b$ Perimeter= $2* (l+b)$ 7) Write a C program to calculate the value of the third angle of a triangle if two angles are given as input.($a+b+c=180$) 8) Write a C program to read the consumer number and number of units consumed and the cost per unit and print the amount to be paid.($Amt=num\ of\ units*cost$) 9) Write a C Program to calculate area and perimeter of a triangle. a. Perimeter= $(a+b+c)$ b. $s=(a+b+c)/2$ c. $Area=sqrt(s*(s-a)(s-b)*(s-c))$	

10) Write a C program to read five Subject marks and find the average. 11) Write a C program to Calculate Compound interest ($CI=p(1+r/100)^n$)	
Week-4	CONDITIONAL STATEMENTS
1. Write a C program to find largest and smallest of given numbers. 2. Write a C program which takes two integer operands and one operator from the user(+, -, *, /, % use switch) 3. Write a program to compute grade of students using if else ladder. The grades are assigned as followed: marks<50 F $50 \leq \text{marks} < 60$ C $60 \leq \text{marks} < 70$ B $70 \leq \text{marks}$ B+ $80 \leq \text{marks} < 90$ A $90 \leq \text{marks} \leq 100$ A+ 4. Write a C program to whether given year leap year or not. 5. Write a C program to find whether given triangle is scalene or isosceles or equilateral.	
Week-5	LOOPING STATEMENTS
1) Write a C program to find Sum of individual digits of given integer 2) Write a C program to generate first n terms of Fibonacci series 3) Write a C Program to find the Sum of Series $SUM=1-x^2/2! + x^4/4! - x^6/6! + x^8/8! - x^{10}/10!$ 4) Write a C program to print the Fibonacci sequence up to given value of n. 5) Write a C program to print the multiplication table to the given value of n. 6) Write a C program to check whether given number is palindrome or not 7) Write a C program to check whether given number is perfect or not	
Week-6	NESTED LOOPING STATEMENTS
1) Write a C program to generate prime numbers between 1 and n 2) Write a C program to generate Pascal's triangle. 3) Write a C program to generate the following pyramid of numbers.	

1 1 3 1 1 3 5 3 1 4) Write a C program to generate the pattern <pre> ***** ***** *** ** *</pre>	
Week-7	ARRAYS
1) Write a C Program to implement following searching methods - Binary Search - Linear Search 2) Write a C program to find largest and smallest number in a list of integers 3) Write a C program - To add two matrices - To multiply two matrices 4) Write a C program to find Transpose of a given matrix	
Week-8	FUNCTIONS
1) Write a C program to find the factorial of a given integer using non recursive functions 2) Write a C program to find GCD of given integers using non recursive functions 3) Write a C Program to find the power of a given number using non recursive functions 4) Write a C program to find sum of natural numbers using non recursive function. 5) Write a C program to reverse a given integer number using non recursive functions 6) Write a C Program to find binary equivalent of a given decimal number using recursive functions. 7) Write a C Program to print Fibonacci sequence using recursive functions. 8) Write a C Program to find LCM of 3 given numbers using recursive functions 9) Write a C program to find the factorial of a given integer using recursive functions 10) Write a C program to print fibonacci series till n terms using recursion.	
Week-9	STRINGS

1) Write a C program using to Insert a sub string into a given main string from a given position 2) Write a C program using to Delete n characters from a given position in a string 3) Write a C program to determine if given string is palindrome or not 4) Write C Programs to demonstrate the following string handling functions. a. strcat() b. strcmp() c. strrev() d. strcpy() e. strlen() f. strstr() g. strncpy() h. strncat() i. strncmp()	
Week-10	POINTERS
1) Write a C program to read the elements of 1-d array using pointers and print them in reverse order using pointers. 2) Write a C Program to read two elements dynamically using malloc() function and interchange the two numbers using call by reference. 3) Write a C Program to read and print the elements of 1-D array using calloc() memory allocation function and reallocate memory for the array by increasing the size of the array, read and print the elements of reallocated array. 4) Write a C Program to print 2-D array using pointers.	
Week-11	STRUCTURES
1) Write a C Program using functions to a) Reading a complex number b) Writing a complex number c) Add two complex numbers d) Multiply two complex numbers Note: represent complex number using structure 2) Write a C program to read employee details employee number, employee name, basic salary, hra and da of n employees using structures and print employee number, employee name and gross salary of n employees.	
Week-12	FILES

1) Write a C program to read and print the content of a file. 2) Write a C program copy the content of one file to another file 3) Write a C program to merge two file into third file. 4) Write a C Program to find the number of lines in a text file
Text Books:
1) Riley DD, Hunt K.A. Computational Thinking for the Modern Problem Solver. CRC press, 2014 Mar 27. 2) B.A. Forouzan and R.F. Gilberg C Programming and Data Structures, Cengage Learning, (3rd Edition) Yashavant Kanetkar, “Let Us C”, BPB Publications, New Delhi, 13th Edition, 2012.
Reference Books:
1) Ferragina P, Luccio F. Computational Thinking: First Algorithms, Then Code. Springer; 2018 2) King KN, “C Programming: A Modern Approach”, Atlantic Publishers, 2nd Edition, 2015. 3) Kochan Stephen G, “Programming in C: A Complete Introduction to the C Programming Language”, Sam’s Publishers, 3rd Edition, 2004. 4) Linden Peter V, “Expert C Programming: Deep C Secrets”, Pearson India, 1st Edition, 1994.
Web References:
1. http://www.flowgorithm.org/documentation/ 2. http://www.sanfoundry.com/c-programming-examples 3. http://www.geeksforgeeks.org/c 4. http://www.cprogramming.com/tutorial/c

Electrical Circuits LAB

I Year II Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6EE02	ESC	L	T	P	C	CIA	SEE	Total
		0	0	2	1	40	60	100

Course Objectives:

The course should enable the students to:

1. Apply different techniques used in electric circuit analysis to calculate circuit parameters and two port network parameters.
2. Applications of Network theorems in different electric circuits.

Course Outcomes:

At the end of the lab the students will be able to :

1. Analyze the electrical circuits using Kirchhoffs law and ohm's law.
2. Analyze the electrical circuits using network simplification theorems
3. Justify resonant condition of series and parallel RLC circuits
4. Apply different techniques used in electric circuit analysis to calculate circuit parameters and two port network parameters.

LIST OF EXPERIMENTS

1	a. Identification, Specifications, Testing of R, L, C Components (Color Codes), Bread Boards.. b. Study and operation of Function Generator Regulated Power Supply (RPS) CRO. Verification of Ohms law.
2	Verification of KVL and KCL
3	Verification of Series and Parallel Resonance
4	Calculations and Verification of Impedance and Current of RL, RC and RLC series circuits
5	Verification of Superposition Theorem
6	Verification of Thevenin's and Norton's Theorem
7	Verification of Reciprocity theorem
8	Verification of Milliman's Theorem
9	Verification of Maximum Power Transfer Theorem
10	Verification of Tellegans theorem
11	Determination of Z and Y Parameters
12	Determination of Transmission and Hybrid parameters

Text Books:

1. M. E. Van Valkenburg, "Network Analysis", Prentice Hall, 2006.
2. D. Roy Choudhury, "Networks and Systems", New Age International Publications, 1998.

APPLIED PYTHON PROGRAMMING LABORATORY

I Year II Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6EE03	ESC	L	T	P	C	CIA	SEE	Total
		0	0	2	1	40	60	100

Course Outcomes:

Course Outcomes: Upon completing this course, the students will be able to

1. Build basic programs using fundamental programming constructs
2. Write and execute python codes for different applications
3. Capable to implement on hardware boards

LIST OF EXPERIMENTS

Cycle – 1

1	Downloading and Installing Python and Modules a) Python 3 on Linux Follow the instructions given in the URL https://docs.pythonguide.org/starting/install3/linux/ b) Python 3 on Windows Follow the instructions given in the URL https://docs.python.org/3/using/windows.html (Please remember that Windows installation of Python is harder!) c) pip3 on Windows and Linux Install the Python package installer by following the instructions given in the URL https://www.activestate.com/resources/quick-reads/how-to-install-and-use-pip3/ d) Installing numpy and scipy You can install any python3 package using the command <code>pip3 install <packagename></code> e) Installing jupyterlab Install from pip using the command <code>pip install jupyterlab</code>
2	Introduction to Python3 a) Printing your biodata on the screen b) Printing all the primes less than a given number c) Finding all the factors of a number and show whether it is a <i>perfect</i> number, i.e., the sum of all its factors (excluding the number itself) is equal to the number itself
3	Defining and Using Functions a) Write a function to read data from a file and display it on the screen b) Define a boolean function <i>is_palindrome</i>(<input>) c) Write a function <i>collatz</i>(x) which does the following: if x is odd, $x = 3x + 1$; if x is even, then $x = x/2$. Return the number of steps it takes for $x = 1$ d) Write a function $N(m, s) = \exp(-(x-m)^2/(2s^2))/\sqrt{2\pi}s$ that computes the Normal distribution

4	The package numpy a) Creating a matrix of given order $m \times n$ containing <i>random numbers</i> in the range 1 to 99999 b) Write a program that adds, subtracts and multiplies two matrices. Provide an interface such that, based on the prompt, the function (addition, subtraction, multiplication) should be performed c) Write a program to solve a system of n linear equations in n variables using matrix inverse
5	The package scipy and pyplot a) Finding if two sets of data have the same <i>mean</i> value b) Plotting data read from a file c) Fitting a function through a set a data points using <i>polyfit</i> function d) Plotting a histogram of a given data set
6	The strings package a) Read text from a file and print the number of lines, words and characters b) Read text from a file and return a list of all n letter words beginning with a vowel c) Finding a secret message hidden in a paragraph of text d) Plot a histogram of words according to their length from text read from a file
Cycle – 2	
7	. Installing OS on Raspberry Pi a) Installation using Pilmager b) Installation using image file · Downloading an Image · Writing the image to an SD card · using Linux · using Windows · Booting up Follow the instructions given in the URL https://www.raspberrypi.com/documentation/computers/getting-started.html
8	Accessing GPIO pins using Python a) Installing GPIO Zero library. First, update your repositories list: <code>sudo apt update</code> Then install the package for Python 3: <code>sudo apt install python3-gpiozero</code> b) Blinking an LED connected to one of the GPIO pin c) Adjusting the brightness of an LED Adjust the brightness of an LED (0 to 100, where 100 means maximum brightness) using the in-built PWM wavelength.
9	Collecting Sensor Data a) DHT Sensor interface ◦ Connect the terminals of DHT GPIO pins of Raspberry Pi. ◦ Import the DHT library using <code>import Adafruit_DHT</code> ◦ Read sensor data and display it on screen.

II B.TECH I SEMESTER SYLLABUS

ELECTRICAL CIRCUITS – II

II Year I Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6EE04	ESC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	40	60	100

Course Objectives:

The course should enable the students to

1. Analyze the self and mutual inductance concepts
2. Understand transient response of electrical circuits
3. Evaluate the line, phase values and power measurement in 3-phase circuits.

Course Outcomes:

At the end of the course the student should be able to

1. Calculate the self and mutual inductances
2. Obtain relation between voltages and currents in balanced and unbalanced three phase circuits.
3. Apply Laplace transforms to different electrical networks.
4. Determine the transient response of series and parallel RLC circuits for initial and final conditions.
5. Understand the operation of different filters.

SYLLABUS

UNIT-I	MAGNETIC CIRCUITS	Classes: 10
Faraday's laws of electromagnetic induction - concept of self and mutual inductance - dot convention in coupled circuits- coefficient of coupling - Mutually coupled circuits Series and parallel.		
UNIT-II	THREE PHASE CIRCUITS	Classes: 10
Star and delta connections, phase sequence, relation between line and phase voltages and currents in balanced star and delta circuits, three phase three wire and three phase four wire systems, Analysis of balanced and unbalanced three phase circuits, Measurement of active and reactive power using single wattmeter, two wattmeter.		
UNIT-III	ELECTRICAL CIRCUIT ANALYSIS USING LAPLACE TRANSFORMS	Classes: 10
Application of Laplace Transforms to RL, RC and RLC (series and parallel) Networks for impulse, step, and ramp, exponential and sinusoidal excitations		
UNIT-IV	DC AND AC TRANSIENT ANALYSIS	Classes: 10
Transient response: Initial conditions, transient response of RL, RC and RLC series and parallel circuits. Differential equations and Laplace transform approach with DC and AC excitations.		

UNIT-V	FILTERS	Classes: 10
Classification of filters – Low pass, High pass, Band pass and Band Elimination, Constant-k and M-derived filters-Low pass and High pass Filters and Band pass and Band elimination filters (Elementary treatment only)		
Text Books:		
1. M. E. Van Valkenburg, "Network Analysis", Prentice Hall, 2006. 2. Engineering Circuit Analysis - William Hayt ,Jack E. Kemmerly, S M Durbin, McGraw Hill Companies. 3. Network Analysis – A. Sudhakar, Shyam Mohan S Palli, 3rd Edition, Tata McGraw Hill Publications.		
Reference Books:		
4. D. Roy Choudhury, "Networks and Systems", New Age International Publications,1998. 5. Electric Circuits - A.Chakrabarhty, DhanipatRai& Sons. 6. Network Analysis - N.C Jagan and C. Lakhminarayana, BS publications.		
Web References:		
• www.wikipedia.org • www.pa.msu.edu • www.tutorvista.com • www.globalspec.com • www.ee.bilkent.edu.tr		
E-Text Books:		
• http://www.cl.cam.ac.uk/teaching/1011/SysOnChip/socdam-notes1011.pdf • https://www.doc.ic.ac.uk/~wl/teachlocal/cuscomp/notes/cc11.pdf • https://www.cs.ccu.edu.tw/~chen/arch/SOC-design.pdf		
MOOC Course		
• https://onlinecourses.nptel.ac.in/noc18_ee33/preview • https://onlinecourses.nptel.ac.in/noc18_ee34/preview		

ELECTRONIC DEVICES AND CIRCUITS

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6EC01	ESC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	40	60	100
COURSE OBJECTIVES: The course should enable the students to: 1. To introduce components such as diodes, BJTs and FETs. 2. To know the applications of devices. 3. To know the switching characteristics of devices. COURSE OUTCOMES: Upon successful completion of the course, the student is able to 1. Understand the operation and characteristics of various electronic devices. 2. Analyze few applications of electronic devices. 3. Understand the importance of biasing and stabilization of transistors. 4. Design and Analyze the small signal model for BJT and FET amplifiers. 5. Study operation and characteristics of Rectifiers with filters.								
UNIT-I	DIODES						Classes:09	
Diode-Static and Dynamic resistances, Equivalent circuit, Diffusion and Transition Capacitances, V-I Characteristics, Diode as a switch-switching times.								
UNIT-II	DIODE APPLICATIONS						Classes:09	
Rectifier - Half Wave Rectifier, Full Wave Rectifier, Bridge Rectifier, Rectifiers with Capacitive and Inductive Filters, Clippers-Clipping at two independent levels, Clamper-Clamping Circuit Theorem, Clamping Operation, Types of Clampers.								
UNIT-III	BIPOLAR JUNCTION TRANSISTOR (BJT)						Classes:09	
Principle of Operation, Common Emitter, Common Base and Common Collector Configurations, Transistor as a switch, switching times,								
UNIT-IV	JUNCTION FIELD EFFECT TRANSISTOR (FET)						Classes:09	
Construction, Principle of Operation, Pinch-Off Voltage, Volt-Ampere Characteristic, Comparison of BJT and FET, FET as Voltage Variable Resistor, MOSFET, MOSTET as a capacitor.								
UNIT-V	SPECIAL PURPOSE DEVICES						Classes:09	
Zener Diode-Characteristics, Zener diode as Voltage Regulator, Principle of Operation- SCR, Tunnel diode, UJT, Varactor Diode, Photodiode, Solar cell, LED, Schottky diode.								
Text Books:								

-
-
- | |
|---|
| <ol style="list-style-type: none">1. Jacob Millman-Electronic Devices and Circuits, McGraw Hill Education2. Robert L. Boylestad, Louis Nashelsky-Electronic Devices and Circuits theory, 11th Edition, 2009, Pearson |
| <p>Reference Books:</p> <ol style="list-style-type: none">1. Horowitz-Electronic Devices and Circuits, David A. Bell–5th Edition, Oxford2. Chinmoy Saha, Arindam Halder, Debaati Ganguly-Basic Electronics-Principles and Applications, Cambridge, 2018. |

ELECTRICAL MACHINES – I

II Year I Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6EE05	PCC	L	T	P	C	CIA	SEE	Total
		3	1	-	4	40	60	100

Course Objectives:

The course should enable the students to:

1. Illustrate the theory of electromechanical energy conversion.
2. Demonstrate the working principle of different types of dc machines and transformers.
3. Analyze the losses in dc machines to improve the efficiency by conducting various tests.
4. Outline the principle of operation, construction and testing of single phase transformers.
5. Compare the applications of DC shunt, series and compound generators as well as motors.
6. Judge the performance of various types of DC generators and DC motors.

Course Outcomes:

At the end of the course the student should be able to:

1. Estimate various types of losses, efficiency and performance of DC Machines.
2. Estimate the behavior of generator using parallel operation for power generation.
3. Find losses and efficiency by various testing technique of DC machines.
4. Estimate various types of losses, efficiency and performance of single phase transformers
5. Analyze the operation of various three phase transformers

SYLLABUS

UNIT-I	DC GENERATORS	Classes: 10
Electro mechanical energy conversion principles. Principle of operation, construction, armature windings, lap and wave windings, simplex windings, commutator, Emf equation, types of DC generators, Armature reaction, Load Characteristics and Applications: problems		
UNIT-II	DC MOTORS	Classes: 10
DC motors: Principle of operation, back EMF, torque equation, condition for maximum power developed, types of DC motors, characteristics, numerical problems; Losses and efficiency, condition for maximum efficiency.		
UNIT-III	TESTING AND SPEED CONTROL OF DC MACHINES	Classes: 10
Testing :Swinburne's test, brake test, regenerative testing, Hopkinson's test, fields test, retardation test, problems. Speed control of DC Motors: Armature voltage and field flux control methods. Types of starters.		

UNIT-IV	SINGLE PHASE TRANSFORMERS	Classes: 10
Single phase transformers: Principle of operation, construction, types of transformers, emf equation, operation of transformer under no load and on load, phasor diagram and equivalent circuit, efficiency, regulation. Testing of transformer: polarity test, OC and SC tests, heat run test and parallel operation, problems.		
UNIT-V	THREE PHASE TRANSFORMERS	Classes: 10
Three phase transformer: Principle of operation, star to star, delta to delta, star to delta, delta to star, Parallel operation. Auto transformers: Principles of operation, equivalent circuit, no load and on load tap changers, problems.		
Text Books:		
1. J Nagrath, D P Kothari, —Electrical MachinesII, Tata McGraw Hill publication, 3rd Edition, 2010. 2. P S Bimbra, —Electrical MachinesII, Khanna Publishers, 2nd Edition, 2008.		
Reference Books:		
1. M G Say, E O Taylor, —Direct Current MachinesII, Longman Higher Education, 1st Edition, 1985. 2. M V Deshpande, —Electrical MachinesII, PHI Learning Private Limited, 3rd Edition, 2011. 3. 3.Ian McKenzie Smith, Edward Hughes, —Electrical TechnologyII, Prentice Hall, 10thEdition, 2015 4. J B Gupta, —Theory and Performance of Electrical MachinesII, S K Kataria& Sons Publication, 14th Edition, 2010. 5. A E Fitzgerald, Charles Kingsley, JR., Stephen D Umans, —Electric MachineryII, McGraw Hill,6th Edition, 1985.		

ELECTROMAGNETIC FIELDS

II Year I Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6EE06	PCC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	40	60	100

Course Objectives:

The course should enable the students to:

1. Demonstrate the concept of electrostatic field intensity and electric potential.
2. Learn polarization of dielectrics and the behaviour of conductors and dielectrics in Electric field.
3. Study the expressions of magnetic field intensity and flux density.
4. Study forces in magnetic fields and law of electromagnetic induction.
5. Learn the modified Maxwell's equations in time varying fields.

Course Outcomes:

At the end of the course the student should be able to:

1. Analyze concept of electrostatic field intensity and electric potential.
2. Apply polarization of dielectrics and the behavior of conductors and dielectrics in electric field.
3. Analyze the concept of magnetic field intensity and flux density in various media
4. Evaluate forces in magnetic fields according to laws of electromagnetic induction.
5. Analyze concept of electrostatic field intensity and electric potential.

SYLLABUS

UNIT-I	STATIC ELECTRIC FIELD	Classes: 10
Review of conversion of a vector from one coordinate system to another coordinate system, Coulomb's law, Electric field intensity, Electrical field due to point charges. Line, Surface and Volume charged distributions. Gauss law and its applications. Absolute Electric potential, potential difference, Calculation of potential differences for different configurations. Electric dipole, Electrostatic Energy and Energy density.		
UNIT-II	CONDUCTORS, DIELECTRICS AND CAPACITANCE	Classes: 10
Current and current density, Ohms Law in Point form, Continuity equation, Boundary conditions of conductors and dielectric materials. Capacitance, Capacitance of a two wire line, Poisson's equation, Laplace's equation, Solution of Laplace and Poisson's equation.		
UNIT-III	STATIC MAGNETIC FIELDS AND MAGNETIC FORCES	Classes: 10
Biot-Savart Law, Ampere Law, Magnetic flux and magnetic flux density, Scalar and Vector Magnetic potentials. Steady magnetic fields produced by current carrying conductors. Force on a moving charge, Force on a differential current element, Force between differential current elements, Magnetic boundary conditions, Magnetic circuits, Self inductances and mutual inductances.		
UNIT-IV	TIME VARYING FIELDS AND MAXWELL'S EQUATIONS	Classes: 10
Faraday's law for Electromagnetic induction, Displacement current, Point form of Maxwell's equation, Integral form of Maxwell's equations, Motional Electromotive forces.		

UNIT-V	ELECTROMAGNETIC WAVES	Classes: 10
Derivation of Wave Equation, Uniform Plane Waves, Maxwell's equation in Phasor form, Wave equation in Phasor form, Plane wave in free space and in a homogenous material. Wave equation for a conducting medium, Plane waves in lossy dielectrics, Propagation in good conductors. Poynting theorem.		
Text Books:		
1. William H Hayt, John A Buck, "Engineering Electromagnetics", McGraw Hill Publications, 8 th Edition, 2012. 2. David J Griffiths, "Introduction to Electrodynamics" Pearson Education Ltd., 4 th Edition, 2014. 3. Sunil Bhooshan, "Fundamentals of Engineering Electromagnetics", Oxford University Press, 1 st Edition, 2012.		
Reference Books:		
1. Matthew N O Sadiku, S V Kulkarni, "Principles of Electromagnetics", Oxford University Press, 6 th Edition, 2015. 2. J D Krauss, Fleish, "Electromagnetics with Applications", McGraw Hill Publications, 5 th Edition, 1999 3. Matthew N O Sadiku, "Numerical Techniques in Electromagnetics", CRC Press, 2 nd Edition, 2001.		
Web References:		
1. www.wikipedia.org 2. www.pa.msu.edu 3. www.tutorvista.com 4. www.globalspec.com 5. www.ee.bilkent.edu.tr		
E-Text Books:		
1. http://www.cl.cam.ac.uk/teaching/1011/SysOnChip/socdam-notes1011.pdf 2. https://www.doc.ic.ac.uk/~wl/teachlocal/cuscomp/notes/cc11.pdf 3. https://www.cs.ccu.edu.tw/~chen/arch/SOC-design.pdf		
MOOC Course:		
1. https://onlinecourses.nptel.ac.in/noc18_ee33/preview 2. https://onlinecourses.nptel.ac.in/noc18_ee34/preview		

VECTOR CALCULUS AND COMPLEX ANALYSIS**R22**

II year - I Semester: ECE/EEE								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6BS06	BSC	L	T	P	C	CIA	SEE	Total
		3	1	--	4	40	60	100
Contact Classes: 42	Tutorial Classes: 08	Practical Classes: Nil			Total Classes: 50			
Course Objectives								
1. Verification of vector integral theorems.								
2. Evaluation of Limit, continuity, differentiability, analyticity and finding harmonic function								
3. Evaluation of the line integrals along piece wise smooth paths& Express the given complex function as a power series using Taylor’s series and Maclaurin’s series								
4. Evaluation of Residues and improper integrals.								
5. Concept of bilinear transformation and mapping it to the given points.								
Course Outcomes:								
At the end of the course the student should be able to:								
1. Find Interpolating polynomial for the given tabular data.								
2. Solve the first order ordinary differential equations using numerical techniques.								
3. Calculate line integrals along piece wise smooth paths, interpret such quantities as work done by a force.								
4. Express the given complex function as a power series using Taylor’s series and Maclaurin’s series.								
5. Evaluate Residues by Laurent series								
UNIT-I	VECTOR INTEGRATION AND VECTOR INTEGRAL THEOREMS						Classes: 10	
Line integral - work done - surface integrals - volume integral - Vector integral theorems - Green’s theorem in a plane - Stoke’s theorem - Gauss divergence theorem (all theorem statements and their verification).								
UNIT-II	FUNCTIONS OF COMPLEX VARIABLES						Classes: 08	
Complex functions and its representation on Argand plane, Concepts of limit&Continuity, Differentiability, Analyticity and Cauchy-Riemann conditions, Harmonic functions- Milne-Thompson method.								
UNIT-III	COMPLEX POWER SERIES AND INTEGRATION						Classes: 08	
Line integral-Evaluation along a path and by indefinite integration-Cauchy’s integral theorem- Cauchy’s integral formula -Generalized integral formula. Radius of convergence-expansion in Taylor’s series - Maclaurin’s series and Laurent series - Singular point - Isolated singular point-pole of order m - essential singularity.								
UNIT-IV	CALCULUS OF RESIDUES						Classes:08	

Residue- Evaluation of residue by formula and by Laurent's Series- Residue Theorem. Evaluation of integrals of the type (a) Improper real integral $\int_{-\infty}^{+\infty} f(x)dx$ (b) $\int_{-\infty}^{+\infty} e^{imx} f(x)dx$ (c) $\int_c^{c+2\pi} f(\cos \theta, \sin \theta)d\theta$		
UNIT-V	CONFORMAL MAPPING	Classes: 08
Transformation by e^z , $\log z$, z^2 , z^n (n is a positive integer), $\sin z$, $\cos z$, $\frac{z+a}{z}$, Translation, rotation, inversion and bilinear transformation – fixed point – cross ratio properties – invariance of circles and cross ratio – determination of bilinear transformation mapping three given points.		
Text Books:		
2. Ervin Kreyszig, Advanced Engineering Mathematics, 9th Edition, John Wiley & Sons, 2006. 2. B.S.Grewal, Higher Engineering Mathematics, Khanna publishers, 36th Edition, 2010.		
Reference Books:		
4. G.B.Thomas, calculus and analytical geometry, 9th Edition, Pearson Reprint 2006. 5. N.P Bali and Manish Goyal, A Text of Engineering Mathematics, Laxmi publications, 2008. 6. E.L.Ince, Ordinary differential Equations, Dover publications, 1958.		
Web references:		
1. https://www.efunda.com/math/math_home/math.cfm 2. https://www.ocw.mit.edu/resources/#Mathematics 3. https://www.sosmath.com/ 4. https://www.mathworld.wolfram.com/		
E -Text Books:		
1. https://www.e-booksdirectory.com/details.php?ebook=10166 2. https://www.e-booksdirectory.com/details.php?ebook=10166		
Course Outcomes At the end of the course, student will be able to:		
1. Verify vector integral theorems. 2. Evaluate Limit, continuity, differentiability, analyticity and harmonic function . 3. Evaluation of the line integrals along piece wise smooth paths. 4. Express the given complex function as a power series using Taylor's series and Maclaurin's series and Evaluation of Residues 5. Find the bilinear transformation mapping to three given points.		

ELECTRONIC DEVICES AND CIRCUITS LAB

II Year I Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6EC02	ESC	L	T	P	C	CIA	SEE	Total
		0	0	2	1	40	60	100

Course Objectives:

The course should enable the students to:

1. To identify various components and testing of active devices.
2. To operate various equipments like millimeter, function generators, regulated power supplies and CRO
3. To analyze the characteristics of various active devices.
4. To provide the concepts involved in design of electronic Circuits.

Course Outcomes:

The course should enable the students to:

1. Calculate various parameters of semi conductor devices from their characteristics.
2. Know the role of semi conductor devices in real time applications.
3. Calculate h-parameters of BJT under various configurations.
4. Compute frequency response of various amplifiers.

LIST OF EXPERIMENTS

1	c. Identification, Specifications, Testing of R, L, C Components (Color Codes), Potentiometers, Switches (SPDT, DPDT, and DIP), Coils, Relays, Bread Boards, PCB's. d. Identification, Specifications and Testing of Active Devices: Diodes, BJTs, JFETs, MOSFETs, Power Transistors, LED's, LCD's, SCR, UJT. e. Study and operation of Multimeter (Analog and Digital) Function Generator Regulated Power Supply (RPS) CRO f. Soldering Practice.
2	Forward and Reverse Bias Characteristics of PN junction diode.
3	Zener Diode Characteristics and Zener diode as voltage regulator
4	Half wave rectifier with and without filters
5	Full wave rectifier with and without filters.
6	Input & output characteristics of transistor in CB configuration
7	Input & output characteristics of transistor in CE configuration
8	Input & output characteristics of transistor in CC configuration
9	Drain and Transfer characteristics of JFET
10	Voltage divider bias using BJT

11	UJT characteristics
12	SCR characteristics
Reference Books:	
1. Jacob Milliman, Christos C .Halkias, SatyabrataJit (2015), <i>Electronic Devices and Circuits</i>, 3rd edition, Tata McGraw Hill, New Delhi. 2. G. K. Mittal (1999), <i>Electronic Devices and Circuits</i>, 22nd edition, Khanna Publications, New Delhi. 3. Robert Boylestad, LowisNashelsky (1993), <i>Electronic Devices and Circuit Theory</i>, 5th edition, Prentice Hall of India, New Delhi, India.	
Web References:	
1. https://unacademy.com/course/electronic-devices-gate-ece/JTIAAKX1 2. https://freevideolectures.com/course/2261/basic-electronics-and-lab/2 3. https://unacademy.com/lesson/build-in-potential-and-depletion-width/ALHF5QVM	

ELECTRICAL MACHINES - I LAB**II Year I Semester**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6EE07	PCC	L	T	P	C	CIA	SEE	Total
		0	0	2	1	40	60	100

Course Objectives:

The course should enable the students to:

1. Illustrate the theory of electromechanical energy conversion.
2. Demonstrate the working principle of different types of dc machines and transformers.
3. Analyze the losses in dc machines to improve the efficiency by conducting various tests.
4. Outline the principle of operation, construction and testing of single phase transformers.
5. Compare the applications of DC shunt, series and compound generators as well as motors.
6. Judge the performance of various types of DC generators and DC motors.

Course Outcomes:

At the end of the course the student should be able to:

1. Estimate various types of losses, efficiency and performance of DC Machines.
2. Estimate the behavior of generator using parallel operation for power generation.
3. Find losses and efficiency by various testing technique of DC machines.
4. Analyze the speed control of DC Machines.

LIST OF EXPERIMENTS

1	Magnetization characteristics of DC shunt generator (Determination of critical field resistance and critical speed)
2	Brake test on DC shunt motor (Determination of performance curves)
3	Load test on dc shunt generator (Determination of characteristics)
4	Load test on dc series generator (Determination of characteristics)
5	Load test on dc compound generator (Determination of characteristics)
6	Field test on dc series machines (Determination of efficiency)
7	Swinburne's test on dc shunt motor (Predetermination of efficiencies)
8	Brake test on dc compound motor (Determination of performance curves)
9	Retardation test on dc shunt motor (Determination of losses)
10	Separations of losses in dc shunt motor
11	Speed control of dc shunt motor (Armature and Field control methods)
12	Hopkinsons test (Predetermination of efficiencies)

Text Books:

-
- | |
|--|
| 1. I J Nagrath, D P Kothari, "Electrical Machines", Tata McGraw Hill publication, 3rd Edition, 2010.
2. P S Bimbra, "Electrical Machines", Khanna Publishers, 2nd Edition, 2008. |
| Reference Books: |
| 1. J B Gupta, "Theory and Performance of Electrical Machines", S K Kataria & Sons Publication, 14th Edition, 2010.
2. A E Fitzgerald, Charles Kingsley, JR., Stephen D Umans, "Electric Machinery", McGraw Hill, 6th Edition, 1985. |
| Web References: |
| 1. www.wikipedia.org
2. www.pa.msu.edu |
| E-Text Books: |
| 1. http://www.cl.cam.ac.uk/teaching/1011/SysOnChip/socdam-notes1011.pdf
2. https://www.doc.ic.ac.uk/~wl/teachlocal/cuscomp/notes/cc11.pdf |
| MOOC Course |
| 1. https://onlinecourses.nptel.ac.in/noc18_ee33/preview
2. https://onlinecourses.nptel.ac.in/noc18_ee34/preview |

ELECTRICAL SIMULATION TOOLS LABORATORY

II Year I Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6EE08	PCC	L	T	P	C	CIA	SEE	Total
		0	0	2	1	40	60	100

Course Objectives:

- To understand basic block sets of different simulation platform used in electrical/electronic circuit design.
- To understand use and coding in different software tools used in electrical/ electronic circuit design.
- To understand the simulation of electric machines/circuits for performance analysis.

Course Outcomes:

After learning the contents of this paper the student must be able to

1. Develop knowledge of software packages to model and program electrical and electronics systems.
2. Model different electrical and electronic systems and analyze the results.
3. Articulate importance of software packages used for simulation in laboratory experimentation by analyzing the simulation results.

Students should be encouraged to use open-source software's such as **SCILAB, ORCAD, LTSPICE, Ngspice, Octave, Solve Elec, Simulide, CircuitLab, QElectroTech, Circuit Sims, DcAcLab, Every Circuit, DoCircuitsetc.** for carrying out the lab simulation listed below.

Use of Professional Licensed versions of softwares like **MATLAB, LabVIEW, NI Multisim, PSpice, PowerSim, TINA** etc. is also allowed.

Use of '**Python**' platform for simulating components/ circuit behaviour.

Suggested List of Laboratory Experiments:

The following experiments need to be performed from various subject domains

1	Introduction to basic block sets of simulation platforms. Basic matrix operations, Generation of standard test signals.
2	Solving the linear and nonlinear differential equations
3	Measurement of Voltage, Current and Power in DC circuits.
4	Verification of different network theorems with dependent and independent sources using suitable simulation tools.
5	Verification of performance characteristics of basic Electronic Devices using suitable simulation tools.
6	Analysis of series and parallel resonance circuits using suitable simulation tools
7	Obtaining the response of electrical network for standard test signals using suitable simulation tools.
8	Modeling and Analysis of Low pass and High pass Filters using suitable simulation tools

9	Performance analysis of DC motor using suitable simulation tools
10	Modeling and analysis of Equivalent circuit of transformer using suitable simulation tools.
11	Analysis of single-phase bridge rectifier with and without filter using suitable Simulation tools.
12	Modeling and Verification of Voltage Regulator using suitable simulation tools.
13	Modeling of transmission line using simulation tools.
14	Performance analysis of Solar PV model using suitable simulation tools

CONSTITUTION OF INDIA

II Year II Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6HS06	HSMC	L	T	P	C	CIE	SEE	Total
		2	-	-	-	50	-	50

Course Objectives:

Students will be able to:

1. Understand the need for constitution
2. Appreciate the fundamental duties and rights of the citizens of India.
3. Explain the role and amendments of constitution in a democratic society.
4. Describe the directive principles of state policy and their significance.
5. List the key features of the constitution, union government and state government.

Course Outcomes:

Students will be able to:

1. Create awareness about the constitutional values and objectives written in the Indian constitution.
2. List fundamental rights and fundamental duties of Indian citizens.
3. Identify the division of legislative, executive and financial powers between the union and state governments.
4. Understand the working of Indian democracy, its institutions and processes at the local, state and union levels.
5. Explain the functions and responsibilities of election commission of India and union public service commission.

SYLLABUS

UNIT-I	HISTORY OF MAKING OF THE INDIAN CONSTITUTION	Classes: 10
. Introduction to the constitution of India, the making of the constitution and salient features of the constitution.		
UNIT-II	PHILOSOPHY OF THE INDIAN CONSTITUTION	Classes: 10
Preamble Salient Features, Contours of Constitutional Rights & Duties: Fundamental Rights, Right to Equality, Right to Freedom, Right against Exploitation, Right to Freedom of Religion, Cultural and Educational Rights, Right to Constitutional Remedies, Directive Principles of State Policy, Fundamental Duties, Amendment of the constitutional powers and procedures		
UNIT-III	UNION GOVERNMENT	Classes: 10
Union Government, Union Legislature (Parliament), Lok Sabha and Rajya Sabha (with powers and functions), president of India (with powers and functions), Prime minister of India (With powers and functions), Union judiciary (Supreme court), Jurisdiction of the supreme court.		
UNIT-IV	STATE GOVERNMENT	Classes: 10
State Government, State legislature (Legislative Assembly/ Vidhan Sabha, Legislative council/ Vidhan parishad), powers and functions of the state legislature, State executive, Governor of the state (with powers and functions), The chief Minister of the state (with powers and functions), State Judiciary (High courts)		

UNIT-V	ELECTION COMMISSION	Classes: 10
Election Commission: Role and Functioning, Chief Election Commissioner and Election Commissioners, State Election Commission: Role and Functioning, Institute and Bodies for the welfare of SC/ST/OBC and women.		
Text Books:		
1. M.V. Pylee, Indian Constitution Durga Das Basu, Human Rights in Constitutional Law, Prentice – Hall of India Pvt. Ltd. New Delhi 3. Noorani, A.G., (South Asia Human Rights Documentation Centre), Challenges to Civil Right), Callenges to Civil Rights Guarantees in India, Oxford University Press, 2012 4. P.M.Bakshi, The constitution of India, Universal Law Publishing Co., 5. The Constitution of India, 1950 (Bare Act), Government Publication. 6. Dr. S. N. Busi, Dr. B. R. Ambedkar framing of Indian Constitution, 1st Edition, 2015.		
Reference Books:		
1. M. P. Jain, Indian Constitution Law, 7th Edn., Lexis Nexis, 2014. 2. D.D. Basu, Introduction to the Constitution of India, Lexis Nexis, 2015. 3. Indian constitution at work, NCERT 4. SubashKashyap, Indian Constitution, National Book Trust 5. J.A. Siwach, Dynamics of Indian Government & Politics 6. D.C. Gupta, Indian Government and Politics 7. H.M.Sreevai, Constitutional Law of India, 4th edition in 3 volumes (Universal LawPublication) 8. J.C. Johari, Indian Government and Politics Hans 9. J. Raj Indian Government and Politics		
Web References:		
1. www.hss.iitb.ac.in/en/lecture-details 2. www.iitb.ac.in/en/event/2nd-lecture-institute-lecture-series-indian-constitution		
MOOC Course:		
1. nptel.ac.in/courses/109104074/8 2. nptel.ac.in/courses/109104045/ 3. nptel.ac.in/courses/101104065/		

II B.TECH II SEMESTER SYLLABUS

DIGITAL SYSTEM DESIGN								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6EC12	PCC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	40	60	100
COURSE OBJECTIVES: The course should enable the students to: 1. Learn basic techniques for the design of digital circuits. 2. Understand number representation in digital electronic circuits and to be able to convert between different representations. 3. Implement simple logical operations using logic gates, design of combinational and sequential logic circuits. 4. Analyze sequential systems in terms of state machines and implement synchronous state machines using flip-flops.								
COURSE OUTCOMES: Upon successful completion of the course, the student is able to 1. Manipulate numeric information in different forms 2. Minimize simple Boolean expressions using the theorems and postulates of Boolean algebra 3. Design and analyze combinational circuits and to use standard combinational functions to build more complex circuits. 4. Design and analyze sequential circuits and to use standard sequential functions to build more complex circuits								
UNIT-I	REVIEW OF NUMBER SYSTEMS						Classes:09	
Representation of numbers of different radix, conversion of numbers from one radix to another radix, Complements of Numbers, problem solving for addition and subtraction. 4-bit codes: BCD, EXCESS 3, etc, Error Detecting and Correcting Codes, Parity checking and Hamming Code.								
UNIT-II	BOOLEAN ALGEBRA AND COMBINATIONAL CIRCUITS						Classes:09	
Boolean algebra: Boolean theorems and properties, Minimization of logic functions using theorems, Complement and dual of logical expressions, Canonical and Standard Forms, Digital Logic Gates, Universal Logic Gates, Multilevel NAND/ NOR realizations, Minimization of logic functions using K-maps. Combinational Circuits: Combinational Design, Arithmetic Circuits, Comparator, Decoder, Encoder, Multiplexers, De-Multiplexers, Implementation of Higher Order Multiplexers/ Decoder Using Lower Order Multiplexers/ Decoder								
UNIT-III	SEQUENTIAL MACHINES FUNDAMENTALS						Classes:09	
Introduction, Basic Architectural Distinctions between Combinational and Sequential circuits, Classification of sequential circuits, The binary cell, The S-R-Latch and Flip-Flop The D-Latch and Flip-Flop, The “Clocked T” Flip-Flop, The “Clocked J-K” Flip-Flop, Conversion from one type of Flip-Flop to another, Timing and Triggering Consideration, Clock Skew								

UNIT-IV	SEQUENTIAL CIRCUIT DESIGN AND ANALYSIS	Classes:09
Sequential Circuit Design and Analysis: Introduction, State Diagram, Analysis of Synchronous Sequential Circuits, Approaches to the Design of Synchronous Sequential Finite State Machines, Design Aspects, State Reduction, Design Steps, Realization using Flip-Flops Counters - Design of Ripple Counter, Synchronous counter, Ring Counter, Registers, Shift Register.		
UNIT-V	FINITE STATE MACHINE AND ALGORITHMIC STATE MACHINES	Classes:09
Sequential Circuits: Finite state machine- capabilities and limitations, Mealy and Moore models, minimization of completely specified and incompletely specified sequential machines, Partition techniques and Merger chart methods-concept of minimal cover table. Algorithmic State Machines: Salient features of the ASM chart-Simple examples- Weighing machine and binary multiplier.		
Text Books:		
1. Digital Design- Morris Mano, PHI, 3rd Edition. 2. Switching Theory and Logic Design-A. Anand Kumar, PHI, 2nd Edition. 3. Switching and Finite Automata Theory- ZviKohavi&Niraj K. Jha, 3rd Edition, Cambridge.		
Reference Books:		
1. Introduction to Switching Theory and Logic Design – Fredriac J. Hill, Gerald R. Peterson, 3rd Ed, John Wiley & Sons Inc. 2. Digital Fundamentals – A Systems Approach – Thomas L. Floyd, Pearson, 2013. 3. Switching Theory and Logic Design – BhanuBhaskara –Tata McGraw Hill Publication, 2012 4. Fundamentals of Logic Design- Charles H. Roth, CengageLEarning, 5th, Edition, 2004. 5. Digital Logic Applications and Design- John M. Yarbrough, Thomson Publications, 2006. Digital Logic and State Machine Design – Comer, 3rd, Oxford, 2013		
Web References:		
1. www.wikipedia.org 2. www.pa.msu.edu 3. www.tutorvista.com 4. www.globalspec.com 5. www.ee.bilkent.edu.tr		
E-Text Books:		
1. http://www.site.uottawa.ca/~petriu/Digital-Logic.pdf 2. http://uav.ece.nus.edu.sg/~bmchen/courses/EG1108_Digital.pdf 3. http://info.iet.unipi.it/~luigi/biomedica/sito/cosc205.pdf		
MOOCs Course		
1. https://onlinecourses.nptel.ac.in/noc18_ee33/preview 2. https://onlinecourses.nptel.ac.in/noc18_ee34/preview		

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ELECTRICAL MACHINES –II								
IIYear II Semester								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6EE09	PCC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	40	60	100
Course Objectives: 1. The course should enable the students to: 2. Impart knowledge on Construction, principle of operation and performance of induction machines 3. Understand on Starting and speed control of three phase induction motors 4. Understand Construction and performance of salient and non – salient type synchronous generators 5. Impart knowledge on Principle of operation and performance of synchronous motor 6. Impart knowledge on Construction, principle of operation of single phase induction motors								
Course Outcomes: At the end of the course the student should be able to: 1. Estimate various types of losses, efficiency and performance of AC Machines. 2. Estimate the behavior of AC machines using no load and blocked rotor tests 3. Find losses and efficiency by various testing technique of AC machines. 4. Estimate various types of losses, efficiency and performance of single phase AC Machines. 5. Analyze the operation and characteristics of various synchronous motors								
SYLLABUS								
UNIT-I	THREE PHASE INDUCTION MOTORS						Classes: 10	
Three phase induction motors: Introduction, construction, types of induction motors, slip and frequency of rotor currents, rotor MMF and production of torque, equivalent circuit, power across air gap, torque and power output, torque slip characteristics, generating and braking modes, maximum (breakdown) torque, starting torque, maximum power output, problems								
UNIT-II	TESTING AND SPEED CONTROL OF INDUCTION MOTORS						Classes: 10	
Equivalent circuit model: No load test and blocked rotor test, circuit model, starting methods, speed control of induction motors, induction generator, principle of operation, circle diagram, determination of induction motor parameters from circle diagram, problems								
UNIT-III	ALTERNATORS						Classes: 10	
Synchronous generators: Introduction, principle of operation, constructional features, armature windings, distributed and concentrated windings, winding factors, basic synchronous machine model, circuit model of a synchronous machine, phasor diagrams, determination of synchronous impedance, short circuit ratio, armature reaction ampere turns and leakage reactance								
UNIT-IV	SYNCHRONOUS MOTORS						Classes: 10	

Synchronous motors: Principle of operation, power developed, synchronous motor with different excitations, effect of increased load with constant excitation, effect of change in excitation with constant load, effect of excitation on armature current and power factor, construction of VI and inverted VI curves, starting methods, salient pole synchronous motor, phasor diagrams and analysis, synchronous condenser		
UNIT-V	SINGLE PHASE INDUCTION MOTOR	Classes: 10
Single phase induction motor: Principle of operation, two reaction theory, equivalent circuit analysis, split phase motor, construction, principle of operation, capacitor start motor, shaded pole motor, torque speed characteristics		
Text Books:		
1. P S Bimbra, —Electrical MachinesII, Khanna Publishers, 2nd Edition, 2008 2. I J Nagrath, D P Kothari, —Electrical MachinesII, TMH publication, 3rd Edition, 2010		
Reference Books:		
1. A. E Fitzgerald, Charles Kingsley JR., Stephen D Umans, —Electric MachineryII, McGraw Hill, 6th Edition, 1985 2. M G Say, —Alternating Current MachinesII, Pitman Publishing Ltd, 4th Edition, 1976 3. Bhattacharya, —Electrical MachinesII, TMH publication, 2nd Edition 4. J B Gupta, —Theory and Performance of Electrical MachinesII, S K Kataria & Sons Publication, 14th Edition, 2010		

POWER SYSTEMS –I

II Year II Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6EE10	PCC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	40	60	100

Course Objectives:

The course should enable the students to:

1. To describe and use from technical point of view the various methods of electrical energy production
2. To understand the different curves based on load data and different tariff methods.
3. To understand the transmission line R,L,C parameters
4. To understand the performance of Short, medium and long transmission lines
5. To understand criteria of corona and Sag,Tension calculations of transmission lines

Course Outcomes:

At the end of the course the student should be able to:

1. Analyze from technical point of view the various methods of electrical energy production
2. Analyze the load curves from given load data and different tariff methods.
3. Evaluate the expressions for calculation of transmission line inductance and capacitances.
4. Justify between nominal-T and nominal- π method of representation of transmission lines.
5. Analyze the causes of corona, sag and tension effects.

SYLLABUS

UNIT-I	HYDRO, THERMAL AND NUCLEAR POWER STATIONS	Classes: 10
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Hydro electric power station : Elements, Types and operation, Concept of pumped storage plants, Estimation of power developed from a given catchment area

Thermal power stations: Line diagram of Thermal Power Station (TPS), Brief description of TPS components **Nuclear Power Stations**: Nuclear Fission reaction, Nuclear fuels, Nuclear reactor Operation and components, Types of Nuclear reactors BWR,PWR,FBR

UNIT-II	ECONOMICS OF GENERATION	Classes: 10
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Economics of Generation:Introduction, definitions of connected load, maximum demand, demand factor, load factor, diversity factor, Load duration curve, Base load and peak load plants.

Costs of Generation Fixed, Running Cost. Tariff Methods, Flat Rate, Block-Rate, Two-part, Three –part and power factor tariff methods

UNIT-III	TRANSMISSION LINE PARAMETERS	Classes: 10
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Types of conductors - calculation of resistance for solid conductors - Calculation of inductance for single phase and three phase, single and double circuit lines, concept of GMR & GMD, symmetrical and asymmetrical conductor configuration with and without transposition, Numerical Problems. Bundled conductors. Skin and Proximity effect.

Calculation of capacitance for 2 wire and 3 wire systems, capacitance calculations for symmetrical and asymmetrical single and three phase, single and double circuit lines, effect of ground on capacitance Numerical Problems.

UNIT-IV	PERFORMANCE OF TRANSMISSION LINES	Classes: 10
Classification of Transmission Lines - Short, medium and long lines- their model representations -Nominal-T and Nominal-Pie methods-Long Transmission Lines-Rigorous Solution regulation and efficiency of all types of lines. A, B, C, D Constants of Transmission Lines.Surge Impedance and (SIL) -Ferranti effect.		
UNIT-V	INSULATORS- SAG AND TENSION	Classes: 10
Insulators: Types of Insulators, calculation of String efficiency and Methods for improvement. Sag And Tension : Sag and Tension Calculations with equal and unequal heights of towers, Effect of Wind and Ice , Numerical Problems - Stringing chart and sag template. Corona - Description of the phenomenon, factors affecting corona, critical voltages and power loss, Methods reducing Corona loss. Disadvantages of corona.		
Text Books:		
1. C L Wadhawa, —Generation, Distribution and Utilization of Electrical Energyll, NewAgeInternational Limited, New Delhi, 3rd Edition, 2010. 2. J B Gupta, —A Course in Electrical Powerll, S K Kataria and Sons, New Delhi, 15th Edition, 2013		
Reference Books:		
1. M V Deshpande, —Elements of Power Station designll, Prentice Hall India Learning Private Limited,NewDelhi, 1st Edition, 1992. 2. G N Tiwari, M K Ghosal, —Fundamentals of Renewable Energy Sourcesll, NarosaPublications,NewDelhi, 1st Edition, 2007.		
Web References:		
1. www.pa.msu.edu 2. www.tutorvista.com 3. www.globalspec.com 4. www.tutorvista.com 5. www.globalspec.com		
E Text Book-		
1. http://www.cl.cam.ac.uk/teaching/1011/SysOnChip/socdam-notes1011.pdf 2. https://www.doc.ic.ac.uk/~wl/teachlocal/cuscomp/notes/cc11.pdf 3. https://www.cs.ccu.edu.tw/~chen/arch/SOC-design.pdf		

SIGNALS AND SYSTEMS

II Year II Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6EC06	PCC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	40	60	100

COURSE OBJECTIVES:

1. Learn the different types of signals and systems
2. Know the continuous and discrete systems in time and frequency domain using different transforms
3. Check the properties of continuous and discrete systems
4. Understand the properties to analyze the CT and DT signals and systems
5. Represent the LTI systems in the Time domain and various Transform domains

COURSE OUTCOMES:

After going through this course the student will be able to

1. Analyze the different types of signals and systems
2. Represent continuous and discrete systems in time and frequency domain using different transforms
3. Investigate the stability and causality of systems
4. Apply various transforms and its properties to analyze the CT and DT signals and systems
5. Characterize LTI systems in the Time domain and various Transform domains

SYLLABUS

UNIT-I	SIGNAL ANALYSIS AND FOURIER SERIES	Classes:09
Signals: Continuous and discrete time signals- representations of continuous, discrete and digital signals- Classifications of Signals based on properties - Energy and power signals, Even and Odd-Periodic and non-periodic-Causal Non causal Deterministic and non deterministic-Elementary signals-unit step- unit ramp-unit impulse-sinusoidal-signum and sinc signals - Basic operations on signals with examples. Fourier Series: Representation of Fourier series, Dirichlet's conditions, Trigonometric Fourier Series and Exponential Fourier Series, Properties of Fourier Series, Complex Fourier spectrum.		
UNIT-II	FOURIER TRANSFORMS AND SAMPLING	Classes:09
Fourier Transforms: Deriving Fourier Transform from Fourier Series, Fourier Transform of arbitrary signal, Fourier Transform of standard signals, Fourier Transform of Periodic Signals, Properties of Fourier Transform, and Introduction to Hilbert Transform. Sampling: Sampling theorem – Graphical and analytical proof for Band Limited Signals, Types of Sampling - Impulse Sampling, Natural and Flat top Sampling, Reconstruction of signal, Effect of under sampling		
UNIT-III	SIGNAL TRANSMISSION THROUGH LINEAR SYSTEMS	Classes:09
Systems: Continuous and discrete time Systems - representations of continuous and discrete time systems- Classifications of Systems based on properties – linearity - shift-invariance- causality- stability-static and dynamic Transfer function of a LTI system, Filter characteristics of Linear Systems, Distortion less transmission through a system, Signal bandwidth, System bandwidth, Ideal LPF, HPF, BPF&BSF characteristics, Relationship between Bandwidth and Rise time.		

UNIT-IV	CONVOLUTION AND CORRELATION OF SIGNALS	Classes:09
Concept of convolution in Time domain and Frequency domain, Graphical representation of Convolution, Cross Correlation and Auto Correlation of functions, Properties of Correlation function, Energy density spectrum, Parseval's Theorem, Power density spectrum, Relation between Auto Correlation function and Energy/Power spectral density function, Relation between Convolution and Correlation, Detection of periodic signals in the presence of Noise by Correlation, Extraction of signal from noise by filtering.		
UNIT-V	LAPLACE TRANSFORMS AND Z-TRANSFORMS	Classes:09
Laplace Transforms: Laplace transform, Concept of Region of Convergence (ROC) and its properties, Properties of L.T. Inverse Laplace Transform- problem solving - Relation between L.T and F.T of a signal Z-Transforms: Concept of Z- Transform of a Discrete Sequence, Distinction between Laplace, Fourier and Z Transforms, Region of Convergence in Z-Transform and its properties, properties of Z transform, Inverse Z-transform and problem solving		
Text Books: 1. Oppenheim A. V, Willisky (2009), <i>Signals and Systems</i>, 2nd edition, Prentice Hall of India, India. 2. P. Lathi (2001), <i>Signals, Systems & Communications</i>, BS Publications, New Delhi. Reference Books: 1. Simon Haykin, Van Veen (2007), <i>Signals & Systems</i>, 2nd edition, Wiley publications, India. 2. Hwei Piao Hsu, Schaums (2003), <i>Outline of Theory Problems of Signals and Systems</i>, McGraw Hill, India. 3. Charles L. Phillips, John M. Parr, Eve A. Riskin (2007), <i>Signals, Systems and Transforms</i>, Prentice Hall of India, New Delhi		
Web References: 1. https://www.khanacademy.org/science/electrical-engineering/ee-signals/ee-fourier-series/v/ee-fourier-series-intro 2. https://www.khanacademy.org/science/electrical-engineering/ee-signals 3. https://www.edx.org/course/signals-systems-part-2-iitbombayx-ee210-2x-3 4. https://www.edx.org/course/signals-and-systems-part-1-1		
E-Text Books: 1. http://www.freebookcentre.net/Mathematics/Fourier-Analysis-Books.html 2. http://www.freebookcentre.net/Mathematics/Differential-Equations-Books_1.html		
MOOCS Course 1. http://nptel.ac.in/courses/117101055/ 2. https://ocw.mit.edu/resources/res-6-007-signals-and-systems-spring-2011/lecture-notes/		

ELECTRICAL MEASUREMENTS & INSTRUMENTATION

II Year II Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6EE11	PCC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	40	60	100

Course Objectives:

The course should enable the students to:

1. Demonstrate the construction, working and characteristics of electrical measurement instruments.
2. Illustrate the principles of energy measurement in electrical loads.
3. Outline the use of cathode ray oscilloscope.
4. Evaluate various transducers for electrical measurements.

Course Outcomes:

At the end of the course the student should be able to:

1. Understand the basic definition of electrical instruments
2. Calculate the value of error in an instrument and measure the accuracy.
3. Find the value of unknown resistance by using bridge circuits.
4. Measure the power in a three-phase circuit.
5. Understand the working of energy meter and calculate the energy consumed

SYLLABUS

UNIT-I	INTRODUCTION TO MEASURING INSTRUMENTS	Classes: 10
Introduction: Classification of measuring instruments, deflecting, damping and control torques, types of errors, ammeter and voltmeter: PMMC, MI instruments, expression for deflection and control torque, errors and compensation extension of range using shunts and series resistance; Electro static voltmeter, electro type and attracted type, disc type, extension of range of ES voltmeters.		
UNIT-II	POTENTIOMETERS AND INSTRUMENT TRANSFORMERS	Classes: 10
DC Potentiometers: Principle and operation of Crompton potentiometer, standardization, measurement of unknown resistance, current, voltage; AC potentiometers: polar and coordinate type, standardization, applications; Instrument transformer: CT and PT, ratio and phase angle error.		
UNIT-III	MEASUREMENT OF POWER AND ENERGY	Classes: 10
Measurement of Power: single phase dynamometer type wattmeter, LPF and UPF, double elements and three elements dynamometer wattmeter; Expression for deflection and control torque, extension of range of wattmeter by using instrument transformers, Measurement of active and reactive powers in balanced and unbalanced systems. Measurement of Energy: Single phase induction type energy meter, driving and braking torques, errors and		

compensations, three phase energy meter.		
UNIT-IV	DC AND AC BRIDGES	Classes: 10
Measurement of Resistance: Methods of measuring low, medium, high resistance, Wheatstone bridge, carry foster, Kelvin"s double bridge, loss of charge method; Measurement of Inductance: Maxwell"s bridge, hay"sbridge ,Anderson"s bridge, Owen"s bridge; Measurement of Capacitance: Desauty"s bridge, Wein"s bridge, Schering bridge.		
UNIT-V	TRANSDUCERS AND OSCILLOSCOPES	Classes: 10
Transducers: Definition of transducers, Classification of transducers, Advantages of Electrical transducers, Characteristics and choice of transducers. Principle of operation of LVDT and capacitor transducers, LVDT Applications. Strain gauge and its principle of operation, gauge factor, Cathode ray oscilloscope: Cathode ray tube, time base generator, horizontal and vertical amplifiers, Measurement of phase and frequency, Lissajous patterns, sampling oscilloscope, analog oscilloscope, tubeless oscilloscopes, digital storage oscilloscope.		
Text Books:		
1. A K Sawhney, "Electrical and Electronic measurement and instruments", DhanpatRai and Sons Publications. 2. E W Golding and F C Widdis, "Electrical measurements and measuring instruments" wheeler publishing 5th Edition		
Reference Books:		
1. Buckingham and Price, "Electrical measurements", Prentice Hall. 2. D V S Murthy, "Transducers and Instrumentation", Prentice Hall of India, 2nd Edition, 2009. 3. A S Morris, "Principles of measurement of instrumentation", Pearson/Prentice Hall of India, 2nd Edition, 1994. 4. H S Kalsi, "Electronic Instrumentation", Tata McGraw Hill Edition, 1st Edition 1995		
Web References:		
1. www.wikipediorg 2. www.pa.msu.edu 3. www.tutorvista.com 4. www.globalspec.com 5. www.ee.bilkent.edu.tr		
E-Text Books:		
1. http://www.cl.cam.ac.uk/teaching/1011/SysOnChip/socdam-notes1011.pdf 2. https://www.doc.ic.ac.uk/~wl/teachlocal/cuscomp/notes/cc11.pdf 3. https://www.cs.ccu.edu.tw/~chen/arch/SOC-design.pdf		
MOOC Course:		
1. https://onlinecourses.nptel.ac.in/noc18_ee33/preview 2. https://onlinecourses.nptel.ac.in/noc18_ee34/preview		

DIGITAL SYSTEM DESIGN LAB								
IIYear II Semester								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6EC16	PCC	L	T	P	C	CIA	SEE	Total
		0	0	2	1	40	60	100
Course Objectives: The course should enable the students to 1. Study of logic gates and verify their truth tables. 2. Implementation of the given Boolean function using logic gates. 3. Use K-Map for simplification. 4. Design the combinational and sequential circuits								
Course Outcomes: The course should enable the students to 1.Study of logic gates and verify their truth tables. 2.Implementation of the given Boolean function using logic gates. 3.Use K-Map for simplification. 4.Design the combinational and sequential circuits								
LIST OF EXPERIMENTS								
Week-1	TO STUDY OF LOGIC GATES AND VERIFY THEIR TRUTH TABLES							
Verification and interpretation of truth tables for AND, OR, NOT, NAND, NOR Exclusive OR (EX-OR), Exclusive NOR (EX-NOR) Gates.								
Week-2	TO IMPLEMENT BOOLEAN FUNCTION USING AOI LOGIC							
Implement Exclusive-OR gate using AOI logic and to implement Boolean function $F=xy+x'y'+y'z$ using AOI logic								
Week-3	VERIFICATION OF K-MAP							
To Simplify the Boolean function $F(w, x, y, z) = \Sigma (0, 1, 2, 4, 5, 6, 8, 9, 12, 13, 14)$ using 4-variable K-Map, implement it and verify it using truth-table.								
Week-4	ADDER AND SUBTRACTORS							
To design and construct half adder, full adder half subtractor and full subtractor circuits using logic gates and verify the truth table.								
Week-5	BCD TO EXCESS-3 CONVERTER							

To Design and Implement BCD TO EXCESS-3, 4-bit Binary to gray code converter/ 4-bit Gray to Binary code converter and verify the truth table.	
Week-6	BINARY TO GRAY /GRAY TO BINARY CODE CONVERTER
To Design and Implement 4-bit Binary to gray code converter/ 4-bit Gray to Binary code converter and verify the truth table.	
Week-7	VERIFICATION OF TRUTH TABLES OF R-S FLIP-FLOP , J - K FLIP-FLOP , T FLIP-FLOP AND D FLIP-FLOP USING NAND AND NOR GATES
To Verify the truth tables of RS Flip Flop, JK Flip flop, T Flip flop and D Flip Flop using NAND and NOR gates	
Week-8	IMPLEMENTATION AND VERIFICATION OF DECODER, DE-MULTIPLEXER AND ENCODER USING LOGIC GATES
To Implement and verify truth tables of decoder, de-multiplexer and encoder, using logic gates.	
Week-9	IMPLEMENTATION OF 4X1 MULTIPLEXER, USING LOGIC GATES
To Implement and verify truth tables of 4x1 multiplexer, using logic gates	
Week-10	IMPLEMENTATION OF 4-BIT PARALLEL ADDER, USING 7483 IC
To Implement and verify the truth table of 4-bit parallel adder, using 7483 IC.	
Week-11	DESIGN AND VERIFY THE 4-BIT SYNCHRONOUS COUNTER.
To Design, and verify the 4-bit synchronous counter.	
Week-12	TO DESIGN AND VERIFY 4 BIT RIPPLE (ASYNCHRONOUS COUNTER)
To design and verify a 4 bit ripple (asynchronous counter).	
Reference Books:	
1. R.P. Jain, "Modern digital Electronics", Tata McGraw Hill, 4th edition, 2009. 2. W.H. Gothmann, "Digital Electronics- An introduction to theory and practice", PHI, 2nd edition, 2006. 3. D.V. Hall, "Digital Circuits and Systems", Tata McGraw Hill, 1989	
Web References:	
1. https://www.youtube.com/watch?v=M0mx8S05v60&list=PLBlnK6fEyqRjMH3mWf6kwqiTbT798eAOm 2. https://www.youtube.com/watch?v=CeD2L6KbtVM&list=PL018B3BB2E6FE781D 3. https://www.youtube.com/watch?v=CeD2L6KbtVM&list=PL803563859BF7ED8C	

ELECTRICAL MACHINES - II LAB

II Year II Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6EE12	PCC	L	T	P	C	CIA	SEE	Total
		0	0	2	1	40	60	100

Course Objectives:

1. To prepare the students to have a basic knowledge of induction motors.
2. To prepare the students to have a basic knowledge of alternators.
3. The ability to conduct testing and experimental procedures on different types of electrical machines.
4. To expose the students to the operation synchronous and induction machines and give them experimental skills.

Course Outcomes:

Upon completion of the course, students will be able to:

1. Estimate various types of losses, efficiency and performance of AC Machines.
2. Estimate the behavior of AC machines using no load and blocked rotor tests
3. Find losses and efficiency by various testing technique of AC machines.
4. Estimate various types of losses, efficiency and performance of single phase AC Machines.

LIST OF EXPERIMENTS

1	Load Test on Single phase Transformer
2	Sumpner's test on a pair of single phase transformers
3	Scott connection of transformers
4	No-load & Blocked rotor tests on three phase Induction motor
5	Regulation of a three –phase alternator by synchronous impedance & m.m.f. methods
6	Heat run test on single phase transformers
7	Equivalent Circuit of a single phase induction motor
8	Determination of X_d and X_q of a salient pole synchronous machine
9	Parallel operation of Single phase Transformers
10	Separation of core losses of a single phase transformer
11	Polarity test on Transformer.
12	Brake test on three phase Induction motor

Reference Books:

1. P S Bimbra, —Electrical MachinesII, Khanna Publishers, 2nd Edition, 2008
2. I J Nagrath, D P Kothari, —Electrical MachinesII, TMH publication, 3rd Edition, 2010

ELECTRICAL MEASUREMENTS AND INSTRUMENTATION LAB

II Year II Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6EE13	PCC	L	T	P	C	CIA	SEE	Total
		0	0	2	1	40	60	100

Course Objectives:

The course should enable the students to:

1. Understand various measurement techniques used in electrical engineering.
2. Demonstrate the use of sensors and transducers in electrical and nonelectrical measurements.

Course Outcomes:

Upon completion of the course, students will be able to:

1. Able to know the basic electrical instruments.
2. Calculate the value of error in an instrument and measure the accuracy.
3. Find the value of unknown resistance by using bridge circuits.
4. Measure the power in a three phase circuit.

LIST OF EXPERIMENTS

1	Calibration and Testing of single phase energy Meter.
2	Calibration of dynamometer power factor meter.
3	Crompton D.C. Potentiometer – Calibration of PMMC ammeter and PMMC voltmeter.
4	Kelvin's double Bridge – Measurement of resistance – Determination of Tolerance.
5	Dielectric oil testing using H.T. testing Kit.
6	Schering bridge & Anderson bridge.
7	Measurement of 3 – Phase reactive power with single-phase wattmeter.
8	Measurement of parameters of a choke coil using 3 – voltmeter and 3 – ammeter methods
9	Calibration LPF wattmeter – by Phantom loading.
10	Measurement of 3-phase power with single watt meter
11	Resistance strain gauge – strain measurements and Calibration.
12	Transformer turns ratio measurement using AC bridges. & LVDT Trainer kit

Text Books:
1. ELECTRICAL & ELECTRONICS INSTRUMENTATION – A.K .SAWHANI
Reference Books:
1. ELECTRICAL & ELECTRONICS INSTRUMENTATION – KALSI
Web References:
1. www.pa.msu.edu 2. www.tutorvista.com 3. www.globalspec.com 4. www.tutorvista.com 5. www.globalspec.com
E Text Books:
1. http://www.cl.cam.ac.uk/teaching/1011/SysOnChip/socdam-notes1011.pdf 2. https://www.doc.ic.ac.uk/~wl/teachlocal/cuscomp/notes/cc11.pdf 3. https://www.cs.ccu.edu.tw/~chen/arch/SOC-design.pdf
MOOC Course:
1. https://onlinecourses.nptel.ac.in/noc18_ee33/preview 2. https://onlinecourses.nptel.ac.in/noc18_ee34/preview

GENDER SENSITIZATION

GENDER SENSITIZATION								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6HS05	HSMC	L	T	P	C	CIE	SEE	Total
		-	--	2	Nil	50		50
COURSE OUTCOMES: 1. Develop a better understanding of important issues related to gender in contemporary India. 2. Sensitized to basic dimensions of the biological, sociological, psychological and legal aspects of gender 3. Attain a finer grasp of how gender discrimination works in our society and how to counter it 4. Men and Women students and professionals will be better equipped to work and live together as equals 5. Create a sense of appreciation of women in all walks of life								
UNIT-I								
UNDERSTANDING GENDER: Gender: Why Should We Study It? (<i>Towards a World of Equals</i> : Unit -1) Socialization: Making Women, Making Men (<i>Towards a World of Equals</i> : Unit -2) Introduction. Preparing for Womanhood. Growing up Male. First lessons in Caste. Different Masculinities.								
UNIT-II								
GENDER AND BIOLOGY: Missing Women: Sex Selection and Its Consequences (<i>Towards a World of Equals</i> : Unit -4)Declining Sex Ratio. Demographic Consequences. Gender Spectrum: Beyond the Binary (<i>Towards a World of Equals</i> : Unit -10)Two or Many? Struggles with Discrimination								
UNIT-III								
GENDER AND LABOUR: Housework: the Invisible Labour (<i>Towards a World of Equals</i> : Unit -3) “My Mother doesn’t Work.” “Share the Load.” Women’s Work: Its Politics and Economics (<i>Towards a World of Equals</i> : Unit -7) Fact and Fiction. Unrecognized and Unaccounted work. Additional Reading: Wages and Conditions of Work.								
UNIT-IV								
ISSUES OF VIOLENCE: Sexual Harassment: Say No! (<i>Towards a World of Equals</i> : Unit -6) Sexual Harassment, not Eve-teasing- Coping with Everyday Harassment- Further Reading:“ <i>Chupulu</i> ”. Domestic Violence: Speaking Out (<i>Towards a World of Equals</i> : Unit -8) Is Home a Safe Place? -When Women Unite [Film]. Rebuilding Lives. Additional Reading: New Forums for Justice. Thinking about Sexual Violence (<i>Towards a World of Equals</i> : Unit -11) Blaming the Victim-“I Fought for my Life....” - Additional Reading: The Caste Violence.								
UNIT-V								
GENDER: CO – EXISTENCE: Just Relationships: Being Together as Equals (<i>Towards a World of Equals</i> : Unit -12) Mary Kom and Onler. Love and Acid just do not Mix. Love Letters. Mothers and FathersAdditional Reading: Rosa Parks- The Brave Heart								
TEXT BOOKS: 1. All the five Units in the Textbook, “ <i>Towards a World of Equals: A Bilingual Textbook onGender</i> ” written by A. Suneetha, Uma Bhrugubanda, Duggirala Vasanta, Rama Melkote,Vasudha Nagaraj, Asma Rasheed, Gogu Shyamala, Deepa Sreenivas and Susie Tharu and published by Telugu Akademi, Hyderabad ,Telangana State in the year 2015 .								
REFERENCE BOOKS: 1. Menon, Nivedita. Seeing like a Feminist. New Delhi: Zubaan-Penguin Books, 2012								

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2. Abdulali Sohaila. "*I Fought For My Life...and Won.*" Available online at: <http://www.thealternative.in/lifestyle/i-fought-for-my-lifeand-won-sohaila-abdulal/>

III B.TECH I SEMESTER SYLLABUS

POWER ELECTRONICS

III Year I Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6EE15	PCC	L	T	P	C	CIA	SEE	Total
		3	1	-	4	40	60	100

Course Objectives:

The course should enable the students to:

1. Study the operation of power semiconductor devices
2. Study Single phase and three phase rectifiers
3. Describe the operation of AC voltage converters and cyclo-converters
4. Learn design different types of chopper circuits
5. Study the operation of different inverter circuits

Course Outcomes:

At the end of the course the student should be able to:

1. Understand the basics of power electronic devices
2. Analyze the design and control of rectifiers, inverters
3. Design power electronic converters in power control applications
4. Design AC voltage controller and Cyclo Converter.
5. Evaluate Chopper circuits for applications in various industrial applications.

SYLLABUS

UNIT-I	POWER SEMICONDUCTOR DEVICES AND COMMUTATION CIRCUITS	Classes: 10
Power semiconductor devices and commutation circuits: Thyristors, principle of operation of silicon controlled rectifiers (SCR), bipolar junction transistor (BJT), power metal oxide semiconductor field effect transistor (MOSFET), power insulated gate bipolar transistor (IGBT), gate turnoff thyristor (GTO) and characteristics, turn on and turnoff methods, dynamic characteristics of SCR, two transistor analogy, uni-junction transistor firing circuit, series and parallel operation of SCR's, design of snubber circuit; Specifications and ratings: Ratings of SCR, BJT and IGBT, line commutation and forced commutation circuits, numerical problems		
UNIT-II	SINGLE PHASE AND THREE PHASE CONTROLLED RECTIFIERS	Classes: 10
AC - DC converters: Phase control technique, single phase line commutated converters, midpoint and bridge connections, Half controlled and Fully controlled converters: Midpoint and bridge connections with R, RL loads and RLE load, derivation of average load voltage and current, without and with freewheeling diode, derivation of load voltage and current, numerical problems; Three phase converters: Three pulse and six pulse converters, midpoint and bridge connections, average load voltage with R and RL loads, effect of source inductance, operation of single phase and three phase dual converters, numerical problems.		
UNIT-III	AC VOLTAGE CONTROLLERS AND CYCLOCONVERTERS	Classes: 10

AC - AC controllers: Introduction, single phase two SCR's in anti-parallel, with R and RL loads, modes of operation of TRIAC, TRIAC with R and RL loads, derivation of RMS load voltage, current and power factor, wave forms, numerical problems Cyclo converters: Principle of operation of single phase midpoint and bridge type cyclo converters with resistive and inductive loads, continuous and discontinuous mode of operation		
UNIT-IV	DC-DC CONVERTERS	Classes: 10
Principle of operation of choppers, time ratio control and current limit control strategies, types of choppers, derivation of load voltage and currents with R, RL and RLE loads, AC chopper, problems; Switched mode regulators: Study of buck, boost and buck - boost regulators, Cuk regulators		
UNIT-V	INVERTERS	Classes: 10
DC - AC converters: Single phase inverter, basic series inverter, parallel inverter, operation and waveforms, voltage source inverter (VSI), three phase inverters 180, 120 degrees conduction modes of operation, voltage control techniques for inverters, pulse width modulation techniques, reduction of harmonics, current source inverter (CSI) with ideal switches, capacitor commutated type CSI, numerical problems.		
Text Books:		
1.M D Singh, K B Kanchandhani, "Power Electronics", Tata McGraw Hill Publishing Company, 2 nd Edition, 1998 2. Dr. P S Bimbhra, "Power Electronics", Khanna Publishers, 5 th Edition, 2012. 3. Ned Mohan, Tore M Undeland, William P Robbins, "Power Electronics: Converters, Applications, and Design", 3 rd Edition, John Wiley and sons, 2002. 4. M H Rashid, "Power Electronics, Circuits, Devices and Applications", Pearson, 3 rd Edition, 2001.		
Reference Books:		
1.VedamSubramanyam, "Power Electronics", New Age International Limited, 2 nd Edition, 2006. 2. P C Sen, "Power Electronics", Tata McGraw-Hill Publishing, 1 st Edition, 1987. 3. G K Dubey, S R Doradra, A Joshi, R M K Sinha, "Thyristorised Power Controllers", New Age International Limited, 2 nd Edition, 2008 4. V R Moorthi, "Power Electronics Devices", Oxford University Press, 4 th Edition, 2005.		
Web References:		
1. https://www.nptel.iitm.ac.in 2. https://www.bookboon.com/en/introduction-to-power-electronics-ebook		
E-Text Books:		
1. https://www.freebookcentre.net 2. https://www.amazon.in/power-electronics-handbook 3. https://www.circuitstoday.com		

POWER SYSTEMS –II

III Year I Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6EE16	PCC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	40	60	100

Course Objectives:

The course should enable the students to:

1. To understand insulation of cables and its capacitance parameter.
2. To understand the different Distribution Systems in Power Systems
3. To understand the over voltages in power systems
4. To illustrate Per Unit Representation and symmetrical components
5. To understand the Sequence components and fault calculations in power systems.

Course Outcomes:

At the end of the course the student should be able to:

1. Evaluate the load flow study using numerical methods
2. Understand stability constraints in a synchronous grid.
3. Understand the voltage and frequency control methods
4. Analyse the advanced monitoring and control of a power system using SCADA
5. Analyse the basics of power system economics

UNIT-I	UNDERGROUND CABLES	Classes: 10
Types of cables, construction, types of insulating materials, calculations of insulation resistance and stress in insulation, numerical problems, capacitance of single and 3 core belted cables, numerical problems, grading of cables- capacitance grading, inter sheath grading numerical problems		
UNIT-II	DC & AC DISTRIBUTION SYSTEMS	Classes: 10
Classification of Distribution Systems – Comparison of DC and AC, Underground and Overhead Distribution systems – Voltage Drop Calculations: Radial D.C Distributor fed one end and at the both the ends (equal/unequal Voltages) and Ring Main Distributor – Voltage Drop Calculations in A.C. Distributors: Power Factors referred to receiving end voltage and with respect to respective load voltages – Numerical Problems.		
UNIT-III	OVERVOLTAGE PROTECTION AND INSULATION COORDINATION	Classes: 10
Over voltage due to arcing ground and Peterson coil, lightning, horn gaps, surge diverters, rod gaps, expulsion type lightning arrester, valve type lightning arrester, ground wires, ground rods, counter poise, surge absorbers, insulation coordination, volt-time curves.		
UNIT-IV	PER UNIT SYSTEM AND SYMMETRICAL COMPONENTS	Classes: 10

Per Unit System: The one line diagram, impedance and reactance diagrams, per unit quantities, changing the base of per unit quantities, advantages of per unit system.

Symmetrical components: Positive, Negative and Zero sequence components, Average 3-phase power in terms of symmetrical components, sequence impedances and network equations		
UNIT-V	FAULT CALCULATIONS	Classes: 10
Fault calculations: single line to ground (LG), line to line(LL), double line to ground (LLG) faults with and without fault impedance.Numerical Problems.		
Text Books:		
1.J. Grainger and W. D. Stevenson, "Power System Analysis", McGraw Hill Education, 1994. 2. D. P. Kothari and I. J. Nagrath, "Modern Power System Analysis", McGraw Hill Education, 2003.		
Reference Books:		
1.A. R. Bergen and V. Vittal, "Power System Analysis", Pearson Education Inc., 1999. 2. O. I. Elgerd, "Electric Energy Systems Theory", McGraw Hill Education, 1995. 3.B. M. Weedy, B. J. Cory, N. Jenkins, J. Ekanayake and G. Strbac, "Electric Power Systems", Wiley, 2012		
E-Text Books:		
1. http://www.cl.cam.ac.uk/teaching/1011/SysOnChip/socdam-notes1011.pdf 2. https://www.doc.ic.ac.uk/~wl/teachlocal/cuscomp/notes/cc11.pdf		
MOOC Course:		
1. https://onlinecourses.nptel.ac.in/noc18_ee33/preview 2. https://onlinecourses.nptel.ac.in/noc18_ee34/preview		

LINEAR CONTROL SYSTEMS

III Year I Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6EE17	PCC	L	T	P	C	CIA	SEE	TOTAL
		3	0	0	3	40	60	100

Course Objectives:

At the end of the course, students are expected to:

1. Organize modeling and analysis of electrical and mechanical systems.
2. Analyze control systems by block diagrams and signal flow graph technique.
3. Demonstrate the analytical and graphical techniques to study the stability.
4. Illustrate the frequency domain and state space analysis

Course Outcomes:

Upon successful completion of this course, student will be able to:

1. Understand the basic definition of control system, and types of control system.
2. Explain the concept of Servo mechanism and different translational and rotational motions.
3. Understand the concept of order of a system and finding the response.
4. Able to find the steady state and transient state response of first and second order systems.
5. Explain the concept of relative stability and root locus, Nyquist plots

SYLLABUS

UNIT-I	INTRODUCTION	Classes: 10
Concepts of Control Systems- Open Loop and closed loop control systems and their differences- Classification of control systems, Feed-Back Characteristics, Effects of feedback. Mathematical models – Differential equations, Impulse Response and transfer functions – Translational and Rotational mechanical systems, Block diagram representation of systems considering electrical systems as examples -Block diagram algebra – Representation by Signal flow graph - Reduction uses Mason's gain formula.		
UNIT-II	TIME RESPONSE ANALYSIS	Classes: 10
Standard test signals - Time response of first order systems – Characteristic Equation of Feedback control systems, Transient response of second order systems - Time domain specifications – Steady state response - Steady state errors and error constants – Effects of proportional derivative, proportional integral systems, PID controllers.		
UNIT-III	STABILITY ANALYSIS IN S-DOMAIN	Classes: 10
The concept of stability – Routh's stability criterion – qualitative stability and conditional stability – limitations of Routh's stability. Root Locus Technique: The root locus concept - construction of root loci-effects of adding poles and zeros to $G(s)H(s)$ on the root loci.		
UNIT-IV	FREQUENCY RESPONSE ANALYSIS	Classes: 10
Introduction, Frequency domain specifications-Bode diagrams-Determination of Frequency domain specifications and transfer function from the Bode Diagram-Phase margin and Gain margin-Stability Analysis from Bode Plots.		
UNIT-V	STABILITY ANALYSIS IN FREQUENCY DOMAIN	Classes: 10

Polar Plots-Nyquist Plots-Stability Analysis. CLASSICAL CONTROL DESIGN TECHNIQUES: Compensation techniques – Lag, Lead, Lead-Lag Controllers design in frequency Domain.

STATE SPACE ANALYSIS OF CONTINUOUS SYSTEMS: Concepts of state, state variables and state model, derivation of state models from block diagrams, Diagonalization Solving the Time invariant state Equations- State Transition Matrix and its Properties

Text Books:

1. J Nagrath, M Gopal, Control Systems Engineering II, New Age International Publications, 3rd Edition, 2007
2. K Ogata, —Modern Control Engineering II, Prentice Hall, 4th Edition, 2003

Reference Books:

1. A Anand Kumar, —Control Systems II, PHI Learning, 1st Edition, 2007.
2. S Palani, —Control Systems Engineering II, Tata McGraw Hill Publications, 1st Edition, 2001.
3. N K Sinha, —Control Systems II, New Age International Publishers, 1st Edition, 2002
4. N C Jagan, —Control Systems II, BS Publications, 1st Edition, 2007.

Web References:

1. www.pa.msu.edu
2. www.tutorvista.com
3. www.globalspec.com
4. www.tutorvista.com
5. www.globalspec.com

E Text Books:

1. <http://www.cl.cam.ac.uk/teaching/1011/SysOnChip/socdam-notes1011.pdf>
2. <https://www.doc.ic.ac.uk/~wl/teachlocal/cuscomp/notes/cc11.pdf>
3. <https://www.cs.ccu.edu.tw/~chen/arch/SOC-design.pdf>

MOOC Course:

1. https://onlinecourses.nptel.ac.in/noc18_ee33/preview
2. https://onlinecourses.nptel.ac.in/noc18_ee34/preview

CONTROL SYSTEMS LAB

III Year I Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6EE18	PCC	L	T	P	C	CIA	SEE	Total
		0	0	2	1	40	60	100

Course Objectives:

The course should enable the students to:

1. Understand mathematical models of electrical and mechanical systems
2. Analysis of second order control system stability
3. Demonstrate the time domain and frequency domain analysis for linear time invariant systems
4. Apply programmable logic controllers to demonstrate industrial controls in the laboratory

Course Outcomes:

Upon completion of the course, students will be able to:

1. Observe the effect of different controllers
2. Analyse the effect of feedback on servo motor
3. Apply different compensation techniques in control system.
4. Find the steady state and transient state response of first and second order systems

LIST OF EXPERIMENTS

Any Eight of the following experiments are to be conducted

1	Time response of Second order system	Classes: 3
2	Characteristics of Synchro's	Classes: 3
3	Transfer function of DC motor	Classes: 3
4	Transfer function of DC generator	Classes: 3
5	Effect of feedback on DC servo motor	Classes: 3
6	Effect of P, PD, PI, PID Controller on a second order systems	Classes: 3
7	Lag and lead compensation – Magnitude and phase plots	Classes: 3
8	Temperature controller using PID	Classes: 3
9	Characteristics of magnetic amplifiers	Classes: 3
10	Characteristics of AC servo motor	Classes: 3
11	Programmable logic controller – Study and verification of truth tables of logic gates, simple Boolean expressions and application of speed control of motor	Classes: 3

Any two simulation experiments are to be conducted using software tools		
12	Linear system analysis (Time domain analysis, Error analysis)	Classes: 3
13	Stability analysis (Bode, Root Locus, Nyquist) of Linear Time Invariant system	Classes: 3
14	State space model for classical transfer function– Verification	Classes: 3
Text Books:		
1. I J Nagrath, M Gopal, Control Systems EngineeringII, New Age International Publications, 3 rd Edition, 2007 2. K Ogata, —Modern Control EngineeringII, Prentice Hall, 4 th Edition, 2003		

POWER ELECTRONICS LAB

III Year I Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6EE19	PCC	L	T	P	C	CIA	SEE	Total
		0	0	2	1	40	60	100

Course Objectives:

The course should enable the students to:

1. Examine the characteristics of various devices and application of firing circuits used in power electronics.
2. Outline the performance characteristics of AC voltage regulators, choppers, inverters, rectifiers and cyclo converters.
3. Demonstrate the working principle of various power electronic devices and circuits.
4. Design the circuit of switched mode power supplies.
5. Select a suitable power electronic device for different applications.

Course Outcomes:

The student will be able to

1. Design a Commutation circuit of a thyristor..
2. Control a supply voltage using different converters.
3. Select a suitable power electronic device for different applications..
4. Analyse the operation of three phase converters for different loads

LIST OF EXPERIMENTS

Expt. 1 SCR, MOSFET AND IGBT

Study the characteristics of SCR, MOSFET and IGBT.

Expt. 2 GATE FIRING CIRCUITS

Gate firing circuits of SCR.

Expt. 3 HALF CONTROLLED CONVERTER

Single phase half controlled converter with R and RL loads

Expt. 4 FORCED COMMUTATION CIRCUITS

Forced commutation circuits (Class A, Class B, Class C, Class D and Class E).

Expt. 5 FULLY CONTROLLED BRIDGE CONVERTER

Single phase fully controlled bridge converter with R and RL loads.

Expt. 6 SERIES INVERTER

Single phase series inverter with different loads

Expt. 7 PARALLEL INVERTER

Single phase parallel inverter with different loads.

Expt. 8 VOLTAGE CONTROLLER

Single phase AC voltage controller with R and RL loads

Expt. 9 DUAL CONVERTER

Single phase dual converter with R and RL loads

Expt. 10 CYCLOCONVERTER

Single phase cyclo converters with R and RL loads

In addition to the above ten experiments ,at least any one of the experiments from the following list are required to be conducted

Expt. 11 THREE PHASE CONVERTERS

Three phase half converter with R and RL loads.

Expt. 12 MOSFET BASED CHOPPERS

Operation of step down chopper using MOSFET.

Reference Books:

1. M H Rashid, "Power Electronics, Circuits, Devices and Applications", Pearson, 3rd Edition, 2001.
2. M D Singh, K B Kanchandhani, "Power Electronics", Tata McGraw Hill Publishing Company, 7th Edition, 2007.
3. Dr. P S Bimbhra, "Power Electronics", Khanna Publishers, 5th Edition, 2012.

Web References:

1. <https://www.ee.iitkgp.ac.in>
2. <https://www.citchennai.edu.in>

ADVANCED ENGLISH COMMUNICATION SKILLS LAB

III Year I Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6HS03	HSMC	L	T	P	C	CIA	SEE	Total
		-	-	2	1	40	60	100

COURSE OUTCOMES:

1. Acquire vocabulary and use it contextually

2. Listen and speak effectively

3. Develop proficiency in academic reading and writing

4. Communicate confidently in formal and informal contexts to Increase possibilities of job prospects

LIST OF EXPERIMENTS

The following course content to conduct the activities is prescribed for the Advanced Communication Skills (ACS) Lab:

Week -1

Activities on Fundamentals of Inter-personal Communication

Starting a conversation - responding appropriately and relevantly - using the right body language - Role Play in different situations & Discourse Skills- using visuals.

Week -2

Activities on Building Vocabulary

Synonyms and antonyms, word roots, one-word substitutes, prefixes and suffixes, study of word origin, business vocabulary, analogy, idioms and phrases, collocations & usage of vocabulary

Week -3

Activities on Reading Comprehension

General Vs Local Comprehension, Reading for facts, guessing meanings from context, Scanning and Skimming.

Week -4

Activities on Reading for Specific Purposes

Inferring meaning, Critical reading & Effective goggling.

Week-5

Activities on Writing Skills- Technical Reports

Structure and presentation of different types of writing - letter writing/ Resume writing/ e- correspondence

Week-6

Activities on Writing Skills

Technical report writing/ Portfolio writing - planning for writing - improving one's writing.

Week- 7

Activities on Presentation Skills

Oral presentations (individual and group) through JAM sessions and Seminars.

Week- 8

Activities on Presentation Skills Using ICT

PPTs and written presentations through posters/ projects/ reports/ e-mails/ assignments etc.

Week- 9

Activities on Group Discussion

Dynamics of group discussion, intervention, summarizing, modulation of voice, body language, relevance, fluency and organization of ideas and rubrics for evaluation- Concept and process

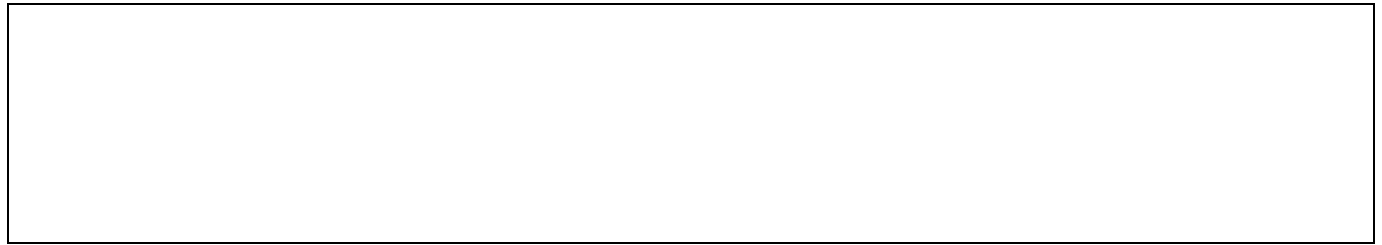
Week-10 Interview

Skills

Pre-interview planning, opening strategies, answering strategies, interview through tele-conference& video-conferencing and Mock Interviews.

Reference Books:

1. Raman, M & Sharma, S. (2009). Technical Communication. Oxford University Press.
 2. Rani. S. (2011). Advanced Communication Skills Laboratory Manual. Pearson Education.
 3. Anderson, V. (2007). Technical Communication. Cengage Learning pvt.Ltd.
 4. Kelly M. Quintanilla & Shawn T. Wahl. (2011). Business and Professional Communication: Keys for Workplace Excellence. Sage South Asia Edition. Sage Publications.
 5. Stev. D & David T. Mc Mahan. (2012). The Basics of Communication: A Relational Perspective. Sage South Asia Edition. Sage Publications.
 6. Mc Murrey. D & Buckley. J. (2012). Handbook for Technical Communication Cengage Learning.
 7. Sen. L. (2009). Communication Skills. PHI Learning Pvt Ltd.
 8. Vishvamohan, A. (2009). English for Technical Communication for Engineering Students. Tata Mc Graw Hill.
 9. Books on TOFEL/ GRE/ GMAT/ CAT/ IELTS by Barron's/ DELTA/ Cambridge University Press.
- Tomalin, B & Thomas, B. (2009). International English for Call Centers. Macmillan Publishers.



III B.TECH II SEMESTER SYLLABUS

SWITCHGEAR AND PROTECTION

III Year II Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6EE21	PCC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	40	60	100
Course Objectives: The course should enable the students to: 1.To introduce all kinds of circuit breakers and relays for protection of Generators, Transformers and feeder bus bars from Over voltages and other hazards. 2.To describe neutral grounding for overall protection. 3.To understand the phenomenon of Over Voltages and it’s classification. Course Outcomes: At the end of the course the student should be able to: 1. Understand and working of circuit breakers 2. Analyze the operation of circuit breaker 3. Understand the usage of the different relays 4. Understand and analyze reflection and refraction coefficients 5. Analyze and understand the power system protection								
UNIT-I	CIRCUIT BREAKERS						Classes: 10	
Circuit Breakers: Elementary principles of arc interruption, Recovery, Restriking Voltage and Recovery voltages, Restriking Phenomenon, Average and Max. RRRV. Current Chopping and Resistance Switching, CB ratings and Specifications, Types. Auto reclosures, Operation of following types of circuit breakers, Minimum Oil Circuit breakers, Air Blast Circuit Breakers, Vacuum and SF6 circuit breakers								
UNIT-II	ELECTROMAGNETIC AND STATIC RELAYS						Classes: 10	
Principle of Operation and Construction of Attracted armature, Balanced Beam, Induction Disc and Induction Cup relays Relays Classification: Instantaneous, DMT and IDMT types. Application of relays: Over current/ Undervoltage relays, Direction relays, Differential Relays and Percentage Differential Relays Universal Torque Equation, Distance Relays: Impedance, Reactance and Mho and Off-Set Mho relays, Comparison. Static Relays: Static Relays verses Electromagnetic Relays								
UNIT-III	GENERATOR, FEEDER AND BUS-BAR PROTECTION						Classes: 10	

Protection of generators against Stator faults, Rotor faults, and Abnormal Conditions. Restricted Earth fault and Inter-turn fault Protection. Numerical Problems on % Winding Unprotected. Protection of Bus bars – Differential protection Protection Of Transformers And Transmission Lines: Percentage Differential Protection, Buchholtz relay Protection. Over Current, Three-zone distance relay protection using Impedance relays. Translay Relay		
UNIT-IV	POWER SYSTEM TRANSIENTS	Classes: 10
Types of System Transients - Travelling or Propagation of Surges - Attenuation, Distortion, Reflection and Refraction Coefficients - Termination of lines with different types of conditions - Open Circuited Line, Short Circuited Line, T-Junction, Lumped Reactive Junctions, Bewley's Lattice Diagrams (for all the cases mentioned with numerical examples).		
UNIT-V	NEUTRAL GROUNDING	Classes: 10
Grounded and Ungrounded Neutral Systems.- Effects of Ungrounded Neutral on system performance. Methods of Neutral Grounding: Solid, Resistance, Reactance - Arcing Grounds and Grounding Practices Protection Against Over Voltages: Generation of Over Voltages in Power Systems.-Protection against Lightning Over Voltages - Valve type and Zinc-Oxide Lightning Arresters		
Text Books:		
1. Badari Ram, D. N. Viswakarma (2007), Power System Protection and Switchgear, 1st edition, Tata McGraw Hill Publications, New Delhi 2. C. L. Wadhwa (2011), Electrical Power Systems, 6th edition, New Age International (P) Limited, New Delhi		
Reference Books:		
1. Sunil S. Rao (1999), Switchgear and Protection, 10th edition, Khanna Publishers, New Delhi. 2. M. L. Soni, P. V. Gupta, U. S. Bhatnagar, A. Chakraborti (1999), A Text Book on Electrical Engineering, 1st edition, Dhanpat Rai & Co. Pvt. Ltd, New Delhi		

MICROPROCESSORS AND MICROCONTROLLERS

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6EC21	PCC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	40	60	100

COURSE OBJECTIVES:

The course should enable the students to:

1. Understand the basic of 8, 16 bit microprocessor architectures and its functionalities.
2. Write an assembly language programming skills of various processors.
3. Interface different peripheral devices with microprocessors and microcontrollers.
4. Develop systems using different microprocessors and microcontrollers.
5. Analyze RISC and ARM microprocessor based systems.

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Understand the architecture of micro processors and micro controller
2. Understand the programming model of micro processors and micro controllers
3. Interface different external peripheral devices with micro processors and micro controllers
4. Analyze a problem and formulate appropriate computing solution for processor or controller based application.
5. Develop an assembly language program for specified application

UNIT-I	MICROPROCESSORS ARCHITECTURE	Classes: 09
Overview of 8085, 8086 architecture- functional diagram, Register organization, memory segmentation, Memory addresses, physical memory organization, Signal descriptions of 8086, timing diagrams, Interrupt structure of 8086, Vector interrupt table, Interrupt service routine.		
UNIT-II	INSTRUCTION SET AND ASSEMBLY LANGUAGE PROGRAMMING OF 8086	Classes: 09
Programming model, Addressing modes, Instruction set, Assembler directives, Programs involving logical, branch and call instructions, Sorting, evaluating arithmetic expressions, and string manipulations.		
UNIT-III	INTERFACING WITH 8086	Classes: 09
8255 PPI, various modes of operation and interfacing to 8086, Stepper motor interfacing, D/A & A/D converter, Memory interfacing to 8086, DMA controller (8257) , 8251 USART architecture and Interfacing. Interfacing With advanced devices: Memory interfacing to 8086, Interrupts of 8086, Serial communication standards, serial data transfer schemes, 8251 USART architecture and Interfacing.		
UNIT-IV	INTRODUCTION TO MICROCONTROLLERS	Classes: 09
overview of 8051 microcontroller, Architecture, I/O ports, Memory organization, addressing modes and instruction set of 8051, Simple programs, Programming 8051 timers/ counters, Programming 8051 interrupts.		
UNIT-V	8051 REAL TIME CONTROL	Classes: 09

Programming Timer Interrupts, Programming External Hardware Interrupts, Interrupts, Programming serial communication interrupts, Programming 8051 timers and counters, Introduction to ARM-Architecture.

Text Books:

- 1.D.V.Hall, Microprocessors and Interfacing. TMGH, 2nd edition 2006.
3. Kenneth.J.Ayala. The 8051 microcontroller, 3rd edition, Cengage learning,2010
4. Advanced microprocessors and peripherals-A.K ray and K.M.Bhurchandani, TMH, 2nd edition2006.
- R. S. Gaonkar, Microprocessor Architecture: Programming and Applications with the 8085/8080A, Penram International Publishing, 1996.

Reference Books:

1. D A Patterson and J H Hennessy, "Computer Organization and Design The hardware and software interface. Morgan Kaufman Publishers.
2. Micro computer system 8086/8088 family architecture, programming and design- By Liu and GA Gibson, PHI, 2nd Ed.,
3. Microcontrollers and application, Ajay.V.Deshmukh, TMGH, 2005.
- The 8051 Microcontroller and Embedded Systems: Using Assembly and C by Muhammad Ali Mazidi, Janice Gillispie Mazidi, Second Edition.

Web References:

- 1.<http://www.freebookcentre.net/electronics-ebooks-download/Microprocessor-and-Microcontroller.html>
- 2.<http://coen.boisestate.edu/sml00/sml00-courses/ece-332-microprocessors-fall07/lecture-notes/>
- 3.<http://www.freebookcentre.net/electronics-ebooks-download/Introduction-to-Microcontrollers-Lecture-Notes.html>

E-Text Books:

- 1.<http://gen.lib.rus.ec/book/index.php?md5=67C5AC79DC8180A7F0641609D0C7800C>
- 2.<http://www.faadooengineers.com/threads/9039-8085-microprocessor-by-RAMESH-GANOKAR-ebook-pdf-download>
- 3.https://e.edim.co/123389964/The_8051_Microcontroller_Architecture_Programming_And_Application_s.pdf
- 4.https://e.edim.co/123389964/A.K._Ray_and_K.M._Bhurchandi-Advanced_Microprocessors_and_Peripherals_3e-Tata_Mcgraw_Hill.pdf

MOOC Course

- 1.<https://www.mooc-list.com/tags/microprocessors>
- 2.<https://www.coursera.org/courses?query=microprocessor>

ELECTRIC DRIVES

III Year II Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6EE22	PCC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	40	60	100

Course Objectives:

1. To obtain control of DC motors by phase controlled converters.
2. To analyse the operation of DC drives in all four quadrants.
3. Obtain the speed control of Induction motor from stator side.
4. Analyse the speed control of induction motor from rotor side.
5. Analyse the different speed control strategies of synchronous motor.

Course Outcomes:

After going through this course the student will be able to

1. Analyse the control of DC motors by phase controlled converters.
2. Understand four quadrant operation and control of DC motors by choppers.
3. Analyse the speed control of Induction motor from stator side.
4. Study the speed control of Induction motor from rotor side.
5. Understand the speed control of Synchronous motors.

SYLLABUS

UNIT - I : CONTROL OF DC MOTORS BY PHASE CONTROLLED CONVERTERSClasses: 10

Introduction to Thyristor controlled Drives, Single Phase semi and Fully controlled converters connected to D.C separately excited and D.C series motors, continuous current operation, output voltage and current waveforms. Speed and Torque expressions, Speed – Torque Characteristics. Three phase semi and fully controlled converters connected to D.C separately excited and D.C series motors, output voltage and current waveforms. Speed and Torque expressions, Speed–Torque characteristics.

UNIT – II: FOUR QUADRANT OPERATION OF DC DRIVESClasses: 10

Introduction to Four quadrant operation–Motoring operations, Electric Braking – Plugging, Dynamic and Regenerative Braking operations. Four quadrant operation of D.C motors by dual converters – Closed loop operation of DC motor (Block Diagram Only)

Control Of Dc Motors By Choppers:

Single quadrant, Two–quadrant and four quadrant chopper fed dc separately excited and series excited motors. Continuous current operation, Output voltage and current wave forms , Speed torque expressions, speed torque characteristics. Problems on Chopper fed D.C Motors, Closed Loop operation (Block Diagram Only)

UNIT – III : CONTROL OF INDUCTION MOTOR FROM STATOR SIDEClasses: 10

Variable voltage characteristics, Control of Induction Motor by Ac Voltage Controllers, Waveforms, speed torque characteristics.

Stator Frequency Control:

Variable frequency characteristics, Variable frequency control of induction motor by Voltage source and current source inverter and cyclo converters. PWM control, Comparison of VSI and CSI operations, Speed

torque characteristics, numerical problems on induction motor drives, closed loop operation of induction motor drives (Block Diagram Only).

UNIT – IV: CONTROL OF INDUCTION MOTOR FROM ROTOR SIDE

Classes:

10

Static rotor resistance control, Slip power recovery, Static Scherbius drive, Static Kramer Drive, their performance and speed torque characteristics, advantages applications.

UNIT – V: CONTROL OF SYNCHRONOUS MOTORS

Classes: 10

Separate control & self control of synchronous motors. Operation of self controlled synchronous motors by VSI and CSI cyclo converters. Load commutated CSI fed Synchronous Motor, Operation, Waveforms, speed torque characteristics, Applications, Advantages. Closed Loop control operation of synchronous motor drives (Block Diagram Only).

Text Books:

1. G. K. Dubey (2002), Fundamentals of Electric Drives, 2nd edition, Narosa Publications, New Delhi.
2. M. H. Rashid (2003), Power Electronic Circuits, Devices and applications, 3rd edition, Prentice Hall of India, New Delhi, India.

Reference Books:

1. M. D. Singh, K. B. Khanchandani (2008), Power Electronics, 2nd Edition, Tata McGraw Hill Publications, New Delhi.
2. Vedam Subramanyam (2008), Thyristor Control of Electric drives, 1st Edition, Tata McGraw Hill Publications, New Delhi, India.
3. S. K. Pillai (2007), A First course on Electrical Drives, 2nd Edition, New Age International (P) Ltd., New Delhi

ELECTRIC DRIVES LAB

III Year II Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A5EE23	PEC	L	T	P	C	CIE	SEE	Total
		0	0	3	1.5	30	70	100

Course Objectives:

The main objective of this course is to know the students about the speed control of DC motors using Thyristors.

Course Outcomes:

The course should enable the students to:

1. Understand about Cycloconverters and its uses.
2. Apply thyristors for Different speed control methods of DC motors.
3. Analyze Choppers and converters.
4. Analyze the operation of BUK BOOST Converters

LIST OF EXPERIMENTS

1. Speed Control of DC Motor using single phase Half Converter.
2. Speed Control of DC Motor using single phase Full Converter.
3. Speed Control of DC Motor using Three phase Half Controlled Bridge Converter
4. Speed Control of DC Motor using Chopper.
5. Study of SCR to drive small load.
6. Speed Control of single phase AC Motor using SCR.
7. Single phase cycloconverter fed AC Motor.
8. Three phase AC Induction Motor drive with Variable Voltage Variable Frequency control.
9. Speed Measurement and closed loop control using PMDC Motor.
10. Thyristorised drive for PMDC motor with speed measurement and closed loop control.
11. Study of series inverter for light load
12. Speed control of PMDC motor using MOSFET based Buck Boost Converter

MICROPROCESSORS AND MICROCONTROLLERS LAB**III Year II Semester**

Course Code	Category	Hours /Week			Credits	Maximum Marks		
		L	T	P		CIA	SEE	Total
A6EC22	PCC	-	-	2	1	40	60	100

COURSE OUTCOMES:**Upon successful completion of the course, the student is able to**

1. Develop the basic skills on hardware and software/programming of microprocessor
2. Enhance assembly language programming skills for simple and complex calculations used in various engineering disciplines
3. Capable to innovative and design intelligent systems, called embedded systems, using microprocessor for special purpose.
4. Involve in verification of functionality, speed and power of microprocessor based system.

LIST OF EXPERIMENTS**1. PROGRAMS FOR 16 BIT ARITHMETIC OPERATIONS (Using various addressing modes)**

- a. Write an ALP to Perform Addition and Subtraction of Multi precision numbers.
- b. Write an ALP to Perform Multiplication and division of signed and unsigned Hexadecimal numbers.
- c. Write an ALP to find square, cube and factorial of a given number.

2. PROGRAMS INVOLVING BIT MANIPULATION INSTRUCTIONS

- a. Write an ALP to find the given data is positive or negative.
- b. Write an ALP to find the given data is odd or even.
- c. Write an ALP to find Logical ones and zeros in a given data.

3. PROGRAMS ON ARRAYS FOR 8086

- a. Write an ALP to find Addition/subtraction of N no's.
- b. Write an ALP for finding largest/smallest no.
- c. Write an ALP to sort given array in Ascending/descending order.

4. PROGRAM FOR STRING MANIPULATIONS FOR 8086

- a. Write an ALP to find String length.
- b. Write an ALP for Displaying the given String.
- c. Write an ALP for Comparing two Strings.
- d. Write an ALP to reverse String and Checking for palindrome.

1. PROGRAM FOR DIGITAL CLOCK DESIGN USING 8086	
	a. Write an ALP for Designing clock using INT 21H Interrupt. b. Write an ALP for Designing clock using DOS Interrupt Functions. c. Write an ALP for Designing clock by reading system time.
6	INTERFACING STEPPER MOTOR WITH 8086
	a. Write an ALP to 8086 processor to Interface a stepper motor and operate it in clockwise by choosing variable step-size. b. Write an ALP to 8086 processor to Interface a stepper motor and operate it in Anti-clockwise by choosing variable step-size.
7	INTERFACING ADC/DAC WITH 8086
	a. Write an ALP to 8086 processor to Interface ADC. b. Write an ALP to 8086 processor to Interface DAC and generate Square Wave/Triangular Wave/Step signal.
8	COMMUNICATION BETWEEN TWO MICROPROCESSORS
	a. Write an ALP to have Parallel communication between two microprocessors using 8255. b. Write an ALP to have Serial communication between two microprocessor kits using 8251.
9	PROGRAMS USING ARITHMETIC AND LOGICAL INSTRUCTIONS FOR 8051
	a. Write an ALP to 8051 Microcontroller to perform Arithmetic operations like addition, subtraction, Multiplication and Division. b. Write an ALP to 8051 Microcontroller to perform Logical operations like AND, OR and XOR. c. Programs related to Register Banks.
10	PROGRAM TO VERIFY TIMERS/COUNTERS OF 8051
	a. Write a program to create a delay of 25msec using Timer0 in mode 1 and blink all the Pins of P0. b. Write a program to create a delay of 50 µsec using Timer1 in mode 0 and blink all the Pins of P2. c. Write a program to create a delay of 75msec using counter0 in mode 2 and blink all the Pins of P1. d. Write a program to create a delay of 80 µsec using counter1 in mode 1 and blink all the Pins of P2.
11	UART OPERATION IN 8051
	a. Write a program to transfer a character serially with a baud rate of 9600 using UART. b. Write a program to transfer a character serially with a baud rate of 4800 using UART. c. Write a program to transfer a character serially with a baud rate of 2400 using UART.
12	INTERFACING LCD WITH 8051
	a. Develop and execute the program to interface 16*2 LCD to 8051. b. Develop and execute the program to interface LCD to 8051 in 4-bit or 8-bit mode.

Reference Books:

1. Kenneth.J.Ayala. The 8051 microcontroller, 3rd edition, Cengage learning,2010.
2. Advanced microprocessors and peripherals-A.K ray and K.M.Bhurchandani, TMH, 2nd edition 2006.
3. The 8051 Microcontroller and Embedded Systems: Using Assembly and C by Muhammad Ali Mazidi, Janice GillispieMazidi, Second Edition.

POWER SYSTEMS AND PROTECTION LAB

III Year II Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6EE24	PCC	L	T	P	C	CIA	SEE	Total
		0	0	2	1	40	60	100

Course Objectives:

The course should enable the students to:

1. Determine the parameters of transmission lines
2. Understand the concept of various transmission line protection schemes
3. Study of feeder protection and generator protection circuits

Course Outcomes:

Upon completion of the course, students will be able to:

1. Evaluate the ABCD parameters transmission lines
2. Analyze the concept of various transmission line protection schemes
3. Understand the concept of feeder protection and earth fault protection
4. Evaluate the sequence impedance parameters of an alternator

LIST OF EXPERIMENTS

1	STUDIES OF CHARACTERISTICS OF FUSE AND THERMAL OVERLOAD PROTECTION Study and plotting of characteristics of fuse and tripping of bimetallic thermal overload protection	Classes: 3
2	CHARACTERISTICS OF AN MCB Plotting the Characteristics of an MCB (miniature circuit breaker)	Classes: 3
3	SINGLE PHASE OVER CURRENT RELAY Plotting the characteristics of single phase over current relay .	Classes: 3
4	IDMT OVERCURRENT RELAY Determination of tripping characteristics of IDMT over current relay	Classes: 3
5	ABCD PARAMETERS OF A TRANSMISSION LINE Determination of ABCD parameters of a transmission line	Classes: 3
6	REGULATION AND EFFICIENCY OF A TRANSMISSION LINE Determination of transmission line efficiency and regulation and Study of impedance relay (distance relay).	Classes: 3
7	FERRANTI EFFECT AND VAR COMPENSATION Study of Ferranti effect and study of VAR compensation	Classes: 3
8	UNDER VOLTAGE AND OVER VOLTAGE PROTECTION TO A FEEDER Study the protection schemes of a feeder under UV/OV conditions	Classes: 3

9	OVER CURRENT AND EARTH FAULT PROTECTION TO A FEEDER Open delta connection of secondary's of three PTs to detect earth fault protection and connection of secondary's of three current transformers to detect earth fault	Classes: 3
10	SEQUENCE COMPONENTS OF 3-Φ ALTERNATOR Determination of positive, negative and zero sequence components of three phase alternator	Classes: 3
11	FAULT ANALYSIS OF A 3-Ø ALTERNATOR Determine the fault currents on an unloaded synchronous generator for L-G, L-L, & LL-G Faults	Classes: 3
Text/Reference Books:		
1. Paithankar, S R Bhide, "Fundamentals of Power System Protection", PHI, 1 st Edition, 2003 2. C L Wadhwa, "Electrical Power Systems", New Age international (P) Limited, 6 rd Edition, 2010 3. VK Mehta, "Principles of power systems", S Chand Publications, 4 th Edition, 2009		

IV B.TECH I SEMESTER SYLLABUS

POWER SYSTEM OPERATION AND CONTROL

IV Year I Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6EE26	PCC	L	T	P	C	CIE	SEE	Total
		3	1	0	4	40	60	100

Course Objectives:

At the end of the course, students are expected to

1. calculate the voltage and power of various busses.
2. Analyse the Stability analysis in power systems
3. Judge the performance of various types of load curves and load duration curves.
4. Build the Transfer Functions of Governor, Turbine and Generator
5. Compare and select the various types of voltage control and frequency control as per the requirement.

Course Outcomes:

Upon successful completion of this course, student will be able to

1. Get an overview of power system operation and control
2. Understand & model power-frequency dynamics
3. Understand & model reactive power-voltage interaction
4. Estimate different methods of control for maintaining voltage profile against varying system load.
5. Analyze different computer based control methods in power systems

SYLLABUS

UNIT –I : POWER FLOW ANALYSIS

Introduction, Bus classification - Formulation of Y bus for a system - Load flow equations - Iterative methods - Gauss Seidel(GS) Method, Newton- Raphson (NR) Methods for the solution of the power flow equations - Merits and demerits of the above methods.

UNIT-II : ECONOMIC OPERATION OF POWER SYSTEMS

Optimal operation of Generators in Thermal Power Stations, Heat rate Curve, Cost Curve, Incremental fuel and Production costs, Input-output characteristics, Optimum generation allocation with line losses neglected. Optimum generation allocation including the effect of transmission line losses, Loss Coefficients, General transmission line loss formula.

UNIT –III : STABILITY ANALYSIS

Concepts of Steady State, Dynamic and Transient Stabilities, synchronous machine connected to an infinite bus, Power Angle Curve and Determination of Steady State Stability and Methods to improve steady state stability, Derivation of Swing Equation, Determination of Transient Stability by Equal Area Criterion, Application of Equal Area Criterion, Critical Clearing Angle Calculation - Solution of Swing Equation: Point-by-Point Method, Methods to improve Stability

UNIT-IV: MODELING OF SPEED GOVERNOR AND EXCITATION SYSTEMS

Mathematical Modeling of Speed Governor, Steam Turbines, Generators and Excitation systems -Transfer function and Block Diagram Representation

UNIT -V: Load Frequency Control and Voltage Control:

Load Frequency Control :Necessity of keeping frequency constant-concept of control area, Single area control, Flat frequency control, Flat tie-line frequency control, Tie-line bias control, Advantages of pool operation, Proportional plus Integral control of single area and its block diagram representation, steady state response – Load Frequency Control and Economic dispatch control.

Voltage Control :Generation and absorption of reactive power by various components of a Power System. Automatic Voltage Regulators, Shunt and series Compensators, Tap Changing Transformers.

Text Books:

1. I. J. Nagrath and D.P. Kothari, Modern Power System Analysis, TMH Publication, 4th Edition 2011
2. O. Elgard, Electric Energy Systems Theory, TMH Publication, 2nd Edition, 2001.
3. A. Chakrabarthy and S. Halder, Power System Analysis Operation & control, PHI Publications, 3rd Edition, 2010

References:

1. Allen. J. Wood and Bruce F. Wollenberg, 'Power Generation, Operation and Control', John Wiley & Sons, Inc., 2003.
2. Kundur P., 'Power System Stability and Control, Tata McGraw Hill Education Pvt. Ltd., New Delhi, 10th reprint, 2010.
3. Hadi Saadat, 'Power System Analysis', Tata McGraw Hill Education Pvt. Ltd., New Delhi, 21st reprint, 2010.

UTILIZATION OF ELECTRICAL ENERGY

IV Year I Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6EE27	PEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100

Course Objectives:

At the end of the course, students are expected to

1. Articulate different types of Electric Heating
2. Able to discuss about electric Welding.
3. Able to know types of Illumination.
4. Ability to discuss mechanics of Train movement.
5. Ability to plot trapezoidal and quadrilateral speed time curves.

Course Outcomes:

At the end of the course student will have ability to:

1. Analyze the different types of Electric Heating
2. Know the types of electric Welding.
3. Understand the types of Illumination.
4. Analyze the mechanics of Train movement.
5. Develop trapezoidal and quadrilateral speed time curves.

SYLLABUS

UNIT – I : ELECTRIC HEATING

Classification of electric heating, Electric Resistance Heating, Properties of good heating material, Different types of heating material, Causes of failure of heating element, Direct Arc Furnace, Indirect Arc Furnace, Induction Heating, Direct Core-type Induction Furnace, Vertical Core-Type Induction Furnace, Indirect Core-Type Induction Furnace, Coreless Induction Furnace, High Frequency Eddy-current Heating, Dielectric Heating.

UNIT – II : ELECTRIC WELDING

Electric Welding: Introduction, Classification of Welding Processes, Formation and Characteristics of Electric Arc, Effect of Arc Length, Electrodes for Metal Arc Welding, Advantages of Coated Electrodes, Types of Joints - Welding Transformer

Electric Arc welding: Carbon Arc Welding, Submerged Arc Welding, Atomic Hydrogen Welding.

Resistance Welding: Spot Welding, Seam Welding, Projection Welding, Butt Welding, Flash Butt Welding, Upset Welding, Electron Beam Welding, Laser Welding

UNIT – III: ILLUMINATION

Introduction, terms used in illumination, laws of illumination, polar curves, photometry, integrating sphere, sources of light.

Types of Lamps : Discharge lamps, MV and SV lamps, comparison between tungsten filament lamps and fluorescent tubes, Basic principles of light control, Types and design of lighting and flood lighting.

UNIT – IV : ELECTRIC TRACTION - I

System of electric traction and track electrification. Review of existing electric traction systems in India. Special features of traction motor, methods of electric braking-plugging rheostat braking and regenerative braking.

UNIT – V : ELECTRIC TRACTION - II

Mechanics of train movement. Speed-time curves for different services, trapezoidal and quadrilateral speed

time curves. Calculations of tractive effort, power, specific energy consumption for given run, effect of varying acceleration and braking retardation, adhesive weight and braking retardation adhesive weight and coefficient of adhesion.

Text Books:

1. G. C. Garg (2005), Utilization of Electrical Power & Electric traction, 8th edition, Khanna publishers, New Delhi.
2. N. V. Suryanarayana (2005), Utilization of Electrical Power including Electric drives and Electric traction, 1st edition
New Age International (P) Ltd., New Delhi.

Reference Books:

1. Partab (2007), Art & Science of Utilization of electrical Energy, 2nd edition, Dhanpat Rai & Sons, New Delhi.
2. C. L. Wadhwa (2005), Generation, Distribution and Utilization of Electrical Energy, 2nd edition, New Age International (P) Ltd., New Delhi.

POWER SYSTEM SIMULATION LAB

IV B.Tech I Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A5EE28	PCC	L	T	P	C	CIE	SEE	Total
		0	0	3	1.5	30	70	100

COURSE OBJECTIVES:

The course should enable the students to:

1. Demonstrate load flow studies using iterative methods.
2. Analyze stability in power systems.
3. To find sequence impedances of 3- Φ synchronous machine and Transformer
4. To understand the economic load dispatch and load frequency control in power system

COURSE OUTCOMES:

Upon completion of the course, students will be able to:

1. Analyze the load flow studies by numerical methods using simulation
2. Evaluate the critical values of transient stability in power systems using simulation
3. Evaluate the Fault parameters in power systems using simulation
4. Understand the Load frequency control in power system using simulation

LIST OF EXPERIMENTS

Any eight simulation experiments are to be conducted using software tools

1. Computation of Impedance parameters Transmission Line Parameters
2. Modeling Of Transmission Lines
3. Formation of YBUS Matrix
4. Formation of Z BUS Matrix
5. Load Flow Analysis using Gauss Seidal (GS) Method
6. Load Flow Analysis using Newton Raphson (NR) Method
7. Transient Stability Analysis Single-Machine Infinite Bus System
8. Fault Analysis
9. Load – Frequency Dynamics of Single- Area Power System
10. Economic Load Dispatch in Power Systems.
11. Load flow analysis in large systems for different load conditions.
12. Small Signal Stability Analysis in Single-Machine Infinite Bus System

REFERENCE BOOKS:

1. M A Pai, "Computer Techniques in Power System Analysis", TMH Publications, 1st Edition, 2010
2. Grainger, Stevenson, "Power System Analysis", Tata McGraw Hill, 1st Edition, 2010

WEB REFERENCES:

1. <https://www.ee.iitkgp.ac.in>

PROFESSIONAL ELECTIVES

WIND AND SOLAR ENERGY SYSTEMS
(PROFESSIONAL ELECTIVE- I)
III Year I Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6EE31	PEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100

Course Objectives:

1. To study the physics of wind power and energy
2. To understand the principle of operation of wind generators
3. To know the solar power resources
4. To analyse the solar photo-voltaic cells
5. To discuss the solar thermal power generation
6. To identify the network integration issues

Course Outcomes:

At the end of this course, students will demonstrate the ability to

1. Understand the basic physics of wind and solar power generation
2. Understand the power electronic interfaces for wind and solar generation
3. Understand the issues related to grid-integration of solar and wind energy systems
4. Understand the issues related to grid-integration of solar and wind energy systems
5. Demonstrate the Solar Thermal and PV applications along with its storage systems.

SYLLABUS

UNIT- I : Physics of Wind

Classes: 10

History of wind power, Indian and Global statistics, Wind physics, Betz limit, Tip speed ratio, stall and pitch control, Wind speed statistics-probability distributions, Wind speed and power-cumulative distribution functions.

UNIT- II : Wind generator topologies

Classes: 10

Review of modern wind turbine technologies, Fixed and Variable speed wind turbines, Induction Generators, Doubly-Fed Induction Generators and their characteristics, Permanent Magnet Synchronous Generators, Power electronics converters. Generator-Converter configurations, Converter Control.

UNIT- III : The Solar Resource

Classes: 10

Introduction, solar radiation spectra, solar geometry, Earth Sun angles, observer Sun angles, solar day length, Estimation of solar energy availability.

UNIT- IV : Solar photovoltaic technologies

Classes: 10

Technologies-Amorphous, mono crystalline, polycrystalline; array, Power Electronic Converters for Solar Systems, Maximum Power Point Tracking (MPPT) algorithms. Converter Control.

UNIT- V : Network Integration Issues

Classes: 10

Overview of grid code technical requirements. Fault ride-through for wind farms - real and reactive power regulation, voltage and frequency operating limits, solar PV and wind farm behaviour during grid disturbances. Power quality issues. Power system interconnection experiences in the world. Hybrid and

isolated operations of solar PV and wind systems.

Solar thermal power generation: Technologies, Parabolic trough, central receivers, parabolic dish, Fresnel, elementary analysis.

Text Books:

1. T. Ackermann, "Wind Power in Power Systems", John Wiley and Sons Ltd., 2005.
2. G. M. Masters, "Renewable and Efficient Electric Power Systems", John Wiley and Sons, 2004.
3. S. P. Sukhatme, "Solar Energy: Principles of Thermal Collection and Storage", McGraw Hill, 1984.

Reference Books:

1. H. Siegfried and R. Waddington, "Grid integration of wind energy conversion systems" John Wiley and Sons Ltd., 2006.
2. G. N. Tiwari and M. K. Ghosal, "Renewable Energy Applications", Narosa Publications, 2004.
3. J. A. Duffie and W. A. Beckman, "Solar Engineering of Thermal Processes", John Wiley & Sons, 1991.

AI TECHNIQUES IN ELECTRICAL ENGINEERING
(PROFESSIONAL ELECTIVE- I)
III Year I Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6EE32	PEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100

Course Objectives:

- To locate soft commanding methodologies, such as artificial neural networks, Fuzzy logic and genetic Algorithms.
- To observe the concepts of feed forward neural networks and about feedback neural networks.
- To practice the concept of fuzziness involved in various systems and comprehensive knowledge of fuzzy logic control and to design the fuzzy control
- To analyze genetic algorithm, genetic operations and genetic mutations.

Course Outcomes: Upon the completion of this course, the student will be able to

- Understand feed forward neural networks, feedback neural networks and learning techniques.
- Analyze fuzziness involved in various systems and fuzzy set theory.
- Develop fuzzy logic control for applications in electrical engineering
- Develop genetic algorithm for applications in electrical engineering.

UNIT – I: Artificial Neural Networks: Introduction-Models of Neural Network – Architectures – Knowledge representation – Artificial Intelligence and Neural networks – Learning process – Error correction learning – Hebbian learning – Competitive learning – Boltzman learning – Supervised learning – Unsupervised learning – Reinforcement learning – learning tasks.

UNIT- II: ANN Paradigms : Multi – layer perceptron using Back propagation Algorithm-Self – organizing Map – Radial Basis Function Network – Functional link, network – Hopfield Network.

UNIT – III: Fuzzy Logic: Introduction – Fuzzy versus crisp – Fuzzy sets – Membership function – Basic Fuzzy set operations – Properties of Fuzzy sets – Fuzzy cartesian Product – Operations on Fuzzy relations – Fuzzy logic – Fuzzy Quantifiers – Fuzzy Inference – Fuzzy Rule based system – Defuzzification methods.

UNIT – IV: Genetic Algorithms: Introduction-Encoding – Fitness Function-Reproduction operators – Genetic Modeling – Genetic operators – Crossover – Single-site crossover – Two-point crossover – Multi point crossover-Uniform crossover – Matrix crossover – Crossover Rate – Inversion & Deletion – Mutation operator –Mutation – Mutation Rate-Bit-wise operators – Generational cycle-convergence of Genetic Algorithm.

UNIT-V: Applications of AI Techniques: Load forecasting – Load flow studies – Economic load dispatch – Load frequency control – Single area system and two area system – Small Signal Stability (Dynamic stability) Reactive power control – speed control of DC and AC Motors.

TEXT BOOK:

- S. Rajasekaran and G. A. V. Pai, "Neural Networks, Fuzzy Logic & Genetic Algorithms"- PHI, New Delhi, 2003.

REFERENCES:

- P. D. Wasserman, Van Nostrand Reinhold, "Neural Computing Theory & Practice" – New York, 1989.
- Bart Kosko, "Neural Network & Fuzzy System" Prentice Hall, 1992.
- G. J. Klir and T. A. Folger, "Fuzzy sets, Uncertainty and Information"-PHI, Pvt.Ltd,1994.
- D. E. Goldberg," Genetic Algorithms"- Addison Wesley 1999

SPECIAL ELECTRICAL MACHINES
(PROFESSIONAL ELECTIVE –I)

III Year I Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6EE33	PEC	3	0	0	3	30	70	100

Course Objectives:

1. To make the learner to be aware of latest special machines which are in vogue
2. To impart knowledge on Construction, principle of operation and performance of synchronous reluctance motors.
3. To impart knowledge on the Construction, principle of operation, control and performance of stepping motors& switched reluctance motors.
4. To impart knowledge on the Construction, principle of operation, control and performance of permanent magnet brushless D.C. motors & permanent magnet synchronous motors.
5. To be familiar with design features of special electrical machines

Course Outcomes:

1. Ability to model and analyze electrical apparatus and their application to power system
2. Identify appropriate machine for a specific application
3. Recognize the principle of operation and characteristics of the given special machine
4. Distinguish the special machine with the obtained characteristics

SYLLABUS

UNIT- I : SYNCHRONOUS RELUCTANCE MOTORS

Classes: 10

Constructional features – Types – Axial and Radial flux motors – Operating principles – Variable Reluctance Motors – Voltage and Torque Equations - Phasor diagram - performance characteristics – Applications.

UNIT - II : STEPPER MOTORS

Classes: 10

Constructional features – Principle of operation – Variable reluctance motor – Hybrid motor – Single and multi-stack configurations – Torque equations – Modes of excitation – Characteristics – Drive circuits – Microprocessor control of stepper motors – Closed loop Control-Concept of lead angle– Applications.

UNIT- III : SWITCHED RELUCTANCE MOTORS (SRM)

Classes: 10

Constructional features – Rotary and Linear SRM - Principle of operation – Torque production – Steady state performance prediction- Analytical method -Power Converters and their controllers – Methods of Rotor position sensing – Sensor less operation – Characteristics and Closed loop control – Applications.

UNIT - IV : PERMANENT MAGNET BRUSHLESS D.C. MOTORS

Classes: 10

Permanent Magnet materials – Minor hysteresis loop and recoil line-Magnetic Characteristics – Permeance coefficient -Principle of operation – Types – Magnetic circuit analysis – EMF and torque equations –Commutation - Power Converter Circuits and their controllers – Motor characteristics and control– Applications.

UNIT - V : PERMANENT MAGNET SYNCHRONOUS MOTORS (PMSM)

Classes: 10

Principle of operation – Ideal PMSM – EMF and Torque equations – Armature MMF – Synchronous Reactance – Sine wave motor with practical windings - Phasor diagram – Torque/speed characteristics - Power controllers - Converter Volt-ampere requirements– Applications.

Text Books:

1. K.Venkataratnam, 'Special Electrical Machines', Universities Press (India) Private Limited, 2008.
2. T.J.E. Miller, 'Brushless Permanent Magnet and Reluctance Motor Drives', Clarendon Press, Oxford, 1989.
3. T. Kenjo, 'Stepping Motors and Their Microprocessor Controls', Clarendon Press London, 1984.

Reference Books :

1. R.Krishnan, 'Switched Reluctance Motor Drives – Modeling, Simulation, Analysis, Design and Application', CRC Press, New York, 2001.
2. P.P. Aearnley, 'Stepping Motors – A Guide to Motor Theory and Practice', Peter Perengrinus London, 1982.
3. T. Kenjo and S. Nagamori, 'Permanent Magnet and Brushless DC Motors', Clarendon Press, London, 1988.
4. E.G. Janardanan, 'Special electrical machines', PHI learning Private Limited, Delhi, 2014.

DIGITAL SIGNAL PROCESSING
(PROFESSIONAL ELECTIVE –I)

III Year I Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6EE34	PEC	3	0	0	3	40	60	100

COURSE OBJECTIVES:

1. To provide background and fundamental material for the analysis and processing of digital signals.
2. To familiarize the relationships between continuous-time and discrete-time signals and systems.
3. To study fundamentals of time, frequency and z-plane analysis and to discuss the inter-relationships of these Analytic method
4. To study the designs and structures of digital (IIR and FIR) filters from analysis to synthesis for a given Specifications
5. To introduce a few real-world signal processing applications.

COURSE OUTCOMES:

1. Analyze digital to analog (D/A) and analog to digital (A/D) converters operate on a signal and be able to model these operations mathematically.
2. Understand the inter-relationship between DFT and various transforms
3. Analyze the Z-transforms, Properties of Z-transform
4. Explain the linear phase FIR Digital Filters by Frequency Sampling method
5. Analyze Multi stage Implementation of sampling rate conversion.

SYLLABUS

UNIT-I

INTRODUCTION TO DIGITAL SIGNAL PROCESSING: Discrete time signals & systems, linear shiftinvariant systems, stability and causality, discrete time systems described by difference equations, Frequency domain representation of discrete time signals and systems.

UNIT-II

FOURIER SERIES AND FOURIER TRANSFORMS: Discrete Fourier series representation of periodic sequences, Properties of discrete Fourier series, Discrete Fourier transforms: frequency domain sampling, , linear convolution of sequences using DFT, Computation of DFT, Relationship of DFT to other transforms, Properties of DFT, Fast Fourier transforms (FFT) - Radix-2 FFT algorithm, Radix-4 FFT algorithms, Inverse FFT.

UNIT-III

Z-TRANSFORMS: Review of Z-transforms, Properties of Z-transform, Rational Z-transforms, and Inversion of Z- transforms, stability and causality.

REALIZATION OF DIGITAL FILTERS: Structures for FIR systems: Direct form structure, Cascade form structures, Structures for IIR systems: Direct form structures, Signal flow graphs and transposed structures, cascade form structures, Parallel form structures.

UNIT-IV

DESIGN OF FIR DIGITAL FILTERS: Symmetric and antisymmetric FIR filters, Design of linear phase FIR Digital Filters using Windows, Design of linear phase FIR Digital Filters by Frequency Sampling method.

DESIGN OF IIR DIGITAL FILTERS: IIR filter design by Approximation of Derivatives, IIR filter design by impulse invariance, IIR filter design by bilinear transformation, Characteristics of commonly used analog filters (Butterworth and Chebyshev), Frequency transformations, comparison of IIR & FIR filters.

UNIT- V

MULTIRATE DIGITAL SIGNAL PROCESSING: Decimation by a factor D , interpolation by a factor L , sampling rate conversion by a rational factor L/D , Filter Design & Implementation for sampling rate conversion, Multi stage Implementation of sampling rate conversion.

TEXT BOOKS:

1. John G. Proakis, Dimitris G. Manolakis (2007), Digital Signal Processing, Principles, Algorithms, and Applications, Pearson Education / PHI, India.
2. A.V. Oppenheim, R. W. Schaffer (2009), Discrete Time Signal Processing, Prentice Hall of India, New Delhi.

REFERENCE BOOKS:

1. Andreas Antoniou (2006), Digital Signal Processing, Tata McGraw Hill, New Delhi.
2. M. H. Hayes (2007), Schaums Outlines of Digital Signal Processing, Tata McGraw Hill, India.

LINEAR AND DIGITAL IC APPLICATIONS
(PROFESSIONAL ELECTIVE –II)

III Year II Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6EE35	PEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100

COURSE OBJECTIVES:

1. To introduce the basic building blocks of linear integrated circuits.
2. To teach the linear and non-linear applications of operational amplifiers.
3. To introduce the theory and applications of PLL
4. To introduce the concepts of waveform generation and introduce some special function ICs...
5. Exposure to digital IC's

COURSE OUTCOMES:

Up on successful completion of this course, student will be able to:

1. Analyze the DC and AC characteristics of operational amplifiers compensation techniques.
2. Understand the linear and non-linear applications of an op-amp
3. Explain and compare the working of FILTERS, TIMER AND PHASE LOCKED LOOPS
4. Classify and comprehend the working principle of data converters.
5. Analyze TTL and CMOS logic families

SYLLABUS

UNIT–I INTEGRATED CIRCUITS AND OPERATIONAL AMPLIFIER:

Introduction, Classification of IC's, IC chip size and circuit complexity, basic information of Op-Amp IC741 Op-Amp and its features, the ideal Operational amplifier, Op-Amp internal circuit, Op-Amp characteristics - DC and AC.

UNIT–II LINEAR APPLICATIONS OF OP-AMP & NON-LINEAR APPLICATIONS OF OP-AMP:

Inverting and non-inverting amplifiers, adder, subtractor, Instrumentation amplifier, AC amplifier, V to I and I to V converters, Integrator and differentiator. Sample and Hold circuit, Log and Antilog amplifier, multiplier and divider, Comparators, Schmitt trigger, Multivibrators, Triangular and Square waveform generators, Oscillators

UNIT–III ACTIVE FILTERS, TIMER AND PHASE LOCKED LOOPS:

Introduction, Butterworth filters – 1st order, 2nd order low pass and high pass filters, band pass, band reject and all pass filters. Introduction to IC 555 timer, description of functional diagram, monostable and astable operations and applications, Schmitt trigger, PLL - introduction, basic principle, phase detector/comparator,

voltage controlled oscillator (IC 566), low pass filter, monolithic PLL and applications of PLL.

UNIT-IV VOLTAGE REGULATOR, D to A AND A to D CONVERTERS::

Introduction, Series Op-Amp regulator, IC Voltage Regulators, IC 723 general purpose regulators, Switching Regulator. Introduction, basic DAC techniques - weighted resistor DAC, R-2R ladder DAC, inverted R-2R DAC, A to D converters - parallel comparator type ADC, counter type ADC, successive approximation ADC and dual slope ADC, DAC and ADC Specifications.

UNIT-V CMOS LOGIC, COMBINATIONAL & SEQUENTIAL CIRCUITS USING TTL 74XX ICs:

CMOS logic levels, MOS transistors, Basic CMOS Inverter, NAND and NOR gates, CMOS AND-OR-INVERT and OR-AND-INVERT gates, implementation of any function using CMOS logic. Study of logic gates using 74XX ICs, Four-bit parallel adder (IC 7483), Comparator (IC 7485), Decoder (IC 74138, IC 74154), BCD-to-7-segment decoder (IC 7447), Encoder (IC 74147), Multiplexer (IC 74151), Demultiplexer (IC 74154).

Flip Flops (IC 7474, IC 7473), Shift Registers, Universal Shift Register (IC 74194), 4-bit asynchronous binary counter (IC 7493).

TEXT BOOKS:

1. D. Roy Choudhury, Shail B. Jain (2012), Linear Integrated Circuit, 4th edition, New Age International Pvt.Ltd., New Delhi, India
2. Ramakant A. Gayakwad, (2012), OP-AMP and Linear Integrated Circuits, 4th edition, Prentice Hall / Pearson Education, New Delhi.
3. Floyd, Jain (2009), Digital Fundamentals, 8th edition, Pearson Education, New Delhi.

REFERENCE BOOKS:

1. Sergio Franco (1997), Design with operational amplifiers and analog integrated circuits, McGraw Hill, New Delhi.
2. Gray, Meyer (1995), Analysis and Design of Analog Integrated Circuits, Wiley International, New Delhi.

INDUSTRIAL AUTOMATION
(PROFESSIONAL ELECTIVE –II)

III Year II Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6EE36	PEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100

Course Outcome:

After learning the course the student should be able to:

1. Analyze the PLC and supervisory control and data acquisition (SCADA)
2. Explain Actuators, process control valves, power electronics devices DIAC, TRIAC, power MOSFET and IGBT
3. Analyze the Computer aided measurement and control systems, Internet of things (IoT) for plant automation
4. Analyze the application of PLC to process control industries
5. Analyze the DCS software configuration, DCS communication, DCS Supervisory Computer Tasks, DCS integration with PLC and Computers

SYLLABUS

Unit I : Introduction: Automation overview, Requirement of automation systems, Architecture of Industrial Automation system, Introduction of PLC and supervisory control and data acquisition (SCADA). Industrial bus systems: modbus & profibus

Unit II : Automation components: Sensors for temperature, pressure, force, displacement, speed, flow, level, humidity and pH measurement. Actuators, process control valves, power electronics devices DIAC, TRIAC, power MOSFET and IGBT. Introduction of DC and AC servomotors for motion control.

Unit III : Computer aided measurement and control systems: Role of computers in measurement and control, Elements of computer aided measurement and control, man-machine interface, computer aided process control hardware, process related interfaces, Communication and networking, Industrial communication systems, Data transfer techniques, Computer aided process control software, Computer based data acquisition system, Internet of things (IoT) for plant automation

Unit IV : Programmable logic controllers: Programmable controllers, Programmable logic controllers, Analog digital input and output modules, PLC programming, Ladder diagram, Sequential flowchart, PLC Communication and networking, PLC selection, PLC Installation, Advantage of using PLC for Industrial automation, Application of PLC to process control industries.

Unit V : Distributed Control System: Overview of DCS, DCS software configuration, DCS

communication, DCS Supervisory Computer Tasks,DCSintegration withPLCandComputers, FeaturesofDCS, AdvantagesofDCS.

ReferenceBooks:

[1]IndustrialInstrumentationandControlBy.S.K.SinghTheMcGrawHillCompanies

[2]ProcessControlInstrumentationTechnologyBy.C.D.Johnson,PHI

[3]Industrialcontrolhandbook,Parr,Newnem

[4]Programmablelogiccontroller,Dunning,Delmar

SOLAR POWER BATTERIES

Unit 1: Introduction to Solar power Systems and Energy Storage

Components of a solar power systems, Role and importance of batteries in solar systems, Fundamentals of Energy storage, Types of batteries used in solar systems(lead acid, lithium ion,etc) Battery chemistry and performance characteristics.

Unit 2: Battery Management Systems

Function and importance of BMS in solar battery systems, Balancing,monitoring and protection mechanisms, BMS design considerations and implementation

Unit 3 : Battery Configuration and Charging Techniques

Battery Storage calculations for solar power systems, Series and parallel connection of batteries, Charging algorithms and methods for solar batteries,

Unit 4 : Battery Life span and Maintenance

Maintenance and optimization of charging processes, Battery degrading and aging issues, Maintenance practices to extend battery lifespan.

Unit 5 : Integration and Safety

Integration of batteries into solar power systems, Safety considerations in Battery Storage, Regulatory and compliance aspects in battery installation.

ELECTRICAL ENERGY CONSERVATION AND AUDITING
(PROFESSIONAL ELECTIVE –II)

III Year II Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6EE38	PEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100

Course Objectives:

1. Identify the demand supply gap of energy in Indian scenario
2. Carry out energy audit of an industry/Organization
3. Draw the energy flow diagram of an industry and identify the energy wasted or a waste stream.
4. Select appropriate energy conservation method to reduce the wastage of energy
5. Evaluate the techno economic feasibility of the energy conservation technique adopted.

Course Outcomes:

At the end of this course, students will demonstrate the ability to

1. Understand the electrical wiring systems for residential, commercial and industrial consumers, representing the systems with standard symbols and drawings, SLD
2. Understand various components of industrial electrical systems.
3. Analyze and select the proper size of various electrical system components

SYLLABUS

UNIT-1 : Energy Scenario

Classes: 10

Commercial and Non-commercial energy, primary energy resources, commercial energy production, final energy consumption, energy needs of growing economy, long term energy scenario, energy pricing, energy sector reforms, energy and environment, energy security, energy conservation and its importance, restructuring of the energy supply sector, energy strategy for the future, air pollution, climate change. Energy Conservation Act-2001 and its features.

UNIT-2: Basics of Energy and its various forms

Classes: 8

Electricity tariff, load management and maximum demand control, power factor improvement, selection & location of capacitors, Thermal Basics-fuels, thermal energy contents of fuel, temperature & pressure, heat capacity, sensible and latent heat, evaporation, condensation, steam, moist air and humidity & heat transfer, units and conversion.

UNIT-3: Energy Management & Audit

Classes: 10

Definition, energy audit, need, types of energy audit. Energy management (audit) approach understanding energy costs, bench marking, energy performance, matching energy use to requirement, maximizing system efficiencies, optimizing the input energy requirements, fuel & energy substitution, energy audit instruments. Material and Energy balance: Facility as an energy system, methods for preparing process flow, material and energy balance diagrams.

UNIT- 4: Energy Efficiency in Electrical Systems

Classes: 10

Electrical system: Electricity billing, electrical load management and maximum demand control, power factor improvement and its benefit, selection and location of capacitors, performance assessment of PF capacitors, distribution and transformer losses. Electric motors: Types, losses in induction motors, motor efficiency, factors affecting motor performance, rewinding and motor replacement issues, energy saving opportunities with energy efficient motors.

UNIT- 5: Energy Efficiency in Industrial Systems**Classes: 12**

Compressed Air System: Types of air compressors, compressor efficiency, efficient compressor operation, Compressed air system components, capacity assessment, leakage test, factors affecting the performance and savings opportunities in HVAC, Fans and blowers: Types, performance evaluation, efficient system operation, flow control strategies and energy conservation opportunities. Pumps and Pumping System: Types, performance evaluation, efficient system operation, flow control strategies and energy conservation opportunities. Cooling Tower: Types and performance evaluation, efficient system operation, flow control strategies and energy saving opportunities, assessment of cooling towers.

Energy Efficient Technologies in Electrical Systems : Maximum demand controllers, automatic power factor controllers, energy efficient motors, soft starters with energy saver, variable speed drives, energy efficient transformers, electronic ballast, occupancy sensors, energy efficient lighting controls, energy saving potential of each technology.

Text/Reference Books:

1. Guide books for National Certification Examination for Energy Manager / Energy Auditors Book-1, General Aspects (available online)
2. Guide books for National Certification Examination for Energy Manager / Energy Auditors Book-3, Electrical Utilities (available online)
3. S. C. Tripathy, "Utilization of Electrical Energy and Conservation", McGraw Hill, 1991.
4. Success stories of Energy Conservation by BEE, New Delhi (www.bee-india.org)

MODELLING AND ANALYSIS OF ELECTRICAL MACHINES
(PROFESSIONAL ELECTIVE – III)

IV Year I Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6EE39	PEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100

Course Objectives:

1. To learn about the basic concepts of AC/ DC machine modelling.
2. To study about the dynamic modelling and phase transformation.
3. To understand the modelling of synchronous machine modelling.
4. To learn the performance and dynamic modelling of synchronous machines

Course Outcomes:

After going through this course the student will have the

- I. Ability to understand the various electrical parameters in mathematical form.
- II. Ability to understand the different types of reference frame theories and transformation relationships.
- III. Ability to find the electrical machine equivalent circuit parameters and modelling of electrical machines.

SYLLABUS

UNIT I: PRINCIPLES OF ELECTROMAGNETIC ENERGY CONVERSION **Classes: 10**

Magnetic circuits, permanent magnet, stored magnetic energy, co-energy – force and torque in singly and doubly excited systems – machine windings and air gap mmf – winding inductances and voltage equations.

UNIT II: DC MACHINES **Classes: 10**

Elementary DC machine and analysis of steady state operation – Voltage and torque equations – dynamic characteristics of permanent magnet and shunt D.C motors – Time domain block diagrams – solution of dynamic characteristic by Laplace transformation

UNIT III: REFERENCE FRAME THEORY **Classes: 10**

Historical background – phase transformation and commutator transformation – transformation of variables from stationary to arbitrary reference frame – variables observed from several frames of reference.

UNIT IV: INDUCTION MACHINES **Classes: 10**

Three phase induction machine, equivalent circuit and analysis of steady state operation – free acceleration characteristics – voltage and torque equations in machine variables and arbitrary reference frame variables – analysis of dynamic performance for load torque variations – digital computer simulation.

UNIT V: SYNCHRONOUS MACHINES **Classes: 10**

Three phase synchronous machine and analysis of steady state operation – voltage and torque equations in machine variables and rotor reference frame variables (Park's equations) – analysis of dynamic performance for load torque variations – Generalized theory of rotating electrical machine and Krons primitive machine.

Text Books / References:

1. Paul C.Krause, Oleg Wasyzcuk, Scott S, Sudhoff, "Analysis of Electric Machinery and Drive

Systems", John Wiley, Second Edition, 2010..

2. P S Bimbhra, "Generalized Theory of Electrical Machines", Khanna Publishers, 2008

3. A.E, Fitzgerald, Charles Kingsley, Jr, and Stephan D, Umanx, " Electric Machinery", Tata McGraw Hill, 5th Edition, 1992

4. R. Krishnan, Electric Motor & Drives: Modeling, Analysis and Control, New Delhi, Prentice Hall of India, 2001

Machine Learning Applications to Electrical Engineering

Unit 1 : Basics of Electrical Engineering for Machine Learning

Machine learning concepts and algorithms, Introduction to electrical engineering applications, Ethical considerations and limitations in ML applications, Signal processing fundamentals, Feature extraction and engineering techniques, Data representation in electrical systems.

Unit 2: Regression Techniques in Electrical Engineering

Linear regression and its applications in electrical systems, Polynomial regression and curve fitting, Regularization techniques and model evaluation, Logistic regression, Decision trees and ensemble methods, Support Vector Machines and their electrical engineering applications.

Unit 3 : Neural Networks and Deep Learning

Basics of neural networks, Feedforward, convolutional, and recurrent neural networks, Applications in signal processing and image analysis

Unit 4 : Time series analysis and Forecasting

Time series data analysis techniques, Autoregressive integrated moving average.(ARIMA) models and their applications in electrical engineering, Forecasting methods using machine learning.

Unit 5 : Unsupervised Learning in Electrical Engineering

Clustering algorithms and their applications, Dimensionality reduction techniques, Anomaly detection in electrical systems.

ELECTRICAL DISTRIBUTION SYSTEMS
(PROFESSIONAL ELECTIVE – III)

IV Year I Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6EE41	PEC	3	0	0	3	40	60	100

Course Objectives:

- 1.To distinguish between transmission and distribution systems
- 2.To understand design considerations of feeders
- 3.To compute voltage drop and power loss in feeders
- 4.To understand protection of distribution systems
- 5.To examine the power factor improvement and voltage control

Course Outcomes:

After completion of this course, the student able to

- 1.distinguish between transmission and distribution line and design the feeders
- 2.compute power loss and voltage drop of the feeders
- 3.design protection of distribution systems
- 4.understand the importance of voltage control and power factor improvement

SYLLABUS

UNIT-I: DISTRIBUTION SYSTEMS

Introduction to distribution system, Distribution system planning, Factors effecting the Distribution system planning, Load modelling and characteristics. Coincidence factor - contribution factor – Loss factor - Relationship between the load factor and loss factor. Load growth, Classification of loads (Residential, commercial, Agricultural and Industrial) and their characteristics.

DISTRIBUTION FEEDERS:

Design Considerations of Distribution Feeders: Radial, loop and network types of primary feeders, Introduction to low voltage distribution systems (LVDS) and High voltage distribution systems (HVDS), voltage levels, Factors effecting the feeder voltage level, feeder loading, Application of general circuit constants (A,B,C,D) to radial feeders, basic design practice of the secondary distribution system, secondary banking, secondary network types, secondary mains.

UNIT-II: SUBSTATIONS

Location of Substations: Rating of distribution substation, service area with 'n' primary feeders. Benefits derived through optimal location of substations. Optimal location of Substations (Perpendicular bisector rule and X, Y co-ordinate method).

System Analysis: Voltage drop and power-loss calculations: Derivation for voltage drop and power loss in lines, manual methods of solution for radial networks, three phase balanced primary lines, analysis of non-three phase systems, method to analyze the distribution feeder cost.

UNIT-III: PROTECTION

Objectives of distribution system protection, types of common faults and procedure for fault calculations, over current Protective Devices: Principle of operation of Fuses, Auto-Circuit Recloser - and Auto-line sectionalizers, and circuit breakers.

COORDINATION:

Coordination of Protective Devices: Objectives of protection co-ordination, general coordination procedure, Types of protection coordination: Fuse to Fuse, Auto-Recloser to Fuse, Circuit breaker to Fuse, Circuit breaker to Auto-Recloser.

UNIT-IV: COMPENSATION FOR POWER FACTOR IMPROVEMENT

Capacitive compensation for power-factor control - Different types of power capacitors, shunt and series capacitors, effect of shunt capacitors (Fixed and switched), effect of series capacitors, difference between shunt and series capacitors, Calculation of Power factor correction, capacitor allocation – Economic

justification of capacitors - Procedure to determine the best capacitor location.

UNIT-V: VOLTAGE CONTROL

Voltage Control: Importance of voltage control, methods of voltage control, Equipment for voltage control, effect of shunt capacitors, effect of series capacitors, effect of AVB/AVR on voltage control, line drop compensation, voltage fluctuations.

Text Books:

1. TuranGonen, Electric Power Distribution system Engineering, CRC Press, 3rd Edition 2014.
2. V. Kamaraju, Electrical Power Distribution Systems, Tata Mc Graw Hill Publishing Company, 2nd edition, 2010.

References:

1. G. Ram Murthy, Electrical Power Distribution hand book, 2nd edition, University press 2004.
2. A.S. Pabla, Electric Power Distribution, Tata McGraw Hill Publishing company, 6th edition, 2013

FLEXIBLE ALTERNATING CURRENT TRANSMISSION SYSTEMS
(PROFESSIONAL ELECTIVE –III)

IV Year I Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6EE42	PEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100

Course Objectives:

1. To understand concepts of various FACTS devices and controllers
2. To study the various converter topologies used in FACTS
3. To study the principles of operation and control of shunt series and combined FACTS controllers

Course Outcomes:

After completion of the course students will be able to:

1. Understand the operations of different FACTS devices
2. Analyse the design, modeling and applications of SVC
3. Analyse the different FACTS devices in different stability conditions
4. Analyse power flow control and stability issues in power system applications
5. Extend the knowledge of active & reactive power and voltage control with FACTS devices.

SYLLABUS

Unit- I : Transmission Lines and Series/Shunt Reactive Power Compensation

Basics of AC Transmission. Analysis of uncompensated AC transmission lines. Passive Reactive Power Compensation. Shunt and series compensation at the mid-point of an AC line. Comparison of Series and Shunt Compensation.

Unit- II : Thyristor-based Flexible AC Transmission Controllers (FACTS)

DPCCription and Characteristics of Thyristor-based FACTS devices: Static VAR Compensator (SVC), Thyristor Controlled Series Capacitor (TCSC), Thyristor Controlled Braking Resistor and Single Pole Single Throw (SPST) Switch. Configurations/Modes of Operation, Harmonics and control of SVC and TCSC. Fault Current Limiter.

Unit- III : Voltage Source Converter based (FACTS) controllers

Voltage Source Converters (VSC): Six Pulse VSC, Multi-pulse and Multi-level Converters, Pulse-Width Modulation for VSCs. Selective Harmonic Elimination, Sinusoidal PWM and Space Vector Modulation. STATCOM: Principle of Operation, Reactive Power Control: Type I and Type II controllers, Static Synchronous Series Compensator (SSSC) and Unified Power Flow Controller (UPFC): Principle of Operation and Control. Working principle of Interphase Power Flow Controller. Other Devices: GTO Controlled Series Compensator. Fault Current Limiter.

Unit- IV : Application of FACTS

Application of FACTS devices for power-flow control and stability improvement. Simulation example of power swing damping in a single-machine infinite bus system using a TCSC. Simulation example of voltage regulation of transmission mid-point voltage using a STATCOM.

Unit-V : DSTATCOM

Reactive Power Compensation, Harmonics and Unbalance mitigation in Distribution Systems using DSTATCOM and Shunt Active Filters. Synchronous Reference Frame Extraction of Reference Currents. Current Control Techniques in for DSTATCOM.

Dynamic Voltage Restorer and Unified Power Quality Conditioner: Voltage Sag/Swell mitigation: Dynamic Voltage Restorer – Working Principle and Control Strategies. Series Active Filtering. Unified Power Quality Conditioner (UPQC): Working Principle. Capabilities and Control Strategies

Text/References:

1. N. G. Hingorani and L. Gyugyi, "Understanding FACTS: Concepts and Technology of FACTS Systems", Wiley-IEEE Press, 1999.
2. K. R. Padiyar, "FACTS Controllers in Power Transmission and Distribution", New Age International (P) Ltd. 2007.
3. T. J. E. Miller, "Reactive Power Control in Electric Systems", John Wiley and Sons, New York, 1983.
4. R. C. Dugan, "Electrical Power Systems Quality", McGraw Hill Education, 2012. 5. G. T. Heydt, "Electric Power Quality", Stars in a Circle Publications, 1991

HVDC TRANSMISSION SYSTEMS
(PROFESSIONAL ELECTIVE-IV)

IV Year I Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6EE43	PEC	3	0	0	3	40	60	100

Course Objectives:

- 1.To compare EHV AC and HVDC systems
- 2.To analyze Graetz circuit and also explain 6 and 12 pulse converters
- 3.To control HVDC systems with various methods and to perform power flow analysis in AC/DC systems
- 4.To describe various protection methods for HVDC systems and Harmonics

Course Outcomes :

After completion of this course the student is able to

- 1.Compare EHV AC and HVDC system and to describe various types of DC links
- 2.Analyze Graetz circuit for rectifier and inverter mode of operation
- 3.Describe various methods for the control of HVDC systems and to perform power flow analysis in AC/DC systems
- 4.Describe various protection methods for HVDC systems and classify Harmonics and design Different types of filters

SYLLABUS

UNIT - I : INTRODUCTION

Classes: 10

DC Power transmission technology – Comparison of AC and DC transmission – Application of DC transmission – Description of DC transmission system – Planning for HVDC transmission – Modern trends in HVDC technology – DC breakers – Operating problems – HVDC transmission based on VSC – Types and applications of MTDC systems.

UNIT - II : ANALYSIS OF HVDC CONVERTERS

Classes: 10

Line commutated converter - Analysis of Graetz circuit with and without overlap - Pulse number – Choice of converter configuration – Converter bridge characteristics – Analysis of a 12 pulse converters – Analysis of VSC topologies and firing schemes.

UNIT - III : CONVERTER AND HVDC SYSTEM CONTROL

Classes: 10

Principles of DC link control – Converter control characteristics – System control hierarchy – Firing angle control – Current and extinction angle control – Starting and stopping of DC link – Power control – Higher level controllers – Control of VSC based HVDC link.

UNIT - IV : REACTIVE POWER AND HARMONICS CONTROL

Classes: 10

Reactive power requirements in steady state – Sources of reactive power – SVC and STATCOM – Generation of harmonics – Design of AC and DC filters – Active filters.

UNIT - V : POWER FLOW ANALYSIS IN AC/DC SYSTEMS

Classes: 10

Per unit system for DC quantities – DC system model – Inclusion of constraints – Power flow analysis – case study.

Text Books:

1. Padiyar, K. R., "HVDC power transmission system", New Age International (P) Ltd., New Delhi, Second Edition, 2010.
2. Edward Wilson Kimbark, "Direct Current Transmission", Vol. I, Wiley inter science, New York, London, Sydney, 1971.
3. Rakosh Das Begamudre, "Extra High Voltage AC Transmission Engineering", New Age International (P)

Ltd., New Delhi, 1990.

References:

1. Kundur P., "Power System Stability and Control", McGraw-Hill, 1993.
2. Colin Adamson and Hingorani N G, "High Voltage Direct Current Power Transmission", Garraway Limited, London, 1960.
3. Arrillaga, J., "High Voltage Direct Current Transmission", Peter Pregrinus, London, 1983

POWER SYSTEM DYNAMICS AND CONTROL
(PROFESSIONAL ELECTIVE –IV)

IV Year I Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6EE44	PEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100

Course Objectives:

- 1.To remember the dynamic characteristics of power system equipment
- 2.To recognize dynamic performance of power systems
- 3.To illustrate the system stability and controls.

Course Outcomes:

At the end of this course, students will demonstrate the ability to

- 1.Understand the problem of power system stability and its impact on the system.
- 2.Model different power system components for the study of stability.
- 3.Analyse linear dynamical systems and use of numerical integration methods
- 4.Understand the methods to improve stability.

SYLLABUS

UNIT-1: Introduction to Power System Operations

Classes: 10

Introduction to power system stability.Power System Operations and Control.Stability problems in Power System.Impact on Power System Operations and control.

UNIT-2: Analysis of Linear Dynamical System and Numerical Methods

Classes: 09

Analysis of dynamical System, Concept of Equilibrium, Small and Large Disturbance Stability.Modal Analysis of Linear System.Analysis using Numerical Integration Techniques. Issues in Modelling: Slow and Fast Transients, Stiff System.

UNIT-3: Modelling of Synchronous Machines and Associated Controllers

Classes: 11

Modelling of synchronous machine: Physical Characteristics.Rotor position dependent model.D-Q Transformation.Model with Standard Parameters.Steady State Analysis of Synchronous Machine.Short Circuit Transient Analysis of a Synchronous Machine.Synchronization of Synchronous Machine to an Infinite Bus.Modelling of Excitation and Prime Mover Systems.Physical Characteristics and Models.Excitation System ControlAutomatic Voltage Regulator.Prime Mover Control Systems. Speed Governors.

UNIT-4: Modelling of other Power System Components:

Classes: 10

Modelling of Transmission Lines and Loads. Transmission Line Physical Characteristics. Transmission Line Modelling. Load Models - induction machine model. Frequency and Voltage Dependence of Loads. Other Subsystems – HVDC and FACTS controllers, Wind Energy Systems.

UNIT- 5: Stability Analysis

Classes: 10

Angular stability analysis in Single Machine Infinite Bus System.Angular Stability in multimachine systems – Intra-plant, Local and Inter-area modes. Frequency Stability: Centre of Inertia Motion. Load Sharing: Governor droop. Single Machine Load Bus System: Voltage Stability. Introduction to Torsional Oscillations and the SSR phenomenon. Stability Analysis Tools: Transient Stability Programs, Small Signal Analysis Programs.
Enhancing System Stability Planning Measures.Stabilizing Controllers (Power System

Stabilizers).Operational Measures-Preventive Control.Emergency Control.

Text/Reference Books:

1. K.R. Padiyar, "Power System Dynamics, Stability and Control", B. S. Publications, 2002.
2. P. Kundur, "Power System Stability and Control", McGraw Hill, 1995.
3. P. Sauer and M. A. Pai, "Power System Dynamics and Stability", Prentice

SOFT COMPUTING TECHNIQUES
(PROFESSIONAL ELECTIVE –IV)

IV Year I Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6EE45	PEC	3	0	0	3	40	60	100

Course Objectives:

- 1.To understand the fundamental theory and concepts of neural networks, neuro-modeling, several neural network paradigms and its applications.
2. To understand the concepts of fuzzy sets, knowledge representation using fuzzy rules, approximate reasoning, fuzzy inference systems, and fuzzy logic control and other machine intelligence applications of fuzzy logic.
- 3 To understand the basics of an evolutionary computing paradigm known as genetic algorithms and its application to engineering optimization problems.

Course Outcomes:

The students should be able to

- 1.Comprehend the concepts of feed forward neural networks
- 2.Analyze the various feedback network.
- 3.Understand the concept of fuzziness involved in various systems and fuzzy set theory.
- 4.Comprehend the fuzzy logic control and adaptive fuzzy logic and to design the fuzzy control using genetic algorithm.
- 5.Analyze the application of fuzzy logic control to real time systems.

SYLLABUS

UNIT-I : INTRODUCTION & ESSENTIALS TO NEURAL NETWORKS

Classes: 12

Introduction, Humans and Computers, Organization of the Brain, Biological Neuron, Biological and Artificial Neuron Models, Hodgkin-Huxley Neuron Model, Integrate-and-Fire Neuron Model, Spiking Neuron Model, Characteristics of ANN, McCullochPiUs Model, Historical Developments, Potential Applications of ANN. Artificial Neuron Model, Operations of Artificial Neuron, Types of Neuron Activation Function, ANN Architectures, Classification Taxonomy of ANN — Connectivity, Neural Dynamics (Activation and Synaptic), Learning Strategy (Supervised, Unsupervised, Reinforcement), Learning Rules, Types of Application

UNIT-II: SINGLE & MULTI LAYER FEED FORWARD NEURAL NETWORKS

Classes: 10

Introduction, Perceptron Models: Discrete, Continuous and Multi-Category, Training.

Algorithms:

Discrete and Continuous Perceptron Networks, Perceptron Convergence theorem, Limitations of the Perceptron Model, Applications.Credit Assignment Problem, Generalized Delta Rule, and Derivation of Back-propagation (BP) Training, Summary of Back-propagation Algorithm, Kolmogorov Theorem, Learning Difficulties and Improvements.

UNIT-III ; ASSOCIATIVE MEMORIES-I

Classes: 08

Paradigms of Associative Memory, Pattern Mathematics, Hebbian Learning, General Concepts of Associative Memory (Associative Matrix, Association Rules, Hamming Distance, The Linear Associator, Matrix Memories, Content Addressable Memory).

UNIT-IV: ASSOCIATIVE MEMORIES-II

Classes: 10

Bidirectional Associative Memory (BAM) Architecture, BAM Training Algorithms: Storage and Recall Algorithm, BAM Energy Function, Proof of BAM Stability Theorem. Architecture of Hopfield Network: Discrete and Continuous versions, Storage and Recall Algorithm, Stability Analysis, Capacity of the Hopfield Network Summary and Discussion of Instance/Memory Based Learning Algorithms, Applications

UNIT-V: FUZZY LOGIC

Classes: 10

Classical & Fuzzy Sets: Introduction to classical sets – properties, Operations and relations; Fuzzy sets, Membership, Uncertainty, Operations, properties, fuzzy relations, cardinalities, membership functions. Fuzzy Logic System Components: Fuzzification, Membership value assignment, development of rule base and decision making system, Defuzzification to crisp sets, Defuzzification methods.

Text Books:

1. Neural Networks, Fuzzy logic, Genetic algorithms: synthesis and applications, Rajasekharan and Pal, PHI.
2. Neural Networks and Fuzzy Logic, C. Naga Bhaskar, G. Vijay Kumar, BS Publications.

Reference Books:

1. Artificial Neural Networks, B. Yegnanarayana, PHI.
2. Artificial Neural Networks, Zaruda, PHI.
3. Neural Networks and Fuzzy Logic System, Bail Kosko, PHI.
4. Fuzzy Logic and Neural Networks, M. Amirithavalli, Scitech Publications India Pvt. Ltd.
5. Neural Networks, James A Freeman and Davis Skapura, Pearson Education.
6. Neural networks by Satish Kumar, TIVIH, 2004
7. Neural Networks, Simon Haykin, Pearson Education.
8. Neural Engineering, C. Elasmith and C. H. Anderson, PHI.

HIGH VOLTAGE ENGINEERING
(PROFESSIONAL ELECTIVE –IV)

IV Year I Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks			
		L	T	P		CIE	SEE	Total	
A6EE46	PEC	3	0	0	3	40	60	100	

Course Objectives:

- 1.To deal with the detailed analysis of Breakdown occurring in gaseous, liquids and solid dielectrics
- 2.To inform about generation and measurement of High voltage and current
- 3.To introduce High voltage testing methods

Course Outcomes:At the end of the course, the student will demonstrate

1. Understand and analyze power system operation, stability, control and protection.
2. Understand the concept of Dielectric breakdown
3. Understand the generation of high voltages and high currents
4. Analyse High voltage testing & insulation coordination
5. Analyze different methods to measure high voltage and currents.

SYLLABUS

UNIT-I: BREAKDOWN IN GASES

Classes: 12

Ionization processes and de-ionization processes, Types of Discharge, Gases as insulating materials, Breakdown in Uniform gap, non-uniform gaps, Townsend's theory, Streamer mechanism, Corona discharge

Breakdown In Liquid And Solid Insulating Materials

Breakdown in pure and commercial liquids, Solid dielectrics and composite dielectrics, intrinsic breakdown, electro mechanical breakdown and thermal breakdown, Partial discharge, applications of insulating materials.

UNIT-II: GENERATION OF HIGH VOLTAGES

Classes: 08

Generation of high voltages, generation of high D. C. and A.C. voltages, generation of impulse voltages, generation of impulse currents, tripping and control of impulse generators.

UNIT-III: MEASUREMENTS OF HIGH VOLTAGES AND CURRENTS

Classes: 10

Peak voltage, impulse voltage and high direct current measurement method, cathode ray oscillographs for impulse voltage and current measurement, measurement of dielectric constant and loss factor, partial discharge measurements.

UNIT-IV: LIGHTNING AND SWITCHING OVER-VOLTAGES

Classes: 08

Charge formation in clouds, Stepped leader, Dart leader, Lightning Surges. Switching overvoltages, Protection against over-voltages, Surge diverters, Surge modifiers.

UNIT-V:

Classes: 12

HIGH VOLTAGE TESTING OF ELECTRICAL APPARATUS AND HIGH VOLTAGE LABORATORIES:

Various standards for HV Testing of electrical apparatus, IS, IEC standards, Testing of insulators and bushings, testing of isolators and circuit breakers, testing of cables, power transformers and some high voltage equipment, High voltage laboratory layout, indoor and outdoor laboratories, testing facility requirements, safety precautions in H. V. Labs.

Text Books:

1. M. S. Naidu and V. Kamaraju, "High Voltage Engineering", McGraw Hill Education, 2013.
2. C. L. Wadhwa, "High Voltage Engineering", New Age International Publishers, 2007.

References:

1. D. V. Razevicius (Translated by Dr. M. P. Chourasia), "High Voltage Engineering Fundamentals", Khanna

Publishers, 1993.

2. E. Kuffel, W. S. Zaengl and J. Kuffel, "High Voltage Engineering Fundamentals", Newnes Publication, 2000.

3. R. Arora and W. Mosch "High Voltage and Electrical Insulation Engineering", John Wiley & Sons, 2011.

4. Various IS standards for HV Laboratory Techniques and Testing

POWER SYSTEM RELIABILITY
(PROFESSIONAL ELECTIVE – V)

IV Year II Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6EE47	PEC	3	0	0	3	40	60	100

Course Objectives:

- 1.To describe the generation system model and recursive relation for capacitive model building
- 2.To explain the equivalent transitional rates, cumulative probability and cumulative frequency
- 3.To develop the understanding of risk, system and load point reliability indices
- 4.To explain the basic and performance reliability indices

Course Outcomes:

Upon the completion of this course, the student will be able to

- 1.Estimate loss of load and energy indices for generation systems model
- 2.DPCCrbe merging generation and load models
- 3.Apply various indices for distribution systems
- 4.Evaluate reliability of interconnected systems

SYLLABUS

UNIT-I: BASIC PROBABILITY THEORY

Elements of probability, probability distributions, Random variables, Density and Distribution functions- Binomial distribution- Expected value and standard deviation - Binomial distribution, Poisson Distribution, normal distribution, exponential distribution.

DEFINITION OF RELIABILITY: Definition of terms used in reliability, Component reliability, Hazard rate, derivation of the reliability function in terms of the hazard rate. Hazard models - Bath tub curve, Effect of preventive maintenance. Measures of reliability: Mean Time to Failure and Mean Time between Failures.

UNIT-II: GENERATING SYSTEM RELIABILITY ANALYSIS

Generation system model – capacity outage probability tables – Recursive relation for capacitive model building – sequential addition method – unit removal – Evaluation of loss of load and energy indices – Examples. Frequency and Duration methods – Evaluation of equivalent transitional rates of identical and non-identical units – Evaluation of cumulative probability and cumulative frequency of non-identical generating units

UNIT-III: OPERATING RESERVE EVALUATION

Basic concepts - risk indices – PJM methods – security function approach – rapid start and hot reserve units – Modeling using STPM approach.

BULK POWER SYSTEM RELIABILITY EVALUATION:

Basic configuration – conditional probability approach – system and load point reliability indices – weather effects on transmission lines – Weighted average rate and Markov model – Common mode failures.

INTER CONNECTED SYSTEM RELIABILITY ANALYSIS

Probability array method – Two inter connected systems with independent loads – effects of limited and unlimited tie capacity - imperfect tie – Two connected Systems with correlated loads – Expression for cumulative probability and cumulative frequency

UNIT-IV: DISTRIBUTION SYSTEM RELIABILITY ANALYSIS

Basic Techniques – Radial networks –Evaluation of Basic reliability indices, performance indices – load point and system reliability indices – customer oriented, loss and energy oriented indices – Examples.

UNIT-V: SUBSTATIONS AND SWITCHING STATIONS

Effects of short-circuits - breaker operation – Open and Short-circuit failures – Active and Passive failures – switching after faults – circuit breaker model – preventive maintenance – exponential Maintenance times.

Text Books:

1. Reliability Evaluation of Power systems by R. Billinton, R.N.Allan, BS Publications, 2007.
2. Reliability Modeling in Electric Power Systems by J. Endrenyi, John Wiley and Sons, 1978

REFERENCES:

1. Reliability Engineering: Theory and Practice by Alessandro Birolini, Springer Publications.
2. An Introduction to Reliability and Maintainability Engineering by Charles Ebeling, TMH Publications.
3. Reliability Engineering by E. Balaguruswamy, TMH Publications.
4. Reliability Engineering by Elsayed A. Elsayed, Prentice Hall Publications.

SMART GRID TECHNOLOGIES

(PROFESSIONAL ELECTIVE – V)

IV Year II Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6EE48	PCC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100

Course Objectives:

- 1.To understand various aspects of smart grid
- 2.To study various smart transmission and distribution technologies
- 3.To appreciate distribution generation and smart consumption
- 4.To analyze basics of electricity, electricity generation, economics of supply and demand, and the various aspects of electricity market operations in both regulated and deregulated environment

Course Outcomes:

Upon the completion of the subject, the student will be able to

1. Analyze different smart grid technologies
2. Apply the smart grid technologies for efficient power transmission and economic operation
3. Appreciate the smart transmission and distribution systems
4. Realize the distribution, generation and smart consumption technologies
5. Know the regulations and market models for smart grid

SYLLABUS

UNIT – I: Introduction to Smart Grids

Definition, justification for smart grids, smart grid conceptual model, smart grid architectures, Interoperability, communication technologies, role of smart grids standards, intelligrid initiative, national smart grid mission (NSGM) by Govt. of India

UNIT – II: Smart Transmission Technologies

Substation automation, Supervisory control and data acquisition (SCADA), energy management system (EMS), phasor measurement units (PMU), Wide area measurement systems (WAMS)

UNIT – III: Smart Distribution Technologies

Distribution automation, outage management systems, automated meter reading (AMR), automated metering infrastructure (AMI), fault location isolation and service restoration (FLISR), Outage Management Systems (OMS), Energy Storage, Renewable Integration

UNIT – IV: Distributed Generation and Smart Consumption

Distributed energy resources (DERs), smart appliances, low voltage DC (LVDC) distribution in homes / buildings, home energy management system (HEMS), Net Metering, building to Grid B2G, Vehicle to Grid V2G, Solar to Grid, Micro grid

UNIT – V: Regulations and Market Models for Smart Grid

Demand Response, Tariff Design, Time of the day pricing (TOD), Time of use pricing (TOU), Consumer privacy and data protection, consumer engagement etc. Cost benefit analysis of smart grid projects.

Text Books:

Clark W Gellings, “The Smart Grid, Enabling Energy Efficiency and Demand Side Response”- CRC Press, 2009.

Jean Claude Sabonnadière, NouredineHadjsaïd, “Smart Grids”, Wiley-ISTE, IEEE Press, May 2012

References:

- JanakaEkanayake, KithsiriLiyanage, Jianzhong. Wu, Akihiko Yokoyama, Nick Jenkins, “Smart Grid: Technology and Applications”- Wiley, 2012.
- James Momoh, “Smart Grid: Fundamentals of Design and Analysis” – Wiley, IEEE Press, 2012.
- India Smart Grid Knowledge Portal

ADVANCED ELECTRIC DRIVES
(PROFESSIONAL ELECTIVE – V)

IV Year II Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6EE49	PEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100

Course Objective:

1. The main objective of this course is to train the students to meet the industrial needs.

Course Outcomes:

At the end of this course, students will demonstrate the ability to

1. Understand the basic concepts of electric drives.
2. Understand DSP based motion controls.
3. Apply the techniques to simple real-life problems.

SYLLABUS

Unit- 1: Power Converters for AC drives

Classes: 10

PWM control of inverter, selected harmonic elimination, space vector modulation, current control of VSI, Different topologies, SVM for 3 level inverter, PWM converter as line side rectifier, current fed inverters with self-commutated devices. Control of CSI, H bridge as a 4-Q drive.

Unit- 2: Induction motor drives

Classes: 10

Different transformations and reference frame theory, modelling of induction machines, voltage fed inverter control-v/f control, vector control, direct torque and flux control(DTC).

Unit- 3: Synchronous motor drives

Classes: 10

Modelling of synchronous machines, open loop v/f control, vector control, direct torque control, CSI fed synchronous motor drives.

Unit- 4: Permanent magnet motor drives

Classes: 10

Introduction to various PM motors, BLDC and PMSM drive configuration, comparison, block diagrams, Speed and torque control in BLDC and PMSM.

Unit- 5: Switched reluctance motor drives

Classes: 10

Evolution of switched reluctance motors, various topologies for SRM drives, comparison, Closed loop speed and torque control of SRM. DSP based motion control: Use of DSPs in motion control, various DSPs available, realization of some basic blocks in DSP for implementation of DSP based motion control.

Text / References:

1. B. K. Bose, "Modern Power Electronics and AC Drives", Pearson Education, Asia, 2003.
2. P. C. Krause, O. Wasynczuk and S. D. Sudhoff, "Analysis of Electric Machinery and Drive Systems", John Wiley & Sons, 2013.
3. H. A. Taliyat and S. G. Campbell, "DSP based Electromechanical Motion Control", CRC press, 2003.
4. R. Krishnan, "Permanent Magnet Synchronous and Brushless DC motor Drives", CRC Press, 2009

ELECTRICAL AND HYBRID VEHICLES
(PROFESSIONAL ELECTIVE – V)

IV Year II Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P		CIE	SEE	Total
A6EE50	PEC	3	0	0	3	40	60	100

Course Objectives:

1. Explain the basics of electric and hybrid electric vehicles, their architecture, technologies and fundamentals.
2. Analyze various electric drives suitable for hybrid electric vehicles
3. Discuss different energy storage technologies used for hybrid electric vehicles and their control.
4. Demonstrate different configurations of electric vehicles and its components, hybrid vehicle configuration by different techniques, sizing of components and design optimization and energy management.

Course Outcomes :

At the end of this course, students will demonstrate the ability to

1. Understand the models to describe hybrid vehicles and their performance.
2. Understand the different possible ways of energy storage.
3. Understand the different strategies related to energy storage systems.

SYLLABUS

UNIT- I : Introduction To Conventional Vehicles

Classes: 10

Basics of vehicle performance, vehicle power source characterization, transmission characteristics, mathematical models to describe vehicle performance.

UNIT- II : Introduction to Hybrid Electric Vehicles

Classes: 10

History of hybrid and electric vehicles, social and environmental importance of hybrid and electric vehicles, impact of modern drive-trains on energy supplies. Hybrid Electric Drive-trains: Basic concept of hybrid traction, introduction to various hybrid drive-train topologies, power flow control in hybrid drive-train topologies, fuel efficiency analysis.

UNIT- III : Electric Drive-trains

Classes: 10

Power flow control in electric drive-train topologies, fuel efficiency analysis. Electric Propulsion unit: Introduction to electric components used in hybrid and electric vehicles, Configuration and control of DC Motor drives, Configuration and control of Induction Motor drives, configuration and control of Permanent Magnet Motor drives, Configuration and control of Switch Reluctance Motor drives, drive system efficiency.

UNIT-IV : Energy Storage

Classes: 10

Introduction to Energy Storage Requirements in Hybrid and Electric Vehicles, Battery based energy storage and its analysis, Fuel Cell based energy storage and its analysis, Super Capacitor based energy storage and its analysis, Hybridization of different energy storage devices. Sizing the drive system: Matching the electric machine and the internal combustion engine (ICE), Sizing the propulsion motor, sizing the power electronics, selecting the energy storage technology, Communications, supporting subsystems

UNIT-V : Energy Management Strategies

Classes: 10

Energy Management Strategies: Introduction to energy management strategies used in hybrid and electric vehicles, classification of different energy management strategies, comparison of different energy management strategies, implementation issues of energy management strategies. Case Studies: Design of

a Hybrid Electric Vehicle (HEV), Design of a Battery Electric Vehicle (BEV).

Text Books:

1. C. Mi, M. A. Masrur and D. W. Gao, "Hybrid Electric Vehicles: Principles and Applications with Practical Perspectives", John Wiley & Sons, 2011.
2. S. Onori, L. Serrao and G. Rizzoni, "Hybrid Electric Vehicles: Energy Management Strategies", Springer, 2015.

Reference Books:

1. M. Ehsani, Y. Gao, S. E. Gay and A. Emadi, "Modern Electric, Hybrid Electric, and Fuel Cell Vehicles: Fundamentals, Theory, and Design", CRC Press, 2004.
2. T. Denton, "Electric and Hybrid Vehicles", Routledge, 2016.

REACTIVE POWER CONTROL & MANAGEMENT
(PROFESSIONAL ELECTIVE –VI)

IV Year II Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6EE51	PEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100

Course Objectives:

1. Learn the necessity of reactive power compensation
2. Study and describe load compensation
3. Describe various types of reactive power compensation in transmission systems
4. Learn reactive power coordination system
5. To characterize distribution side and utility side reactive power management.

Course Outcomes:

Upon the completion of the subject, the student will be able to

1. Distinguish the importance of load compensation in symmetrical as well as un symmetrical loads
2. Observe and apply various compensation methods in transmission lines
3. Construct model for reactive power coordination
4. Determine demand side reactive power management & user side reactive power management based upon the dynamic characteristics of load and supply.

SYLLABUS

UNIT-I:

Load Compensation:

Objectives and specifications – reactive power characteristics – inductive and capacitive approximate biasing – Load compensator as a voltage regulator – phase balancing and power factor correction of unsymmetrical loads- examples.

UNIT-II:

Steady – State Reactive Power Compensation in Transmission System:

Uncompensated line – types of compensation – Passive shunt and series and dynamic shunt compensation – examples Transient state reactive power compensation in transmission systems: Characteristic time periods – passive shunt compensation – static compensations- series capacitor compensation – compensation using synchronous condensers – examples

UNIT-III:

Reactive Power Coordination:

Objective – Mathematical modeling – Operation planning – transmission benefits – Basic concepts of quality of power supply – disturbances- steady –state variations – effects of under voltages – frequency – Harmonics, radio frequency and electromagnetic interferences

UNIT-IV:

Demand Side Management:

Load patterns – basic methods load shaping – power tariffs- KVAR based tariffs penalties for voltage flickers and Harmonic voltage levels Distribution side Reactive power Management:: System losses –loss reduction methods – examples – Reactive power planning – objectives – Economics Planning capacitor placement – retrofitting of capacitor banks

UNIT-V:

User Side Reactive Power Management:KVAR requirements for domestic appliances – Purpose of using capacitors – selection of capacitors – deciding factors – types of available capacitor, characteristics and Limitations Reactive power management in electric traction systems and arc furnaces: Typical layout of traction systems – reactive power control requirements – distribution transformers- Electric arc furnaces – basic operations- furnaces transformer –filter requirements – remedial measures –power factor of an arc furnace

TEXT BOOKS:

- Reactive power control in Electric power systems by T.J.E. Miller, John Wiley and sons, 1982.
- Reactive power Management by D. M. Tagare, Tata McGraw Hill, 2004.

REFERENCES:

- Wolfgang Hofmann, Jurgen Schlabbach, Wolfgang Just “Reactive Power Compensation: A Practical Guide, April, 2012, Wiley publication.

ADVANCED CONTROL SYSTEMS
(PROFESSIONAL ELECTIVE –VI)

IV Year II Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6EE52	PEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100

Course Objectives:

The course should enable the students to,

- 1.Learn phase plane analysis to linear and nonlinear control systems.
- 2.Study and describe the stability of the systems using different techniques.
- 3.Learn the design of optimal controller.
- 4.Understand state variable analysis, non-linear systems and optimal control.

Course Outcomes:

After going through this course, the student will be able to

- 1.Apply phase plane analysis to linear and nonlinear control systems.
- 2.Analyse the stability of the systems using different techniques.
- 3.Illustrate the design of optimal controller.
- 4.Demonstrate state variable analysis for evaluating system stability, non-linear systems and optimal control techniques.

SYLLABUS

UNIT- I : STATE VARIABLE ANALYSIS:

Classes: 10

Concept of state , state variable and state model, state models for linear and continuous time systems, solution of state and output equation, controllability and observability, pole placement, state observer design of control systems with observers

UNIT-II : PHASE LINE ANALYSIS

Classes: 10

Features of linear and nonlinear systems, common physical non linearity's, methods of linearizing on linear systems, concept of phase portraits, singular points, limit cycles, construction of phase portraits, phase plane analysis of linear and nonlinear systems, isoclines method.

UNIT -III: DPCCRIBING FUNCTION ANALYSIS

Classes: 10

Basic concepts, derivation of dPCCribing functions for common non-linearities. DPCCribing function analysis of non-linear systems, Conditions for stability, Stability of oscillation

UNIT – IV: STABILITY ANALYSIS

Classes: 10

Introduction, Liapunov"s stability concept, Liapunov"s direct method, Lure"s transformation, Aizerman's and Kaman's conjecture, Popov's criterion

UNIT – V: OPTIMAL CONTROL

Classes: 10

Introduction, decoupling, time varying optimal control, linear quadratic regulator (LQR), steady state optimal control, optimal estimation

Text Books:

1. J Nagrath and M Gopal, „Control Systems Engineering“, New Age International Publishers
- 2.Ashish Tewari, „Modern control Design with Matlab and Simulink“, John Wiley, 2nd Edition, 2002

Reference Books:

- 1.George J Thaler, “Automatic Control Systems”, Jaico Publishers, 1st Edition, 1993.
- 2.MGopal, “Modern control system theory”, New Age International Publishers, 1st Edition, 2002.
- 3.Gene F Franklin, J David Powell, Abbasemami-Naeini, “Feedback Control of Dynamic Systems”, 4th Edition, Pearson Education, 1st Edition 2002.

POWER QUALITY
(PROFESSIONAL ELECTIVE –VI)

IV Year II Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6EE53	PEC	L	T	P	C	CIE	SEE	Total
		3	0	0	3	40	60	100

COURSE OBJECTIVES:

1. Analyze performance of Flickers and Transient Voltages.
2. Outline various Sag and Wave Form Distortions.
3. Explain Power Quality Monitoring.
4. Understand interruptions, Wave Form Distortion.

COURSE OUTCOMES:

After going through this course the student will be able to

1. Analyze the Transient Voltages and Symptoms of poor power quality
2. Analyze performance of Flickers
3. Analyze the various Sag and Wave Form Distortions.
4. Understand the Power Quality Monitoring
5. Understand interruptions, Wave Form Distortion

SYLLABUS

UNIT – I INTRODUCTION:

Importance of power quality, terms and definitions of power quality as per IEEE std. 1159 such as transients, short and long duration voltage variations, interruptions, short and long voltage fluctuations, imbalance, flickers and transients. Symptoms of poor power quality: Definitions and terminology of grounding, Purpose of groundings, Good grounding practices and problems due to poor grounding.

UNIT – II FLICKERS AND TRANSIENT VOLTAGES:

RMS voltage variations in power system and voltage regulation per unit system, complex power. Principles of voltage regulation, Basic power flow and voltage drop, Various devices used for voltage regulation and impact of reactive power management. Various causes of voltage flicker and their effects, Short term and long term flickers. Various means to reduce flickers. Transient over voltages, sources, impulsive transients, switching transients, Effect of surge impedance and line termination, control of transient voltages.

UNIT – III VOLTAGE SAG, SWELLS AND INTERRUPTIONS:

Definitions of voltage sag and interruptions, Voltage sags versus interruptions. Economic impact of voltage sag, Major causes and consequences of voltage sags, Voltage sag characteristics, Voltage sag assessment. Influence of fault location and fault level on voltage sag, Areas of vulnerability.

LIMITS AND MEASURES FOR VOLTAGE SAG:

Assessment of equipment sensitivity to voltage sags, Voltage sag limits for computer equipment, CBEMA, ITIC, SEMI F 42 curves. Representation of the results of voltage sags analysis, Voltage sag indices, Mitigation measures for voltage sags, such as UPS, DVR, SMEs, CVT etc., utility solutions and end user solutions.

UNIT – IV WAVEFORM DISTORTION:

Definition of harmonics, inter-harmonics, sub-harmonics. Causes and effect of harmonics, Voltage versus current distortion. Overview of Fourier analysis, Harmonic indices, A.C. quantities under non-sinusoidal conditions. Triplen harmonics, characteristics and non-characteristics harmonics, harmonics series and parallel resonances, Consequences of harmonic resonance. Principles for controlling harmonics, Reducing harmonic currents in loads. K-

rated transformer. Harmonic study procedure. Computer tools for harmonic analysis, Locating sources of harmonics, Harmonic filtering, passive and active filters. Modifying the system frequency response

UNIT – V POWER QUALITY MONITORING:

Need of power quality monitoring and approaches followed in power quality monitoring. Power quality monitoring objectives and requirements. Initial site survey. Power quality Instrumentation. Selection of power quality monitors, selection of monitoring location and period. System wide and discrete power quality monitoring. Setting thresholds on monitors, data collection and analysis. Selection of transducers. Harmonic monitoring, Transient monitoring, event recording and flicker monitoring.

TEXT BOOKS:

1. M. H. J. Bollen (2000), *Understanding Power Quality Problems, voltage sag and interruptions*, 1st edition, IEEE Press, New Delhi.
2. Roger. C. Dugan, Mark. F. McGranaghan, Surya Santoso, H. Wayne Beaty (2008), *Electrical Power Systems Quality*, 2nd edition, Tata McGraw Hill Publications, New Delhi.

REFERENCE BOOKS:

1. J. Arrillaga, M. R. Watson, S. Chan (2007), *Power system quality assessment*, 1st edition, John Wiley and sons, New Delhi.

Power Electronic Applications to Renewable Energy Systems

Unit 1 : Introduction to Renewable Energy Systems:

Wind power, Hydropower, Solar energy-Biomass, Bio-fuel, Geothermal Heat energy, Solar-thermal plants, Applications.

Unit 2: Solar Energy :

Introduction to PV-Cells, Array, Solar power extraction using PV-Cells, I-V Characteristics, PV-Inverters without D.C. to D.C. converters, Grid interfacing-with isolation, without isolation, Maximum power point tracking-Methods, PV-Inverters with D.C. to D.C. converters-on low frequency side and high frequency side with isolation, without isolation.

Unit 3 :Wind Energy:

Sources and potentials, Evaluation of Wind Intensity, Topography, General Classification of Wind Turbines-Rotor Turbines, Multiple-Blade Turbines, Drag Turbines, Lifting Turbines, System TARP–WARP, Generators and speed control used in wind power energy

Unit 4: Wind Power Control:

Fixed speed with capacitor bank, Rotor resistance control, DFIG, Synchronous Generator-external magnetized, Synchronous Generator-permanent magnets.

Unit 5 :Fuel Cells:

Fuel cells, Commercial Technologies for Generation of Electricity, Constructional Features of Solid Oxide Fuel Cells, Constructional Features of Proton Exchange Membrane Fuel Cells, Load Curve Peak Sharing with Fuel Cells, Advantages and Disadvantages of Fuel Cells, voltage step-up using D.C.-D.C. converter-with and without battery storage, Voltage controller for Fuel cell using D.C. – D.C. converter, Inverter interaction with fuel cell for A.C. loads, A.C. Voltage build-up and controller for fuel cells- using power converters and transformers (isolation).

References

1. Non-Conventional Energy Sources /G.D. Rai
2. Renewable Energy Technologies /Ramesh & Kumar /Narosa
3. Integration of alternative sources of energy /Felix A. Farret, M. Godoy simoes

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4. Wind power plants and projects developments, Joshua Earnest and T Wizelius, PHI, New Delhi, 2011.
 5. Handbook of renewable energy technology, World Scientific, Singapore, 2011

**OPEN ELECTIVES
OFFERED BY DEPARTMENT OF
ELECTRICAL AND ELECTRONICS
ENGINEERING**

ELECTRICAL WIRING AND SAFETY MEASURES								
(OPEN ELECTIVE – I)								
III Year II Semester								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6EE55	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	40	60	100
Course Objectives: 1. To Study the wiring diagram of residential. 2. To understand the Safety measures of Electrical wiring 3. Apply the Different types of safety rules in Electrical Systems. 4. Design the Different types of Residential Electrification Methods. 5. Identifying Different types of Substations.								
Course Outcomes: The students should be able to 1. To Analyze the safety measures & state safety precautions. 2. To Understand the Different Methods of Earthlings. 3. Apply the Different types of safety rules in Electrical Systems. 4. Design the Different types of Residential Electrification Methods. 5. Identifying Different types of Substations.								
SYLLABUS								
UNIT-I	BASICS OF ELECTRICAL INSTALLATIONS						Classes: 12	
Electric Supply System, Three phase four wire distribution system, Protection of Electric Installation against over load, short circuit and Earth fault, General requirements of electrical installations, testing of installations, Types of loads, Systems of wiring, Service connections, Service Mains, Sub-Circuits, Location of Outlets, Location of Control Switches, Location of Main Board and Distribution board, Guide lines for Installation of Fittings, Load Assessment, Permissible voltage drops and sizes of wires, estimating and costing of Electric installations.								
UNIT-II	EARTHING						Classes: 08	
Introduction & importance, Factors affecting Earth Resistance, Methods of earthing Substation and Transmission tower earthing, Neutral and Earth wire, Transformer Neutral Earthing.								
UNIT-III	SAFETY & PREVENTION OF ACCIDENTS						Classes: 08	
Definition of terminology used in safety, I.E. Act & statutory regulations for safety of persons & equipments working with electrical installation. Dos & don'ts for substation operators as listed in IS. Meaning & causes of electrical accidents factors on which severity of shock depends.								

UNIT-IV	RESIDENTIAL BUILDING ELECTRIFICATION	Classes: 10
General rules guidelines for wiring of Residential Installation and positioning of equipments. Principles of circuit design in lighting and power circuits. Procedures for designing the circuits and deciding the number of sub- circuits. Method of drawing single line diagram & wiring diagram.		
UNIT-V	RULES : SUBSTATION AND METERS	Classes: 12
Rule 28: Voltage level definitions. Rule 30: Service lines & apparatus on consumer premises. Rule 31: Cut-out on consumer's premises. Rule 46: Periodical inspection & testing of consumer's installation. Rule 47: Testing of consumer's installation. Rule 54: Declared voltage of supply to consumer. Rule 55: Declared frequency of supply to consumer. Rule 56: Sealing of meters & cut-outs. Rule 77: Clearances above ground of the lowest conductor. Rule 79: Clearances between conductors & trolley wires. Rule 87: Lines crossing or approaching each other. Rule 88: Guarding.		
Text Books:		
1. K.B. Raina, S.K.Bhattacharya Electrical Design; Estimating and costing New Age International (p) Limited, New Delhi Surjit Singh. 2. Electrical Estimating and costing Dhanpat Rai and company, New Delhi .J.B.Gupta 3. A course in Electrical Installation, Estimating & costing S.K.Kataria & sons, S.L. Uappal . 4. Electrical wiring Estimating and costing Khanna Publication. ,A.K.Sawhney		
Reference Books:		
1. Electrical Machine Design Danpat Rai & co. 2. The Electricity Rule 2005 Universal Law Publishing Co. Pvt. Ltd. N. Alagapan S. Ekambaram 3. Electrical Estimating and costing Tata Mc Graw Hill Publication, New Delhi , Surjit Singh 4. Tarlok Singh Installation, Commissioning & Maintenance of Electrical Equipment S.K.Kataria & Sons 5. B.V.S.Rao Operation & Maintenance of Electrical Machines Vol I & II Media Promoters & Publisher Ltd. Mumbai		
Web References:		
1. https://electrical-engineering-portal.com › Technical Articles 2. https://www.st-andrews.ac.uk/staff/policy/healthandsafety/publications/electricalsafety/ 3. https://www.cpwd.gov.in/Publication/Internal2013.pdf		
E-Text Books:		
1. https://books.google.co.in/books?isbn=0323170064 2. https://www.jove.com/science.../electrical-safety-precautions-and-basic-equipment		
MOOC Course		
1. https://nptel.ac.in/courses/103106071/5 2. https://nptel.ac.in/courses/108108099/28 3. https://nptel.ac.in/courses/124107001/		

ELECTRICAL MATERIALS**(OPEN ELECTIVE – I)****III Year II Semester**

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6EE56	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	40	60	100

Course Objectives:

1. To Analyze the various Conducting materials.
2. To Explain the various Semi conducting materials.
3. To Explain The Dielectrics and Insulators.
4. To Explain the Different Magnetic Materials.

5. To Describe the Optical properties of Fibers

Course Outcomes:

The students should be able to

1. Analyze the various Conducting materials.
2. Explain the various Semi conducting materials.
3. Explain The Dielectrics and Insulators.
4. Explain the Different Magnetic Materials.
5. Describe the Optical properties of Fibers.

SYLLABUS

UNIT-I	CONDUCTORS	Classes: 10
Classification: High conductivity, high resistivity materials, fundamental requirements of high conductivity materials and high resistivity materials, mobility of electron in metals, commonly used high conducting materials, copper, aluminum, bronze brass, properties, characteristics, constantan, platinum, nichrome, properties, characteristics and applications, materials used for contacts.		
UNIT-II	SEMICONDUCTORS	Classes: 08
General concepts, energy bands, types of semiconductors, Fermi Dirac distribution, intrinsic Semi-conductors, extrinsic Semi-conductors, hall effect, drift, mobility, diffusion in Semiconductors, and their applications, superconductors.		
UNIT-III	DIELECTRICS AND INSULATORS	Classes: 12
Properties of gaseous, liquid and solid dielectric, dielectric as a field medium, electric conduction in gaseous, liquid and solid dielectric, breakdown in dielectric materials,		

mechanical and electrical properties of dielectric materials, effect of temperature on dielectric materials, polarization, loss angle and dielectric loss, petroleum based insulating oils, transformer oil, capacitor oils, properties, solid electrical insulating materials, fibrous, paper boards, yarns, cloth tapes, sleeving wood, impregnation, plastics, filling and bounding materials, fibrous, film, mica, rubber, mica based materials, ceramic materials, classification of insulation (solid) and application in AC and DC machines.		
UNIT-IV	MAGNETIC MATERIALS	Classes: 10
Soft and hard magnetic materials, diamagnetic, paramagnetic and ferromagnetic materials, electric steel, sheet steel, cold rolled grain oriented silicon steel, hot rolled grain oriented silicon steel, hot rolled silicon steel sheet, hysteresis loop, hysteresis loss, magnetic susceptibility, coercive force, curie temperature, magneto-striction.		
UNIT-V	OPTICAL PROPERTIES OF SOLIDS	Classes: 10
Photo emission, photo emission materials, electro luminous junction diode, photo emitters, photo transistor, photo resistors, injection lasers, optical properties of semiconductors, application of photo sensitive materials (CRT, Tube light, photo panels etc.).		
Text Books:		
1. "Electrical Engineering Materials", Dekker, PHI Pbs. 2. "Electrical Engineering Materials", Indulkar, S. Chand		
Reference Books:		
1. "Electrical Engineering Materials", Tareev 2. "Electrical Engineering Materials", Yu. Koritsky. 3. "Electrical Engineering Materials", R.K. Rajput, Laxmi Pbs		
Web References:		
1. https://physics.info/dielectrics/ 2. https://www.oxfordreference.com/view/10.1093/oi/authority.20110803095631265 3. web.mit.edu/course/6/6.732/www/6.732-pt2.pdf		
E-Text Books:		
1. https://easyengineering.net/electrical-engineering-materials-by-dekker/ 2. https://www.oreilly.com/library/view/dielectric-materials-for/9781118619780/		
MOOC Course		
1. https://nptel.ac.in/courses/108108076/ 2. https://nptel.ac.in/courses/112104203/3 3. https://onlinecourses.nptel.ac.in/noc18_ee14/		

NEURAL NETWORKS AND FUZZY LOGIC

(OPEN ELECTIVE – II)

IV Year I Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6EE57	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	40	60	100

Course Objectives:

1. To understand the fundamental theory and concepts of neural networks, neuro-modeling, several neural network paradigms and its applications.
2. To understand the concepts of fuzzy sets, knowledge representation using fuzzy rules, approximate reasoning, fuzzy inference systems, and fuzzy logic control and other machine intelligence applications of fuzzy logic.
3. To understand the basics of an evolutionary computing paradigm known as genetic algorithms and its application to engineering optimization problems.

Course Outcomes:

The students should be able to

1. Comprehend the concepts of feed forward neural networks
2. Analyze the various feedback networks.
3. Understand the concept of fuzziness involved in various systems and fuzzy set theory.
4. Comprehend the fuzzy logic control and adaptive fuzzy logic and to design the fuzzy control using genetic algorithm.
5. Analyze the application of fuzzy logic control to real time systems.

SYLLABUS

UNIT-I	INTRODUCTION & ESSENTIALS TO NEURAL NETWORKS	Classes: 12
Introduction, Humans and Computers, Organization of the Brain, Biological Neuron, Biological and Artificial Neuron Models, Hodgkin-Huxley Neuron Model, Integrate-and-Fire Neuron Model, Spiking Neuron Model, Characteristics of ANN, McCullochPiUs Model, Historical Developments, Potential Applications of ANN. Artificial Neuron Model, Operations of Artificial Neuron, Types of Neuron Activation Function, ANN Architectures, Classification Taxonomy of ANN — Connectivity, Neural Dynamics (Activation and Synaptic), Learning Strategy (Supervised, Unsupervised, Reinforcement), Learning Rules, Types of Application		
UNIT-II	SINGLE & MULTI LAYER FEED FORWARD NEURAL NETWORKS	Classes: 10

Introduction, Perceptron Models: Discrete, Continuous and Multi-Category, Training.		
Algorithms: Discrete and Continuous Perceptron Networks, Perceptron Convergence theorem, Limitations of the Perceptron Model, Applications. Credit Assignment Problem, Generalized Delta Rule, and Derivation of Back-propagation (BP) Training, Summary of Back-propagation Algorithm, Kolmogorov Theorem, Learning Difficulties and Improvements.		
UNIT-III	ASSOCIATIVE MEMORIES-I	Classes: 08
Paradigms of Associative Memory, Pattern Mathematics, Hebbian Learning, General Concepts of Associative Memory (Associative Matrix, Association Rules, Hamming Distance, The Linear Associator, Matrix Memories, Content Addressable Memory).		
UNIT-IV	ASSOCIATIVE MEMORIES-II	Classes: 10
Bidirectional Associative Memory (BAM) Architecture, BAM Training Algorithms: Storage and Recall Algorithm, BAM Energy Function, Proof of BAM Stability Theorem. Architecture of Hopfield		
Network: Discrete and Continuous versions, Storage and Recall Algorithm, Stability Analysis, Capacity of the Hopfield Network Summary and Discussion of Instance/Memory Based Learning Algorithms, Applications		
UNIT-V	FUZZY LOGIC	Classes: 10
Classical & Fuzzy Sets: Introduction to classical sets – properties, Operations and relations; Fuzzy sets, Membership, Uncertainty, Operations, properties, fuzzy relations, cardinalities, membership functions.		
Fuzzy Logic System Components: Fuzzification, Membership value assignment, development of rule base and decision making system, Defuzzification to crisp sets, Defuzzification methods.		
Text Books:		
1. Neural Networks, Fuzzy logic, Genetic algorithms: synthesis and applications, Rajasekharan and Pal, PHI. 2. Neural Networks and Fuzzy Logic, C. Naga Bhaskar, G. Vijay Kumar, BS Publication-is.		
Reference Books:		
1. Artificial Neural Networks, B. Yegnanarayana, PHI. 2. Artificial Neural Networks, Zaruda, PHI. 3. Neural Networks and Fuzzy Logic System, Bail Kosko, PHI. 4. Fuzzy Logic and Neural Networks, M. Amirthavalli, Scitech Publications India Pvt. Ltd. 5. Neural Networks, James A Freeman and Davis Skapura, Pearson Education. 6. Neural networks by satish Kumar, TIVIH, 2004 7. Neural Networks, Simon Hakens, Pearson Education. 8. Neural Engineering, C.Eliasmith and CH.Anderson, PHI.		
Web References:		
1. users.monash.edu/~app/CSE5301/Lnts/LaD.pdf 2. https://engineering.purdue.edu/~tsoukala/rational.html 3. https://pdfs.semanticscholar.org/5e31/c55a00eb3945e3e483caa2e146a95c12f5aa.pdf		

E-Text Books:

1. <https://www.mheducation.co.in/computer.../neural-networks-fuzzy-systems/text-book>
2. www.crectirupati.com/sites/default/files/lecture_notes/NNFL.pdf
3. www.vssut.ac.in/lecture_notes/lecture1423723637.pdf

MOOC Course:

1. https://nptel.ac.in/noc/individual_course.php?id=noc19-ge07
2. <https://nptel.ac.in/courses/108104049/16>
3. <https://nptel.ac.in/courses/117105084/>
4. <https://nptel.ac.in/syllabus/127105006/>

ENERGY STORAGE SYSTEMS								
(OPEN ELECTIVE – II)								
IV Year I Semester								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6EE58	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	40	60	100
Course Objectives: 1. To enable the student to understand the need for energy storage, devices and technologies available and their applications. 2. To Analyze the characteristics of energy from various sources and need for storage 3. To Classify various types of energy storage and various devices used for the purpose 4. To Explain the Pumped hydro Storage System.								
Course Outcomes: The students should be able to 1. Analyze the characteristics of energy from various sources and need for storage 2. Classify various types of energy storage and various devices used for the purpose 3. Explain the Pumped hydro Storage System. 4. Identify various real time applications.								
SYLLABUS								
UNIT-I	ELECTRICAL ENERGY STORAGE TECHNOLOGIES						Classes: 12	
Characteristics of electricity, Electricity and the roles of EES, High generation cost during peak-demand periods, Need for continuous and flexible supply, Long distance between generation and consumption, Congestion in power grids, Transmission by cable.								
UNIT-II	NEEDS FOR ELECTRICAL ENERGY STORAGE						Classes: 08	
Emerging needs for EES, More renewable energy, less fossil fuel, Smart Grid uses, The roles of electrical energy storage technologies, The roles from the viewpoint of a utility, The roles from the viewpoint of consumers, The roles from the viewpoint of generators of renewable energy.								
UNIT-III	FEATURES OF ENERGY STORAGE SYSTEMS						Classes: 08	
Classification of EES systems , Mechanical storage systems, Pumped hydro storage (PHS), Compressed air energy storage (CAES), Flywheel energy storage (FES),Electrochemical storage systems, Secondary batteries, Flow batteries, Chemical energy storage, Hydrogen (H2),Synthetic natural gas (SNG).								

UNIT-IV	TYPES OF ELECTRICAL ENERGY STORAGE SYSTEMS	Classes: 10
Electrical storage systems, Double-layer capacitors (DLC) ,Superconducting magnetic energy storage (SMES),Thermal storage systems, Standards for EES,Technical comparison of EES technologies.		
UNIT-V	APPLICATIONS	Classes: 12
Present status of applications, Utility use (conventional power generation, grid operation & service) , Consumer use (uninterruptable power supply for large consumers), New trends in applications ,Renewable energy generation, Smart Grid, Smart Micro grid, Smart House, Electric vehicles, Management and control hierarchy of storage systems, Internal configuration of battery storage systems, External connection of EES systems , Aggregating EES systems and distributed generation (Virtual Power Plant), Battery SCADA– aggregation of many dispersed batteries.		
Text Books:		
1. “James M. Eyer, Joseph J. Iannucci and Garth P. Corey “ , “Energy Storage Benefits and Market Analysis”, Sandia National Laboratories, 2004. 2. The Electrical Energy Storage by IEC Market Strategy Board.		
Reference Books:		
1. “Jim Eyer, Garth Corey”, Energy Storage for the Electricity Grid: Benefits and Market Potential Assessment Guide, Report, Sandia National Laboratories, Feb 2010.		
Web References:		
1. https://onlinelibrary.wiley.com/doi/abs/10.1002/9781118991978.hces212		
E-Text Books:		
1. https://www.pewtrusts.org/~media/.../energy_storage-backs_up_power_supply.pdf 2. https://energy.mit.edu/wp-content/uploads/2018/04/Energy-Storage-for-the-Grid.pdf 3. https://www.adb.org/sites/default/files/.../handbook-battery-energy-storage-system.pdf		
MOOC Course:		
1. nptel.ac.in/courses/112105221/56 2. nptel.ac.in/courses/108108036/9 3. https://nptel.ac.in/courses/108102047/7		

SOLAR ENERGY AND APPLICATIONS								
(OPEN ELECTIVE – III)								
IV Year II Semester								
Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6EE59	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	40	60	100
Course Objectives: 1. To Study about solar modules and PV system design and their applications 2. To Deal with grid connected PV systems 3. To Discuss about different energy storage systems								
Course Outcomes: The students should be able to 1. understanding of principles and technologies for solar thermal energy collection, conversion and utilization 2. Understanding of solar heating systems, liquid based solar heating systems for buildings. 3. Identify, formulate and solve simple to complex problems of solar thermal energy conversion and storage. 4. Identify and understand solar thermal systems'components and their function. 5. Analyze hot water load and solar resource data and use this information to properly size a solar thermal system								
SYLLABUS								
UNIT-I	INTRODUCTION						Classes: 10	
Characteristics of sunlight – semiconductors and P-N junctions –behavior of solar cells – cell properties – PV cell interconnection								
UNIT-II	STAND ALONE PV SYSTEM						Classes: 10	
Solar modules – storage systems – power conditioning and regulation - MPPT- protection – stand alone PV systems design – sizing								
UNIT-III	GRID CONNECTED PV SYSTEMS						Classes: 10	
PV systems in buildings – design issues for central power stations – safety – Economic aspect – Efficiency and performance - International PV programs								
UNIT-IV	ENERGY STORAGE SYSTEMS						Classes: 10	
Impact of intermittent generation – Battery energy storage – solar thermal energy storage – pumped hydroelectric energy storage								

UNIT-V	APPLICATIONS	Classes: 10
Water pumping – battery chargers – solar car – direct-drive applications –Space – Telecommunications.		
Text Books:		
1. Solanki C.S., "Solar Photovoltaics: Fundamentals, Technologies And Applications", PHI Learning Pvt. Ltd.,2015. 2. Stuart R.Wenham, Martin A.Green, Muriel E. Watt and Richard Corkish, "Applied Photovoltaics", 2007,Earthscan, UK. Eduardo Lorenzo G. Araujo, "Solar electricity engineering of photovoltaic systems", Progensa,1994		
Reference Books:		
1. Frank S. Barnes & Jonah G. Levine, "Large Energy storage Systems Handbook", CRC Press, 2011. 2. McNeils, Frenkel, Desai, "Solar & Wind Energy Technologies", Wiley Eastern, 1990 5 S.P. Sukhatme , "Solar Energy", Tata McGraw Hill,1987.		
Web References:		
1. https://www.loc.gov/rr/scitech/tracer-bullets/solar-updatetb.html 2. https://www.oxfordreference.com/view/10.1093/oi/authority.20110803100516798 3. https://link.springer.com/journal/11949		
E-Text Books:		
1. https://courses.edx.org/c4x/DelftX/ET.3034TU/asset/solar_energy_v1.1.pdf 2. bookstore.teri.res.in/books/9788179935736		
MOOC Course:		
1. https://nptel.ac.in/courses/112105051/ 2. https://nptel.ac.in/courses/121106014/18 3. https://nptel.ac.in/courses/112105050/ 4. https://nptel.ac.in/syllabus/112105051/		

INSTRUMENTATION AND CONTROL

(OPEN ELECTIVE – III)

IV Year II Semester

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A6EE65	OEC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	40	60	100

Course Objectives:

1. To introduce the basic principles of all measuring instruments
2. To deal with the measurement of voltage, current measurements
3. To understand students how different types of meters work and their construction
4. To understand the different ways of system representations such as Transfer function representation and state space representations and to assess the system dynamic response
5. To assess the system performance using time domain analysis and methods for improving it

Course Outcomes:

The students should be able to

1. Identify the instruments suitable for typical measurements.
2. Apply the knowledge about transducers and instrument transformers to use them effectively.
3. Improve the system performance by selecting a suitable controller for a specific application
4. Apply various time domain techniques to assess the system performance

SYLLABUS

UNIT-I	INTRODUCTION TO MEASURING INSTRUMENTS	Classes: 10
Classification —deflecting, control and damping torques — Ammeters and Voltmeters — PMMC, moving iron type instruments — expression for the deflecting torque and control torque — Errors and compensations, extension of range using shunts and series resistance. Electrostatic Voltmeters-electrometer type and attracted disc type — Extension of range of E.S. Voltmeters.		
UNIT-II	POTENTIOMETERS & INSTRUMENT TRANSFORMERS	Classes: 08
Principle and operation of D.C. Crompton's potentiometer—standardisation — Measurement of unknown resistance, current, voltage. A.C. Potentiometers: polar and coordinate types standardisation — applications. CT and PT — Ratio and phase angle errors.		
UNIT-III	TRANSDUCERS & OSCILLOSCOPES	Classes: 12
Definition of transducers, Classification of transducers, Advantages of Electrical transducers, Characteristics and choice of transducers; Principle operation of LVDT and capacitor transducers; LVDT Applications, Strain gauge and its principle of operation, gauge factor, Thermistors, Thermo couples, Piezo electric transducers, photovoltaic, photo conductive cells, photo diodes. CRO: Cathode ray oscilloscope-Cathode ray tube-time base generator- horizontal and vertical amplifiers-CRO probes-applications of CRO- Measurement of phase and frequency-lissajous patterns.		

UNIT-IV	CLASSIFICATION OF CONTROL SYSTEMS	Classes: 10
Concepts of Control Systems- Open Loop and closed loop control systems and their differences- Different examples of control systems- Classification of control systems Feed-Back Characteristics, Effects of feedback. Mathematical models – Differential equations – Impulse Response and transfer functions .Block diagram algebra – Representation by Signal flow graph – Reduction using mason's gain formula.		
UNIT-V	TIME RESPONSE ANALYSIS	Classes: 10
Standard test signals – Time response of first order systems –Characteristic Equation of Feedback control systems, Transient response of second order systems – Time domain specifications .Effects of proportional derivative, proportional integral systems.		
Text Books:		
1. Electrical and Electronic Measurements and Instrumentation, R. K. Rajput, S. Chand & Company Ltd. 2. Electrical Measuring Instruments and Measurements, S. C. Bhargava, BS Publications. 3. I. J. Nagrath and M. Gopal", "Control Systems Engineering", New Age International (P) Limited, Publishers, 5th edition, 2009 4. "B. C. Kuo", "Automatic Control Systems", John wiley and sons, 8th edition, 2003.		
Reference Books:		
1. Electrical & Electronic Measurement & Instruments, A.K.Sawhney Dhanpat Rai & Co. Publications. 2. Electrical and Electronic Measurements, G. K. Banerjee, PHI Learning Pvt. Ltd. 3. Electrical Measurements and Measuring Instruments, Golding and Widdis, Reem Publications. 4. N. K. Sinha", "Control Systems", New Age International (P) Limited Publishers, 3rdEdition, 1998. 5. "NISE", "Control Systems Engineering", John wiley, 6th Edition, 2011. 6. "Katsuhiko Ogata", "Modern Control Engineering", Prentice Hall of India Pvt. Ltd., 3rd edition, 1998.		
Web References:		
1. home.mit.bme.hu/~virosztek/docs/mt.../Principles_of_electrical_measurement.pdf 2. https://www.mccdag.com/handbook/chapt_4.aspx		
E-Text Books:		
1. www.vssut.ac.in/lecture_notes/lecture1423813026.pdf 2. www.vssut.ac.in/lecture_notes/lecture1423904331.pdf 3. www.ent.mrt.ac.lk/~rohan/teaching/EN5001/Reading/DORFCH1.pdf		
MOOC Course:		
1. https://nptel.ac.in/syllabus/108106070/ 2. https://nptel.ac.in/courses/108105064/ 3. https://nptel.ac.in/courses/108101037/		